UNIVERSITY OF COIMBRA

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Project Management

OliveTrack

Olive Harvest Management Platform

Intermediate Project Report

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1 Introduction

1.1 Purpose of the Document

This document defines the proposed requirements and architecture of OliveTrack, a platform for managing olive harvest operations.

It establishes a shared understanding of the problem, intended users, functional scope, business rules and expected system behaviour. It also connects the use cases to the functional requirements, quality requirements and architectural decisions developed by the team. Section 10 also records the working delivery-risk assessment and response approach. The concise summary in Section 10.6 connects the specification to effort, delivery planning and progress evidence.

The document supports both product development and project management. An explicit description of the expected behaviour helps the team identify missing work, define implementation and validation tasks, allocate responsibilities and prepare effort estimates.

This working specification complements the estimation workbook, team hours sheet and other course deliverables. It provides a shared requirements and design baseline for review; it does not replace task assignments, an approved delivery plan or acceptance evidence.

1.2 Project Context

Olive harvest management involves information that is produced by different people and becomes available at different times.

Before fieldwork, the owner may optionally plan the expected workers, work areas and, where known, schedules and tasks. During or after the working day, actual attendance, work periods, tasks and breaks must be recorded.

Harvested olives are transported and delivered to a cooperative or another receiving destination. Weighing tickets provide information about the delivered load. Analysis results and monetary values may arrive later.

Worker documents, including contracts and insurance records, must also be accessible. Some documents require a signing interaction. Insurance information is particularly important when work takes place in the field.

When this information is distributed across paper records and separate files, it becomes harder to retrieve, reconcile and summarise the campaign's results. Differences between planned and actual work, missing analysis results and ambiguous weight values can also lead to incorrect conclusions.

OliveTrack proposes a shared digital environment for the owner's business. It supports properties with optional parcels, harvest campaigns covering one or more areas, worker records, operational entry, document consultation and campaign indicators.

Field connectivity may be unreliable. Essential field operations therefore require prepared offline access and local entry, followed by synchronisation when connectivity returns. Administrative operations may require connectivity, as specified in the individual use cases.

The project is being developed within the Project Management course. The proposed scope must remain compatible with the academic schedule, available resources and agreed delivery objectives.

1.3 Objectives

The proposed solution pursues the following objectives:

1. **Organise the harvest context.** Maintain identifiable properties, optional parcels and campaigns without mixing operational records from different areas or harvest periods.
2. **Separate planning from actual work.** Support daily and multi-day planning, while requiring explicit confirmation of attendance and actual work.
3. **Record work consistently.** Distinguish machinery operation from general work, account for explicitly recorded breaks and prevent overlapping actual periods for the same worker.
4. **Track deliveries and source information.** Connect transport, delivery, weighing, analysis and monetary information without inventing missing measurements or applying deductions twice.
5. **Support worker-document workflows.** Provide controlled access to uploaded documents, prepared offline insurance consultation and the agreed signing interaction.
6. **Provide understandable indicators.** Present campaign results with clear units, reporting scope, data completeness and provisional or final status where applicable.
7. **Support work in the field.** Allow essential operations to continue with prepared authorised data when a network connection is unavailable.

Detailed measurable targets are addressed in the non-functional requirements. The objectives above do not constitute verified performance or operational results.

1.4 Sources and Document Status

The current document draws on:

- The *OliveTrack Team Guide*, dated 9 October 2026, describing the business context, illustrative examples and unresolved questions.
- The existing Statement of Work, persona descriptions, general architecture, data architecture and non-functional requirements proposals supplied to the team.
- Initial responses supplied by the teammate acting as an intermediary with the client.
- The subsequent requirements review, including the use-case discussions and proposed resolutions recorded during document development.
- The team's security, privacy and reliability review and the updated performance, usability and availability proposal dated 3 November 2026, reconciled with the current functional decisions.
- The team's *OliveTrack General Architecture*, version 1.1, and *OliveTrack Risk Management Plan*, dated November 2026, reconciled with the current functional/data baseline. Unconfirmed voting, milestone, resource and provider assumptions retain explicit review status.

This working specification consolidates the supplied sources and subsequent requirements, business-rule and technical-decision reviews, including the stack, access model and quality targets adopted during the author's review of 5 November 2026. Illustrative examples, working decisions and

client-provided responses have different validation status. Conflicting statements require an explicit decision; author review does not establish formal approval by the whole team or client, or legal validation of templates or signatures.

Section 11 records current decisions, residual questions and validation responsibilities; the detailed questionnaire is maintained separately. The intermediate report is submitted on 6 November 2026. The presentation on 10 November 2026 uses a browser-based navigable prototype with sample data. Throughout this report, specified behaviour, prototype interaction, implemented functionality and verified results are distinct. The demonstration does not establish production security, durable synchronisation or operational acceptance.

Project Deliverables. The files developed during this project phase are collected in the shared project folder: this report and its LaTeX source, effort estimates, recorded team hours, the supplementary data reference, team roster and navigable demonstration. The presentation slides will be added before the presentation on 10 November 2026.

Open the shared project folder.

How to use this specification. Sections 2 and 3 establish stakeholders and boundaries; Sections 4–7 define behaviour, shared constraints and quality criteria. Sections 8 and 9 describe the solution and data design. Section 10 connects risks to delivery planning, and Section 11 records validation, decisions and unresolved matters. Read the full data dictionary only when implementing or checking fields not repeated in the principal model.

Course alignment and AI assistance. The supplied course materials guide requirement organisation/change control, risk assessment and planning. Templates are used selectively, without claiming that every taught process has been performed. AI assistance was used in drafting/restructuring this report, preparing LaTeX and developing the navigable prototype. These contributions require author verification; they are not client approval, executed acceptance tests or evidence of production readiness. The team-wide inventory of tools, contributions and review responsibilities remains to be completed under Q-PM-02 in accordance with the course disclosure requirements.

2 Users, Personas and Permissions

2.1 Stakeholders

OliveTrack affects people who use the application, people whose information is managed and parties that supply information or assess the project.

Farm Owner and Client. Explains the operational needs, validates business behaviour and uses the application to manage the harvest and inspect its results.

Field Manager. A delegated operational user included in the working specification, acting within the scope and permissions assigned by the owner.

Workers. Their profiles, attendance, work periods and documents are managed. Workers may participate in signing without holding application accounts.

Cooperative or Receiving Organisation. Supplies weighing information, analysis results and monetary values. Direct access to OliveTrack is not assumed in the current scope.

Project Team. Analyses requirements, proposes the solution, implements and validates the agreed scope, and maintains the project documentation.

Course Faculty and Evaluators. Establish academic expectations and assess the project deliverables, management process and demonstrated results.

Business validation and academic acceptance are related but distinct. Client agreement does not replace the course's assessment requirements.

2.2 User Personas

The profiles below describe working assumptions for requirements discussion. They represent roles and usage contexts; they are not evidence of a completed user study.

2.2.1 Farm Owner

Role and context. Responsible for the business's properties, workers and harvest decisions. May prepare administrative information with connectivity and consult or record operational information in the field.

Main goals.

- Organise properties, parcels and campaigns.
- Prepare work without repeatedly entering the same information.
- Understand who attended and what work was actually performed.
- Access the correct worker documents.
- Follow deliveries and later results.
- Review campaign totals and investigate their supporting records.

Main difficulties. Information arrives at different times, manual records can be fragmented, and field connectivity is unreliable.

Design implications. Use clear business terminology, group entry with individual adjustments, explicit save states and understandable summaries with access to detail.

2.2.2 Field Manager

Status. Included operational persona, invited by the owner.

Role and context. May coordinate workers and field operations on behalf of the owner. Uses the application within an explicitly authorised operational scope.

Main goals.

- Consult prepared schedules.
- Record attendance and actual work.
- Record transport and delivery information.
- Access relevant worker insurance during field operations.

Access limitations. Work-entry permission does not automatically include correction, contract or configuration rights. Managers initially have financial consultation and complete-ticket access, but financial entry and correction require additional grants.

Design implications. Prioritise essential field tasks, prepared offline information and clear permission boundaries.

2.2.3 Worker as a Participant

Role and context. A person whose work and documents are managed. May review and sign a document in person on the owner's or authorised manager's device.

Main needs.

- Be correctly identified in records.
- Review the document intended for signing.
- Review, redraw or cancel before confirming a signature.
- Avoid unintended access to other workers' information.

No worker self-service application or management account is assumed.

2.3 System Actors

The formal actor name is *Farm Owner*; running text uses *owner*.

Farm Owner. Primary application actor using an account within the owner's business workspace.

Field Manager. Application actor with owner-delegated permissions.

Worker. Reviews and signs documents through an authorised account holder's device, without an application account.

The cooperative is currently an information source, rather than a direct application actor. Likewise, identifying a worker as a driver does not create another application role.

The owner and manager are user roles. A worker record is a business record; creating it does not create an account. These concepts are distinct identities.

Visitor. A Visitor is a person who is not authenticated. Visitors may register as owners, verify their email, log in or initiate account recovery. They cannot access business information.

Authenticated account holders. An owner holds an application account associated with the business created through registration.

A manager receives business access through an owner invitation and acts within the assigned scope and permissions. Creating this invited account does not create an owner workspace.

Worker participation follows the account/worker distinction above.

2.4 Access Permissions

The owner assigns properties and independently configurable operation permissions. The table gives the manager defaults, which the owner reviews on invitation and may subsequently change. Each operation grouped in a row is configured separately. Requests and offline acceptance validate membership, scope and operation authority.

Table 1: Initial permissions for the Field Manager.

Manager capability	Initial setting
Consult properties, parcels and campaigns.	Enabled.
Consult worker profiles and contact details.	Enabled.
Consult, create and edit work plans, subject to plan-freezing rules.	Enabled.
Confirm attendance and record actual assignments, periods and breaks.	Enabled.
Record departures, deliveries and arrival confirmations.	Enabled.
Record weighing data and analyses, excluding financial fields unless separately granted.	Enabled.
Consult operational and financial campaign indicators.	Enabled within assigned scope.
Consult insurance for workers under the manager's responsibility.	Mandatory within that responsibility.
Create or maintain worker profiles.	Disabled; owner may grant each operation.
Manage properties and parcels.	Disabled; owner may grant each operation.
Manage campaign creation, coverage and lifecycle.	Disabled; owner may grant each operation.
Correct confirmed work, deliveries, weighing, analyses or financial entries; delete or cancel eligible records.	Disabled; separate correction/deletion authority required.
Consult contracts/other documents; upload, generate documents or initiate signing.	Disabled; owner may grant each operation.
Consult current monetary values and complete tickets/source attachments, including prepared offline copies.	Enabled; owner may revoke consultation.
Enter or correct structured monetary values, penalties and valuations.	Disabled; owner may grant each operation.
Invite managers, assign scopes or administer permissions.	Owner only.

Delegated scope. Property scope includes its parcels and current/future campaigns until revoked; there are no separate campaign/parcel scope grants. Several managers may share a property. Multi-property campaign views disclose only authorised records and attributable indicators, identifying partial scope. Relevant workers and their insurance are available for scoped operations; the owner may explicitly make additional workers available before assignment. This does not reveal those workers' work on other properties.

Daily attendance remains one shared worker/date decision within the business, not one per manager or property; concurrent changes follow revision/conflict rules. Complete mixed-property delivery records and source files require authority over all disclosed property contexts. Partial access exposes only known attributable authorised contributions, without inferred allocation.

Insurance consultation cannot be removed while the manager remains responsible for those workers. This guarantees access; it does not require opening or checking insurance before every field action. Profile creation does not establish documentary readiness and requires its own grant, including for same-day offline creation.

Financial consultation and complete-ticket/source-file access are configured consistently. Structured financial entry and correction require separate grants. Evidence upload follows its operation's entry/attachment permission; worker-document upload is independent. Correction authority applies to confirmed records, not to reviewing unsaved input or editing an unfrozen plan.

Prepared offline information follows the same grants and account/workspace isolation. Access changes and expiry follow BR-002; the local-protection details remain under Q-DOC-07. No instantaneous remote removal from a disconnected device is promised.

3 Scope and Feature Overview

3.1 Included Scope

OliveTrack supports olive-harvest management within one owner's business workspace: account access, delegated management, properties and optional parcels, campaigns, workers, optional planning, attendance and actual work, worker documents and signing, deliveries, cooperative results and campaign indicators.

Table 2 maps these areas to their use cases. Detailed behaviour, permissions and correction conditions are specified in Sections 4–6. Supported field operations may continue offline; connectivity is defined per operation. Uploaded identity and work-permit evidence is not issued or verified by document generation.

3.2 Excluded Scope

Product scope describes supported system capabilities; project scope also includes the work needed to specify, build, verify and deliver them. The scope below is the current working baseline; inclusion does not assign a capability to an approved release. The first functional increment remains subject to Q-SCOPE-01.

The current proposal excludes:

- Payroll execution, wage calculation, worker remuneration rates and payment processing.
- Invoicing, full accounting and commercial olive oil sales.
- Application-calculated price estimates for deliveries.

- Worker self-service accounts for attendance or business management.
- Direct cooperative or laboratory integrations.
- Automatic extraction of values from ticket images or documents.
- Independent transport-trip management beyond the delivery-associated transport record.
- Management of agricultural activities unrelated to the agreed olive-harvest scope.
- Percentage-based allocation of mixed-load weights or amounts in this version.
- Ownership transfer and selectable local-only/cloud-only storage modes.

The following are optional enhancements, rather than committed first-version capabilities:

- Map-based property and parcel boundaries, visual area selection and estimated map areas.
- Cross-campaign or cross-year comparisons.
- Detailed Excel exports.
- Dedicated document-printing workflows.

A dedicated work-attendance export remains pending.

3.3 MVP Proposal Pending Approval

The Minimum Viable Product has not yet been formally prioritised and approved.

The current proposed baseline is an end-to-end harvest workflow in which the owner can:

1. Prepare properties, campaign coverage, workers and necessary documents.
2. Plan work and consult prepared schedules.
3. Confirm attendance and record actual periods and tasks in the field, including without connectivity.
4. Record transport and delivery, then complete ticket, analysis and reported monetary information as it becomes available.
5. Consult relevant insurance, use the agreed signing interaction and inspect campaign results.

Prioritisation remains a team/client decision, not an inference from catalogue order. A small Must/Should/Could/Won't-this-delivery scheme is suitable for this project; no priorities are assigned here without that review. Record the selected requirement/criterion, rationale, dependencies, effort and target increment on the shared backlog.

Use estimates, available resources and operational acceptance needs to select the first functional increment. Optional maps, percentage allocation and unselected templates are not automatically included. Signing, local document protection and recovery require operational validation.

For the intermediate presentation, the deliverable is the navigable prototype described in Section 1.4, rather than a verified functional release. Its demonstration of the main workflow does not settle implementation priorities.

3.4 Feature Overview

Table 2: Functional areas and corresponding use cases.

Area	Main capability	Use cases
Account access	Owner registration, email verification, login, logout and recovery.	UC-01–UC-03
Manager access	Invitations, operational defaults, adjustable permissions and revocation.	UC-04
Land and campaign organisation	Properties, optional parcels, campaign coverage, planned/actual dates and lifecycle.	UC-05–UC-06
Workforce	Identified worker profiles, machinery capability and authorised same-day offline creation.	UC-07
Planning	Daily and multi-day plans, reuse and schedule consultation.	UC-08–UC-09
Actual work	Attendance, task periods, breaks and corrections.	UC-10–UC-11
Documents and signing	Document access, uploads, retention, offline preparation and worker-specific signing.	UC-12–UC-14
Harvest logistics	Transport-associated deliveries, weighing tickets and analyses.	UC-15–UC-17
Reported values	Current cooperative-reported ticket amounts.	UC-18
Campaign review	Indicators, supporting detail and PDF summaries.	UC-19
Document generation	Reviewed template-based generation, followed by version-specific signing.	UC-20

Administrative configuration normally requires connectivity. Essential field entry and consultation use prepared authorised information and explicit synchronisation states.

Offline capability is defined per operation. It does not imply that the entire application or every historical record is available without a connection.

3.5 Assumptions and Dependencies

3.5.1 Business Assumptions

The baseline provides one business workspace per owner. Several independently owned workspaces through the same owner account are outside scope. Managers use owner-assigned property scope, including parcels and current/future campaigns; several managers may share a property under Q-AA-02. Account/business deletion remains under Q-AA-07.

The current requirements review assumes that:

- Reference information and documents are prepared when possible. New workers may be created on the day, including offline, with mandatory identification. Their creation does not establish documentary readiness.
- A worker may perform different tasks and work in several areas during the same day.
- Campaigns may cover several properties or selected parcels.
- Source information may arrive after the physical delivery or campaign closure.
- One transport record is associated with each delivery in the first version.
- Each delivery may have at most one ticket and one analysis, each added later. Yield aggregation uses the ticket's reported clean net weight.
- Monetary values are supplied by the cooperative and associated with weighing tickets.

These assumptions must be revised when validated examples contradict them.

3.5.2 Business Dependencies

The team depends on client or intermediary input to confirm:

- Weighing-ticket terminology, deduction rules and source-value meanings.
- Representative clean-net weights, analysis values and validation ranges.
- Document types, templates and intended signing requirements.
- Concrete responsibility assignments and validation of the default permission profile.
- First-version priorities and operational acceptance expectations.

Real examples should be provided with unnecessary personal information removed.

3.5.3 Technical Dependencies

Registration, invitations and recovery depend on identity management and email delivery. Offline access depends on prior authentication, protected local storage and the authorisation policy in BR-002. Session, password, authentication and logging controls are specified in Sections 7.3 and 7.6. The approved targets require technical proof; residual mechanisms and deployment choices are listed in Section 11.3. Supported financial operations and PDF output use the same grant and preparation conditions as other field operations.

The solution depends on:

- Devices and browsers capable of supporting the selected application approach.
- Successful preparation of authorised reference data before offline work.
- Sufficient local storage for pending operations and prepared documents.
- A central service for validation, persistence and synchronisation.
- Defined recovery behaviour for rejected operations, concurrent changes and interrupted transfers.
- Consistent deletion propagation for document versions removed under the retention rules.

The selected PWA, IndexedDB, modular backend and PostgreSQL stack is described in Section 8. Its selection does not establish feasibility of the unresolved local-security and synchronisation mechanisms; these require the technical validation listed in Section 11.3.

3.5.4 Project Dependencies

The academic schedule, team availability, hosting resources and required course deliverables constrain the implementation plan.

Use-case approval, task ownership, effort estimates and acceptance evidence must be tracked explicitly. The existence of a documented use case does not establish that its implementation has been completed or validated. The working risk register and response plans in Section 10 link these dependencies to task ownership, effort tracking and milestone review. Final delivery dates, resource commitments and assessment validation remain under Q-PM-01.

3.6 Requirements Review and Change Control

Treat the current specification as a reviewable snapshot, not permanently frozen requirements. Record each proposed change with its source, request/review dates, affected IDs and disposition. Use the use-case links, shared business rules and architecture/data dependencies to assess scope, effort, schedule, quality and risk before acceptance.

The owner/client validates business meaning through the intermediary; the team reviews feasibility and delivery impact. Neither a demonstration nor an unanswered question counts as approval. Record the decision and reviewer, update affected artefacts together, and adjust estimates/backlog before committing the change. Identify the baseline revision used for an estimate or demonstration. The exact approval assignment and release priorities remain under Q-SCOPE-01 and Q-PM-02.

4 Use Cases

This section specifies account and business interactions. Actor identities and permissions follow Section 2; workers participate without management accounts. Each use case states its operation-specific conditions.

4.1 Scope and Review Status

The use cases describe the working specification under the review status in Section 1.4. Decisions and residual questions are recorded in Section 11.

4.2 Shared Definitions

Business context. One owner workspace contains properties, campaigns, workers, documents and operations. Account, membership and Worker are distinct identities. Records from different businesses are not automatically shared or merged.

Property and parcel. A property may have named parcels. A whole-property record is distinct from a parcel record; new divisions do not redistribute past data.

Campaign. Planned, In progress and Closed are lifecycle states; archival is independent. Coverage and execution follow BR-008 and BR-009.

Working date. The business-local start date identifies a work period even after midnight. Full instants validate chronology/overlap; work entry uses minutes and totals display hours/minutes without per-period rounding.

Attendance and participation. Daily attendance belongs to a business worker and working date; an actual assignment supplies campaign and area. Only expected or selected workers await a decision; unrelated unselected workers generate no alert.

Confirmation and synchronisation. User confirmation is distinct from central acceptance. State meanings are defined in Section 6.2.5. Offline preparation covers authorised context, not every historical record; its download window is distinct from the authorisation duration in BR-002.

Authorisation. Authentication, membership, responsibility scope and operation permission are checked separately, including files. Denial must not disclose protected information. Manager defaults follow Section 2.4.

Validation, cancellation and retry. Invalid input remains recoverable. Cancelling unsaved input preserves accepted records. Failed or uncertain transmission does not establish rejection: check or resume the same operation without duplicate effects. Reconnection revalidates current grants, revisions, dependencies and rules; rejected/conflicting proposals remain recoverable and distinct from accepted data.

Current values and domain records. Ordinary profile/work/source corrections retain current values rather than permanent field history. Document versions and individual signatures remain domain records subject to their retention policy.

Offline creation. Basic worker creation and explicit campaign start may operate offline with valid grants. Dependent records keep stable local identities and remain pending until their prerequisites are accepted.

Financial information. Consultation, structured entry and correction are separate grants. Complete-file access and source upload follow Section 2.4; there is no mandatory monetary-content classification or automatic redaction.

Task and capability. Profile machinery capability is distinct from performed work. Every complete confirmed actual period has its task; attendance and assignments may be known before periods are complete.

Provenance and collection. Source property/parcel, collection from trees or ground, harvesting method and olive variety are distinct attributes.

Completion and correction. Section 6.3.2 defines initial completion, changes to confirmed work and late entry in closed campaigns.

4.3 Use Case Catalogue

UC-01 Register as a Farm Owner.

UC-02 Log In and Log Out.

UC-03 Recover and Update Account Access.

UC-04 Manage Manager Access and Permissions.

UC-05 Manage Properties and Parcels.

UC-06 Manage Harvest Campaigns.

UC-07 Manage Worker Records.

UC-08 Plan Work.

UC-09 Consult Work Schedules.

UC-10 Record Attendance and Actual Work Periods.

UC-11 Correct Work Records.

UC-12 Consult Worker Documents.

UC-13 Upload and Update Worker Documents.

UC-14 Review and Sign a Document.

UC-15 Record Transport and Olive Deliveries.

UC-16 Record Weighing Data and Adjustments.

UC-17 Record Analysis Results.

UC-18 Manage Cooperative-Reported Ticket Values.

UC-19 Consult Campaign Indicators.

UC-20 Generate Worker Documents.

4.4 UC-01 — Register as a Farm Owner

Purpose Allow a new owner to create an account and establish an independent business workspace.

Actors Visitor.

Preconditions An internet connection is available. The visitor is not authenticated in the current session.

Required Information Name of the owner, email address, password and business name.

Main Flow

1. The visitor opens the registration form.
2. The system requests the required information.
3. The visitor enters the information and submits the registration request.
4. The system validates the submitted information and checks whether the email can be registered.
5. The system creates a pending registration and sends an email verification link.
6. The visitor follows the verification link.
7. The system validates the verification request.
8. The system activates the account and creates its business workspace.
9. The system assigns the owner role for that workspace to the new account.
10. The system confirms activation and provides access to the login process.

Alternative Flows and Exceptions

A1. **Invalid information:** The system identifies the fields requiring correction. Registration is not completed.

A2. **Email unavailable for registration:** The system does not create a duplicate account and provides a route to login or account recovery without unnecessarily disclosing account information.

A3. **Verification link expired or invalid:** The system does not activate the account and allows a new verification message to be requested.

A4. **Email delivery failure:** Registration remains pending. The visitor can retry delivery without creating a second account.

A5. **Repeated activation request:** An already processed verification request does not create another account or business workspace.

A6. **Connection failure:** The system reports the incomplete operation and supports a retry without duplicate creation.

Postconditions on Success A verified account exists. A business workspace exists. The account is its owner. No operational properties, campaigns or workers are created automatically.

Postconditions on Failure No active business access is granted. A pending registration may remain available for verification or retry.

Connectivity Registration and email verification require an internet connection.

Related Requirements and Rules FR-001.

BR-001, BR-002.

4.5 UC-02 — Log In and Log Out

Purpose Authenticate an account holder, establish access to an authorised business workspace and allow the user to terminate the session.

Actors Farm Owner. Field Manager with delegated business access.

Preconditions For login, an activated account exists and an internet connection is available. For logout, a session exists on the device.

Main Flow — Login

1. The user opens the login screen.
2. The user enters an email address and password.
3. The system validates the credentials and account status.
4. The system determines the user's current business access and permissions.
5. The system establishes an authenticated session.

6. The system opens the authorised workspace.
7. The system makes offline preparation available for the operations supported by the application.

Main Flow — Logout

1. The user selects logout.
2. The system identifies any local changes that have not yet been accepted by the server.
3. The system informs the user of pending changes and provides a synchronisation option when online.
4. The user confirms logout.
5. The system terminates the session on the device and blocks access to the workspace.
6. The system preserves pending changes in protected local storage for recovery through the originating account, subject to renewed authorisation.

Alternative Flows and Exceptions

A1. **Invalid credentials:** Access is refused using a response that does not unnecessarily reveal which credential was incorrect.

A2. **Unverified account:** Workspace access is refused and the user is offered a route to complete verification.

A3. **No authorised business access:** Authentication does not grant access to a business from which the account has been removed.

A4. **No connection during login:** A new online authentication cannot be completed. Previously authorised offline use follows the established offline-session policy.

A5. **Expired or invalid session:** The system requests renewed authentication before protected online operations continue.

A6. **Logout while offline:** The local session is terminated and workspace access is blocked. Any server-side invalidation requiring connectivity is completed when communication becomes possible.

A7. **Pending synchronisation failure:** The user may still log out. Pending changes are not silently discarded or exposed to another account.

A8. **Local inactivity cleanup:** On application opening, check account/business/device inactivity before updating last use. After 30 days, remove only eligible centrally accepted copies, preserving pending proposals and their recovery files. Removed copies need connected preparation again. Central records are unchanged; cleanup does not renew offline authorisation.

A9. **Explicit discard:** Only the authenticated originating user may confirm discard of selected pending work after a warning identifying the loss. Logout alone does not discard it.

A10. **Offline authorisation expires:** Apply BR-002 and FR-004: only successful online permissions validation renews access. Expiry blocks consultation and new operations while preserving protected pending dependencies.

Postconditions on Login The user can access only the authorised business information and operations.

Postconditions on Logout The workspace is inaccessible through the terminated local session. Pending data, if any, remains protected and is not transferred to another account.

Connectivity Login and online permission validation require internet. Supported offline work requires previous authentication and locally available authorised data. Logout is available online and offline.

Related Requirements and Rules FR-002, FR-003, FR-004.

BR-002, BR-003, BR-004.

Decision References D-19, D-22, Q-AA-04, Q-AA-05.

4.6 UC-03 — Recover and Update Account Access

Purpose Allow password recovery and authenticated password or verified-email updates, preserving account identity, business ownership and permissions.

Actors Account holder.

Preconditions An internet connection is available. Password recovery requires access to the registered email inbox; authenticated updates require an active session and identity revalidation.

Main Flow

1. The account holder opens the recovery form.
2. The account holder enters the account email.
3. The system displays a generic acknowledgement.
4. For an eligible account, the system sends a password reset link to the registered email.
5. The account holder follows the link.
6. The system validates the reset request.
7. The account holder enters and confirms a new password.
8. The system validates and saves the new password.
9. The system consumes the reset authorisation and invalidates existing online sessions.
10. The system confirms the change and directs the account holder to login.

Alternative Flows and Exceptions

A1. **Unknown or ineligible email:** The system displays the same generic acknowledgement without disclosing whether an account exists.

A2. **Invalid, expired or previously used link:** The password is not changed. A new recovery request may be submitted.

A3. **Invalid password or confirmation mismatch:** The system requests correction before completing the reset.

A4. **Email delivery failure:** The existing credentials remain unchanged. The account holder may submit another request.

A5. **Connection failure:** The system reports that completion could not be confirmed and requires validation before reporting a successful reset.

A6. **No access to the registered inbox:** Automatic recovery cannot be completed. An alternative identity-verification process is outside the currently defined workflow.

A7. **Authenticated password update:** The connected account holder opens account settings and revalidates identity. The user enters and confirms a new password; the system validates and saves it under the configured password policy and applies the recovery/session invalidation policy before confirming success.

A8. **Verified email update:** The connected authenticated user requests a new address after identity revalidation. The system validates normalised-email uniqueness and sends verification to that address. The user follows the verification link; only successful verification activates the new address. Until then, or on invalid/expired verification, the existing address remains active. The system preserves account identity, ownership, memberships, signatures and record authorship across updates and retries.

Postconditions on Success For recovery/password update, the new password is active and online sessions are invalidated under the same security policy; a reset authorisation cannot be reused. For email update, only the verified new address becomes active. Account identity, business ownership, memberships, permissions, signatures and record authorship remain unchanged.

Postconditions on Failure No unsuccessful credential change is activated. Pending/failed email verification leaves the existing address active. Existing identity and permissions are unchanged.

Connectivity Recovery and authenticated email/password updates require an internet connection. Disconnected devices detect session invalidation when they reconnect or reach the offline authorisation limit.

Related Requirements and Rules FR-005, FR-064.

BR-002, BR-042.

Decision References D-16, Q-AA-05, Q-AA-07.

4.7 UC-04 — Manage Manager Access and Permissions

Status Included in the current working specification; owner-controlled permissions apply.

Purpose Allow an owner to invite a Field Manager, define the authorised scope and permissions, and later modify or revoke that access.

Actors Farm Owner.

Supporting Actor Invited Field Manager.

Preconditions The owner is authenticated. An internet connection is available. The owner controls the selected business workspace.

The initial profile follows Section 2.4, including revocable financial/full-ticket consultation. Structured financial writes require additional grants; source upload follows its entry/attachment permission. Mandatory insurance access is not a prerequisite review before each field operation. Worker creation requires a separate grant, including offline creation.

Main Flow — Invite a Manager

1. The owner selects the invitation option.
2. The owner enters the manager's email address.
3. The owner selects one or more properties, covering their parcels and current/future campaigns until revoked, and any additional workers explicitly made available for that manager. The system presents the default operational permissions from Section 2.4; the owner reviews and adjusts them, preserving mandatory insurance access.
4. The system checks the permission configuration.
5. The system creates an invitation and sends an acceptance link.
6. The recipient follows the invitation link.
7. An existing account holder authenticates; a new recipient creates and verifies an account.
8. The system confirms that the authenticated email matches the invitation.
9. The recipient explicitly accepts the invitation.
10. The system activates the manager's access to the owner's business.

Main Flow — Change Permissions

1. The owner selects an existing manager.
2. The system presents the current scope and permissions.
3. The owner changes the configuration.
4. The system validates and saves the changes.
5. Subsequent online requests use the updated authorisation.

Main Flow — Revoke Access

1. The owner selects an existing manager.
2. The owner requests revocation.
3. The system requests confirmation.
4. The owner confirms.
5. The system revokes the manager's business access.
6. Subsequent online requests to that business are refused.

Alternative Flows and Exceptions

A1. **Invitation already pending:** The system allows the owner to review, resend or cancel the existing invitation without creating duplicate memberships.

A2. **Manager already has access:** The system directs the owner to the existing permission configuration.

A3. **Invalid, expired or cancelled invitation:** Access is not granted. The owner may issue a new invitation. Invitations expire after seven days; resending invalidates the preceding link.

A4. **Different authenticated email:** The invitation cannot be accepted by that account.

A5. **Incomplete mandatory permissions:** The system identifies the incompatible configuration. A manager responsible for workers must retain access to their insurance documents within that scope.

A6. **Invitation cancelled before acceptance:** The invitation no longer grants access.

A7. **Manager device offline during revocation:** Central access is revoked immediately. The disconnected device enforces the change upon reconnection or expiry of offline authorisation.

A8. **Pending changes after revocation:** The revoked manager cannot submit new authorised business changes merely because data was previously available offline.

Existing pending operational proposals may be presented to the owner through an authenticated recovery submission that does not restore the manager's business access.

The owner may review, accept or reject each proposal.

Acceptance supplies new explicit owner authorisation and requires validation of campaign status, dates, coverage, revisions, dependencies and conflicts. Device timestamps cannot prove that a proposal predates revocation. Owner approval does not override signing-source eligibility.

The record distinguishes the account that originally submitted the information from the owner who authorised its acceptance.

Postconditions on Success The invitation, permission change or revocation is saved. Access is limited to the owner's business and the authorised responsibility scope.

Postconditions on Revocation The account remains distinct from its business membership. Existing operational records are preserved. Revocation does not delete worker records or historical business information.

Connectivity Invitations, acceptance and permission changes require an internet connection.

Related Requirements and Rules FR-004, FR-006, FR-007, FR-008.

BR-001, BR-002, BR-003, BR-004.

Decision References Q-AA-01, Q-AA-02, D-01, D-08.

Scoped consultation Several managers may share a property. Multi-property campaign views show only authorised property records and partial indicators. Worker access does not disclose other-property work. Overlapping edits require explicit conflict review.

4.8 UC-05 — Manage Properties and Parcels

Purpose Allow an authorised user to create, consult, update, archive, reactivate and, where permitted, delete properties and their optional parcels.

The current agricultural scope concerns olive harvesting.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Additional actor: Field Manager, when explicitly granted property-management permissions by the owner.

Permission to record operational work does not automatically grant permission to manage properties.

Preconditions

- The user has permission for the requested operation.
- Creation, modification, archival, reactivation and deletion require an internet connection.
- For operations on an existing record, that record exists within the authorised scope.
- To create a parcel, its property exists.
- Offline consultation requires previously downloaded authorised information.

Main Flow

1. The user opens property management.
2. The user chooses to create a property or selects an existing property.
3. The system presents the existing information or an entry form.
4. The user enters or updates the property name and textual location.
5. The user may enter the declared area in hectares and one or more olive varieties.
6. The system identifies the owner associated with the property.
7. If internal divisions are needed, the user creates or updates named parcels. Parcel area and olive varieties are optional.
8. The system validates the information and access permissions.
9. The user reviews and confirms the changes.
10. The system saves the records and confirms the result.

For consultation only, the flow ends after the requested information is displayed.

Alternative Flows and Exceptions

A1 — No internal divisions are needed. The user saves the property without parcels. The absence of parcels does not prevent use of the property in operational records.

A2 — Optional information is unknown. The user leaves area or variety information unfilled and may add it later. An unknown area is not treated as zero.

A3 — The user archives a record. The system requests confirmation and archives the selected property or parcel. Historical records remain accessible within the user's permissions. The archived record is unavailable for new operational associations.

A4 — The user reactivates a record. The system validates the request and restores its availability, subject to its parent property's state.

A5 — The user requests deletion. The system checks for associated records. If none exist, the user may confirm deletion. Otherwise, deletion is refused and archival is offered.

A6 — Connectivity is unavailable. The system permits consultation of previously downloaded authorised records and identifies their synchronisation status. Management operations are unavailable until connectivity is restored.

Postconditions **On successful creation or update:** Property and parcel information is saved with the correct ownership and parent associations.

On archival: Historical information is preserved, and new operational use is blocked for the archived scope.

On reactivation: The record becomes available again, subject to the property's availability.

On deletion: Only the eligible record without associated records is removed.

On consultation, cancellation or rejection: Stored information remains unchanged.

Related Requirements and Rules FR-009, FR-010, FR-011, FR-012, FR-013.

BR-005, BR-006, BR-013.

Decision References Q-SCOPE-01, D-06.

4.9 UC-06 — Manage Harvest Campaigns

Purpose Create and configure Planned campaigns, explicitly start them online or offline, consult/update eligible details, close/reopen, archive/restore or delete eligible campaigns. Shared conditions in Section 4.2 apply.

Actors Farm Owner; Field Manager with the specific owner-granted operation and scope.

Preconditions The user has valid business access and operation permission. Selected properties and optional parcels exist and are eligible. Structural administration, closure and reopening require connectivity; an explicit start may be recorded offline with prepared campaign context and unexpired offline authority.

Main Flow — Create a Planned Campaign

1. The user opens campaign management and chooses creation.
2. The user supplies name, planned start and one or more whole properties or selected parcels. New parcels, if needed, are created through connected UC-05.
3. The system validates scope and avoids duplicate parent/child coverage. Other Planned campaigns or current execution do not prevent this preparation.

4. The user reviews and confirms; the system saves the campaign as Planned. No actual start, presence or work is created.

Alternative Flows

A1 — Explicit start.
The user selects Start campaign and confirms the actual start date. Validate permission, chronology, relevant historical activity and exclusive execution coverage. On central acceptance, transition to In progress. Whole-property execution conflicts with each parcel; distinct parcels may coexist.

A2 — Offline start.
Save explicit start and dependent work/transport locally with stable identifiers, marking them pending. Allow authorised local continuation. Revalidate and accept the start before dependent records; a competing accepted start leaves recoverable proposals for resolution, never two accepted executions. Device time alone is not proof of acceptance or event chronology.

A3 — First actual assignment.
If no actual start exists, UC-10 proposes its working date and requires explicit confirmation or correction. A plan, attendance or delivery never silently starts a campaign. Existing dates remain unchanged unless an authorised date correction is confirmed.

A4 — Consult/update.
Prepared details may be consulted offline. Structural changes remain connected. Changes to In progress coverage also validate exclusive execution. Added divisions do not redistribute historical records.

A5 — Remove coverage.
Resolve all dependent work/delivery associations using their permitted workflows; reopen a Closed campaign first. No cascade deletion occurs and unresolved dependencies block removal.

A6 — Close.
For In progress, confirm actual end while connected. Validate recorded activity and preserve records; transition to Closed.

A7 — Complete missing work.
UC-10 may add missing details or previously unrecorded periods within valid actual dates without replacing/removing confirmed values or changing confirmed associations. All attendance/overlap checks still apply.

A8 — Correct confirmed work.
UC-11 requires reopening before correcting or removing confirmed work; no 15-day grace period or correction-age limit applies. Work outside actual dates also requires reopening and date review.

A9 — Late external information.
Existing delivery/ticket information, analyses and reported amounts may be added/corrected after closure without reopening. This is not permission for new trips or exceptional delivery changes.

A10 — Reopen.
A Closed campaign returns to In progress only after connected authority, date review and exclusive-coverage validation. A currently executing conflicting campaign blocks reopening. Preserve the original records.

A11 — Archive/restore.
Change usual-list visibility independently of Planned/In progress/Closed. Archival does not close or release In progress coverage; restoring visibility does not start or reopen.

A12 — Delete.
Require connected authority, confirmation and no associated operational records or blocking pending dependencies; otherwise retain and offer archival.

Postconditions Creation produces Planned; accepted start/reopening produces In progress; closure produces Closed with actual end. Local confirmation remains distinct from central acceptance. Cancellation before confirmation changes no accepted record.

Related Requirements and Rules FR-004, FR-010, FR-014, FR-015, FR-016, FR-017, FR-018, FR-019.

BR-006–BR-013; decisions D-02, D-03, D-04, D-22–D-24.

4.10 UC-07 — Manage Worker Records

Purpose Create, consult and maintain business-specific workers, including same-day offline creation, without creating application accounts for them.

Actors Farm Owner; Field Manager with permission for the requested operation. Profile consultation does not grant creation or modification rights.

Preconditions The user has valid business access, worker scope and the requested operation grant. Offline consultation requires prepared profiles or locally created workers and unexpired offline authorisation. Basic worker creation may operate offline in an authorised business context; existing-profile edits, inactivation/reactivation and deletion require connectivity. The shared conditions in Section 4.2 apply.

Main Flow

1. The user searches the authorised available records or selects Add worker. For consultation, the system displays the permitted profile and its availability or pending status, then ends the flow without changing records. No modification permission is required. Creation continues at Step 2.
2. For creation, the user enters full name, telephone, official identification type, issuing country and number, and records machinery capability and optional notes.
3. The system generates a stable internal identifier independent of the official number, validates required fields and warns about same-business matches by name, telephone or normalised type/country/number.
4. The user reviews possible matches. Selecting an existing profile ends creation and opens that profile within the user's permissions. Confirming a distinct worker continues to Step 5. The system never automatically merges identities or businesses.
5. The system saves the new worker's current values and last-modified information. An offline create retains a stable operation identity and pending central-acceptance status.

Alternative Flows and Exceptions

A1 — Immediate field use.
A local new worker may be selected for permitted attendance, work, driving and suitable document/signing workflows. Dependent records refer to the pending worker and any required pending campaign start.

A2 — Document prerequisites.
Profile creation does not verify identification, require a scanned copy or establish documentary readiness. Signing requires a suitable available source and separate association/signing grants.

A3 — Reconnection and duplicates.
The server validates authority and duplicate warnings before accepting dependent records. Resolution preserves all linked work, files and identities; retries do not create a second worker.

A4 — Update.
From a consulted profile, the user may explicitly select Edit with separate modification permission and connectivity. The system validates and saves the changes, preserving the internal identity and historical associations without permanent field-change history.

A5 — Inactivate/reactivate.
Connected confirmation preserves history, flags future plans and excludes inactive workers from new assignments. Reactivation reuses the identity; daily absence/non-selection does not inactivate a worker.

A6 — Delete.
Connected deletion requires authority, confirmation and no associated records or pending dependencies; otherwise refuse and offer inactivation.

A7 — Missing or invalid information.
The system preserves input and requires all creation fields, including offline. Accepted identification types and country-specific checks remain under Q-ID-01; entry is not verification.

A8 — Unavailable consultation.
The system distinguishes unavailable local data or retrieval failure from a successful search with no matching records.

Postconditions Consultation changes no records. A created/updated profile retains its business and internal identity, with pending state where applicable. Field dependencies remain linked during recovery. Creation alone confirms no attendance, task or documentary eligibility. Cancellation preserves accepted information.

Related Requirements and Rules FR-004, FR-020, FR-021, FR-022, FR-023.

BR-013–BR-016.

Decision References D-25, D-26, Q-ID-01.

4.11 UC-08 — Plan Work

Purpose Allow an authorised user to plan workers, work areas, optional periods and tasks for one or more working days.

Planning may start from a new entry, an existing plan or previously recorded actual work. It never establishes attendance or actual hours for the target dates.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Additional actor: Field Manager, with the relevant planning permissions granted by the owner.

Preconditions

- The user has permission to plan work within the selected scope.
- The campaign, work areas and worker records exist.
- Selected areas are included in the campaign coverage.
- The requested dates comply with the campaign lifecycle rules.
- Offline planning requires the necessary authorised reference data to be available on the device.

Planning is optional and supports Planned or In progress campaigns; UC-10 does not require a prior plan. Workers created locally through UC-07 may enter pending plans, with worker acceptance preceding dependent plan acceptance. Planning neither starts a campaign nor reserves execution.

Main Flow

1. The user selects the campaign and one or more target dates.
2. The user selects a property or an included parcel.
3. The system displays existing plans for the selected context.
4. The user selects the workers expected to work.
5. The user may enter planned periods, task types and optional breaks. Common values can be applied to a group and adjusted individually.
6. The system checks required information, worker availability, area eligibility and known scheduling conflicts.
7. Where a worker is planned for several areas on the same day, the system requires periods for the relevant assignments.
8. The system presents the proposed daily plans and any unresolved issues.
9. The user reviews and confirms.
10. The system saves independent daily plans and displays their local and server status.

Alternative Flows and Exceptions

A1 — A minimal plan is sufficient. For a worker assigned to one area on a working day, the user may save the campaign, date, area and worker without specifying hours or task type.

A2 — Individual details differ. The user adjusts one worker's periods or tasks without unintentionally changing other workers' entries.

A3 — Several areas apply. The user creates the relevant assignments. Periods are required for all assignments of that worker to different areas on the same working day.

A4 — Several target dates are selected. The system applies the reviewed proposal to each selected date. Each date receives an independent plan, validated against its own context.

A5 — An existing plan is copied. The user selects a source plan and the target dates. Workers, areas, periods and tasks are proposed for review. Source plans remain unchanged.

A6 — Actual work is used as a basis. The user selects previously recorded work. Its workers, areas, periods and tasks are proposed as planning information for the target dates. Confirmed attendance and actual work records are not copied to those dates.

A7 — Target dates already contain plans. The system identifies the existing plans. Reuse does not silently overwrite or duplicate them. Changes require explicit review.

A8 — A worker is inactive. The system identifies the inactive worker and requires an eligible selection or authorised reactivation. Existing future plans are flagged rather than silently edited.

A9 — Periods overlap. The system blocks conflicting periods, considering the worker's relevant plans across properties, parcels and campaigns.

A10 — Work crosses midnight. The user records the start and end with sufficient date information. The period belongs to the working date on which it starts. Validation considers the full interval, including its continuation into the next day.

A11 — A worker is removed or replaced. Before actual work has been recorded for the affected daily area context, the user may confirm removal or replacement. Other assignments remain unchanged.

A12 — Actual work already exists. The corresponding plan remains the reference for what was expected. Differences are recorded as actual work or handled through the work-correction process.

A13 — The device is offline. The user may create or adjust plans using prepared authorised data. The system identifies the changes as pending synchronisation.

A14 — Planned breaks are entered. The user identifies break start and end times and their intended counting treatment. The system prevents duplicate deductions in any displayed planned-duration calculation. These values remain subject to actual confirmation.

Postconditions **On success:** Daily plans identify the selected campaign, working dates, areas and workers, with any supplied periods and tasks.

On reuse: Target plans are independent of their source. Editing a target day does not modify the source or other target days.

On pending synchronisation: Local plans remain distinguishable from server-accepted plans.

On cancellation or rejection: Previously accepted information remains unchanged.

In all cases: Planning does not confirm attendance, create actual hours or modify source actual work.

Period Validation Permit a period to start exactly when another period ends for the same worker. Reject zero-duration periods and positive-duration overlap using complete date/time instants. Validate planned and actual intervals separately (D-20).

Related Requirements and Rules FR-024, FR-025, FR-026, FR-027, FR-029.

BR-015, BR-016, BR-017, BR-019, BR-021.

Decision References Q-WF-04, Q-AA-04, Q-WF-02.

4.12 UC-09 — Consult Work Schedules

Purpose Allow an authorised user to consult planned work by day, by worker or by week, within the selected harvest campaign.

Consultation does not modify plans or establish actual attendance.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Additional actor: Field Manager, with the relevant consultation permissions granted by the owner.

Preconditions

- The selected campaign exists.
- The user is authorised to consult the requested information.
- Offline consultation requires the relevant plans and reference information to be available on the device.

An existing plan is not a precondition: a valid query may return no planned work.

Main Flow

1. The user opens schedule consultation and selects a campaign.
2. The user chooses a daily, worker-based or weekly view.
3. The user selects the relevant date, week or worker consultation period.
4. The user may filter by property, parcel, worker and planned task type.
5. The system retrieves the matching authorised plans.
6. The system displays the workers, working dates, properties or parcels, planned periods, task types and any recorded planned breaks.
7. Missing hours or task types are identified as not yet defined.
8. The system identifies relevant synchronisation and freshness information.
9. The user may change the view or filters and repeat the consultation.

Alternative Flows and Exceptions

A1 — No matching plan exists. After a successful query, the system states that no planned work matches the selection. This does not establish worker absence.

A2 — A plan has incomplete details. The system displays the known information. Missing hours or task types are shown as not yet defined, not as zero hours or an assumed task.

A3 — Several areas are included. The system can display all authorised areas of the selected campaign together. Each assignment retains its property and optional parcel identification.

A4 — A period crosses midnight. The system displays the period under its start date and identifies the end date and time.

A5 — A planned worker is inactive. The system identifies the inactive status. The saved assignment is not silently removed. Any change follows UC-08.

A6 — Local changes are pending. The system identifies locally saved plans or changes awaiting synchronisation. They are not presented as server-confirmed.

A7 — The device is offline. The system displays prepared authorised plans for the current and following week, together with the last successful synchronisation information.

A8 — Requested information is not local. The system explains that the information is unavailable without connectivity. It does not claim that no plan exists.

A9 — Retrieval fails. The system explains the failure and offers a retry where appropriate. A failed query is distinguishable from a successful query with no results.

A10 — Access is restricted. The system excludes unauthorised information and identifies the scope being displayed without exposing protected details.

Postconditions **On success:** Matching authorised plans are displayed, or a successful query confirms no matching plans.

On offline or incomplete consultation: The system identifies the available scope and relevant limitations.

In all cases: Consultation does not modify plans, confirm attendance or create actual hours.

Related Requirements and Rules FR-028.

BR-017, BR-019, BR-021.

Decision References Q-AA-04, Q-AA-05.

4.13 UC-10 — Record Attendance and Actual Work Periods

Purpose Allow an authorised user to confirm daily attendance and record actual work periods, tasks, locations and optional breaks.

Planned information may support entry, but becomes actual work only after explicit confirmation.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Additional actor: Field Manager, with the relevant permissions granted by the owner.

The authorised owner or manager records worker attendance. Workers do not need application accounts for this recording.

Preconditions

- The user has permission for the requested operation.
- Worker records exist centrally or were created locally through UC-07. A new worker may be created before returning to attendance/work entry.
- The relevant campaign and work areas exist within the authorised scope.
- Work dates comply with the campaign lifecycle rules.
- Offline entry requires the necessary authorised reference data to be available.

Main Flow

1. The user selects the campaign and working date.
2. The system displays planned workers, available proposed periods and any existing attendance and work records.
3. Only selected/expected workers without a daily decision are shown as awaiting confirmation. Unrelated unselected workers generate no alert.
4. The user marks workers as present or absent. Daily attendance is shared within the business for that worker and working date.
5. For present workers, the user identifies the actual campaign and work-area assignments. These may remain incomplete as to hours or tasks.
6. The user reviews proposed periods or enters actual periods, tasks and areas. Common values may be applied to a group with individual adjustments.
7. The user reviews or records actual breaks, including their start, end and whether they count towards working time. Planned breaks are not accepted automatically.
8. The system validates the entries and calculates available working-time totals.
9. The user reviews and explicitly confirms the submitted information.
10. The system saves the records and displays their local storage and synchronisation status.

Alternative Flows and Exceptions

A1 — Attendance is known before hours. The user saves presence or absence and completes work details later. Missing hours remain incomplete, not a confirmed zero total.

A2 — An unplanned worker attends. The user selects an existing authorised worker record, including a worker created offline with authorisation through UC-07, and records attendance. A planned assignment is not required.

A3 — No plan exists. The user selects existing workers and enters the actual information directly.

A4 — Actual work differs from the plan. The user adjusts workers, periods, tasks or areas before confirmation. The original plan remains unchanged. A complete period requires its actual task before confirmation. Until then, preserve the incomplete assignment rather than inventing hours.

A5 — A worker changes task or area. The user records separate periods for the relevant campaign and area. Each confirmed complete period identifies its actual task; attendance/assignment may be saved earlier without completed periods.

A6 — Work periods exclude a break. The interval between those periods does not contribute to working time. The same excluded interval is not deducted again.

A7 — A break occurs within a period. The user records its start and end. It is deducted only when explicitly marked as not counted towards working time.

A8 — A worker is absent. The system does not convert planned periods into actual work. Contradictory actual work entries require resolution.

A9 — Work crosses midnight. The period retains its start-date association. Its end date and time establish the full interval for calculations and overlap validation.

A10 — Work is entered retrospectively. The user selects the actual working date. The system preserves the distinction between work date and entry date.

A11 — The campaign is closed. The user may complete previously missing information for work performed within the campaign's valid actual dates.

This includes entering missing details or a previously unrecorded actual period, provided that confirmed values and associations remain unchanged.

The system validates attendance consistency, permissions, campaign coverage and time overlaps.

Changing or removing confirmed information requires authorised reopening and follows UC-11.

Work outside the campaign's valid actual dates requires reopening and date review.

A12 — Confirmed information needs correction. The user follows UC-11. Existing values are not silently replaced through another initial submission.

A13 — Invalid or overlapping periods. The system identifies the problem and preserves unaffected input. Validation considers the worker's relevant periods across areas and campaigns.

A14 — The device is offline. The system saves supported entries locally. Their synchronisation status remains visible.

Recording actual work does not silently change campaign dates.

If the campaign has no actual start, the first confirmed actual assignment provides a proposed start date under UC-06.

The user may explicitly confirm the proposed start offline. Start and dependent work remain pending until the server validates them in order; a competing campaign start is preserved for resolution, not silently accepted.

Postconditions **On success:** The submitted attendance and available actual work details are saved in the correct context.

On attendance-only confirmation: Presence is recorded while hours and tasks remain explicitly incomplete where unknown.

On offline save: User-confirmed entries are stored locally and remain pending server acknowledgement.

On cancellation or rejected changes: Previously saved information remains unchanged.

In all cases: Actual work entry does not rewrite the plan.

Period Validation Permit a period to start exactly when another period ends for the same worker. Reject zero-duration periods and positive-duration overlap using complete date/time instants. Validate planned and actual intervals separately (D-20).

Related Requirements and Rules FR-004, FR-016, FR-020, FR-030, FR-031, FR-032, FR-033, FR-034, FR-035.

BR-011, BR-018, BR-019, BR-020, BR-021.

Decision References Q-WF-01, Q-WF-03, D-04.

4.14 UC-11 — Correct Work Records

Purpose Allow an authorised user to correct saved attendance and actual work information, including assignments without hours, periods, tasks, work areas and breaks.

Corrections update the affected totals without rewriting the original work plan.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Additional actor: Field Manager, with explicit correction permissions granted by the owner.

Permission to record work does not automatically grant permission to correct confirmed records.

Preconditions

- The target record exists.
- The user has correction permission for the affected records and contexts.
- Campaigns whose work records are being changed are open.
- Offline correction requires the relevant authorised data to be available on the device.

Main Flow

1. The user selects the worker, working date and relevant context.
2. The system displays current attendance, periods, tasks, areas, breaks and totals.
3. The user proposes the correction.
4. The system validates permissions, campaign status, dates, associations, time intervals and break treatment.
5. The system checks the worker's relevant periods for overlaps.
6. The system presents the corrected information and affected totals for review.
7. The user confirms.
8. The system saves the current values and updates the affected local totals.
9. The system displays whether the change is pending synchronisation or accepted by the server.

Alternative Flows and Exceptions

A1 — A campaign is closed. Correction cannot proceed until an authorised user reopens the campaign through UC-06. No additional correction deadline applies.

A2 — Presence is changed to absence. The system identifies all actual campaign-area assignments, including those without hours, and all periods and breaks associated with that worker and working date within the business.

The user must have the required permissions, and affected campaigns must satisfy the reopening rules.

The system requests confirmation before removing the affected work and changing attendance. It must not leave the worker absent while contradictory confirmed assignments, periods or breaks remain.

A3 — Record associations are incorrect. The user may correct the worker, working date, campaign, property or parcel.

The system validates the origin and destination contexts, campaign status and time overlaps. It also checks daily-attendance consistency.

Only the selected work records are reassigned. Other periods belonging to the original worker remain unchanged.

Affected origin and destination totals are updated after acceptance.

A4 — Periods or breaks are invalid. The system identifies the problem and preserves unaffected input. Corrections must not create overlapping actual periods or duplicate break deductions.

A5 — The device is offline. The system saves the correction locally and identifies it as pending. Local totals reflect the pending change. Server permissions, campaign status and consistency are checked on reconnection.

A6 — The server record has changed. The system presents the current server values and the pending correction to an authorised user for manual resolution. No conflicting update is silently substituted.

A7 — The user resolves a conflict. The user may retain the server values or review and confirm a revised correction. Any submitted correction is validated again.

Postconditions **On local acceptance:** The corrected values and affected local totals are saved with their synchronisation status.

On server acceptance: Central records and dependent totals reflect the accepted correction. Other clients receive the change through synchronisation.

On cancellation or rejection: Authoritative accepted records remain unchanged. Any unresolved local proposal is clearly identified.

In all cases: Original plans and unrelated records are not modified.

Related Requirements and Rules FR-033, FR-036, FR-037, FR-061.

BR-011, BR-018, BR-019, BR-020, BR-021.

Decision References Q-WF-04, D-04.

4.15 UC-12 — Consult Worker Documents

Purpose Allow an authorised user to find, open and download worker documents, including uploaded files.

Uploaded documents are included in the current scope.

Documents generated through UC-20 are available through the same consultation workflow, subject to the applicable permissions and retention rules.

Insurance documents must support prepared offline consultation for field operations.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Additional actor: Field Manager, with the relevant scope and permissions.

The owner may consult documents within their business.

A Field Manager must have access to insurance documents for workers within their authorised operational scope. Contracts and other documents require additional owner-granted permissions.

Preconditions

- The worker record exists.
- The user is authorised to consult the requested document type and worker scope.
- Offline access requires the relevant authorised files and metadata to have been prepared on the device.

An existing document is not a precondition: the document list may be empty.

Main Flow

1. The user selects a worker and opens the document list.
2. The system displays authorised documents, organised as contracts, insurance or other documents.
3. The system identifies each document by its title, type, relevant date or version and signature status where applicable.
4. Insurance entries display the recorded expiry date and validity status.
5. The user selects an available version.
6. The system verifies access and opens the PDF, JPG or PNG file.
7. The user may download the authorised document.

Alternative Flows and Exceptions

A1 — No documents are available.
The system states that no documents are available within the authorised scope.

A2 — A previous version is requested.
For contracts, all retained versions are consultable.

For insurance and other documents, the current and immediately previous versions remain consultable, as do all older signed versions, including externally declared signed uploads. Older unsigned versions may be removed under UC-13; signed retention periods remain pending.

A3 — Insurance validity is incomplete.
If the expiry date is missing, the system displays validity as unconfirmed. It does not infer validity from upload date or version order.

A4 — Insurance is expired.
The system identifies the expired status while allowing authorised consultation. Expiry does not hide the document.

A5 — The device is offline.
The system opens prepared authorised insurance files and displays the last successful synchronisation information. The current and previous versions are available where they exist and have been successfully prepared.

A6 — A required file is not local.
The system explains that the document is unavailable offline. It does not claim that no document exists.

A7 — Additional workers need preparation.
While connected, the authorised user may prepare insurance documents for additional existing workers who may be called to work.

A8 — Retrieval or download fails.
The system explains the failure and offers a retry where appropriate. Failure is distinguishable from an empty document list.

A9 — A shared document is opened.
The document may be associated with several workers without duplicate file storage.

Within the selected worker's view, application-captured signature information is limited to that worker.

A10 — A document was signed outside the application.
The interface identifies the external-signature declaration separately from application signing. It does not claim that the system verified the external signature.

A11 — A retained version is withdrawn.
The interface identifies it as withdrawn and does not present it as available for operational use. Its retained signature information is not changed to unsigned.

A12 — A deleted version remains on an offline device.
The device applies the authorised deletion when it reconnects and synchronises. The interface must not promise knowledge of server changes that have not yet been received.

Postconditions **On success:** The selected authorised document version is displayed or downloaded.

On offline consultation: The available local version and its freshness are identifiable.

On failure: The system explains the limitation without exposing protected information.

In all cases: Consultation does not change document contents, version identity or signature status.

Related Requirements and Rules FR-038, FR-039, FR-041, FR-044.

BR-022, BR-023, BR-024, BR-025, BR-026, BR-027, BR-028, BR-029, BR-030.

Decision References Q-DOC-07, Q-DOC-04, Q-SCOPE-01.

4.16 UC-13 — Upload and Update Worker Documents

Purpose Allow an authorised user to upload documents and create new versions associated with one or more workers.

Document generation and signing are handled through separate use cases.

This use case covers existing files supplied for upload. Document generation is specified separately in UC-20; both workflows use the same version and retention rules.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Additional actor: Field Manager, with upload or update permissions for the relevant document type.

Insurance consultation permission does not automatically permit document modification.

Preconditions

- An internet connection is available.
- The user has permission for the document type and affected workers.
- Associated worker records exist within the same authorised business.
- For a version update, the target document exists.
- A supported file is available.

Main Flow

1. The user opens document management.
2. The user chooses a new document or a new version of an existing document.
3. The user selects or reviews the associated workers.
4. The user selects the document category: contract, insurance or other.
5. The user provides or confirms the document title and selects the file.
6. For insurance, the user enters the expiry date where known.
7. The user may add observations and declare whether the uploaded version was signed outside the application.
8. The system validates permissions, required fields, file format and size.
9. The system presents the submission and relevant version-retention consequences.
10. The user confirms.

11. The system stores the accepted version, records its upload date and version identity, and makes it current.

12. The system applies the retention policy and confirms the outcome.

Alternative Flows and Exceptions

A1 — An insurance expiry date is unknown. The system allows the document to be saved with validity identified as unconfirmed. It does not infer validity from upload date.

A2 — A document covers several workers. The user associates the same document with the relevant authorised workers. The file is not duplicated for each worker.

A3 — A version was signed externally. The user explicitly identifies that status for the uploaded version. It is distinguishable from an application signing event and is not presented as externally verified by the system.

A4 — An existing signed document is updated. The new version becomes current without inheriting an earlier signature. Earlier versions retain their own status while they remain stored.

A5 — Retention removes an older version. For insurance and other documents, the system retains the current and immediately previous versions and every signed version, including externally declared signed uploads. Only older unsigned versions eligible under the retention policy may be removed after successful replacement. Automatic cleanup protects versions with pending signing operations known to the server. A signature captured on a disconnected device may require resolution if its source version has since become unavailable; the local proposal must be preserved under Q-DOC-04. This policy applies to the shared version, not separately to each worker.

A6 — An incorrect upload may be deleted. An authorised user may confirm deletion when the target has no signature and no dependent records.

A7 — An incorrect upload is signed. The user may withdraw it from use, preserving its file and signature association while it remains retained. Signed versions remain retained; the unsigned rolling-retention policy cannot delete them.

If the withdrawn version is current, it remains identified as current and withdrawn until a replacement is accepted.

An earlier version does not automatically become current.

A8 — Validation fails. The system identifies unsupported formats, files exceeding the size limit, missing required information or invalid associations. No version is accepted until corrected.

A9 — Upload or storage fails. The system reports the failure. The previously current version and its retained history remain unchanged. Failed uploads do not trigger retention deletion.

A10 — A submission is repeated. Retrying the same upload operation does not unintentionally publish duplicate versions or trigger extra deletion.

A11 — Connectivity is unavailable. Upload and update operations cannot complete. Prepared offline insurance consultation remains available through UC-12.

Postconditions **On success:** The accepted version becomes current and is available for authorised consultation through the associated worker records.

On insurance or other-document update: The current and immediately previous versions and all signed versions remain stored. Signed retention periods remain pending.

On contract update: All accepted versions remain stored, subject to separately authorised document actions.

On failure or cancellation: Previously accepted versions are unchanged and no retention deletion is performed.

In all cases: A new version does not inherit signatures from another version.

Related Requirements and Rules FR-040, FR-041, FR-063.

BR-022, BR-023, BR-024, BR-025, BR-026, BR-027, BR-028, BR-030.

Decision References Q-DOC-06, D-07, Q-DOC-07, Q-DOC-01, Q-DOC-04.

4.17 UC-14 — Review and Sign a Document

Purpose Allow a worker to review a specific document version and confirm a drawn signature in person on an authorised user's device.

The signing result belongs to that worker and the reviewed version.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Initiating actor:** Farm Owner.

Additional initiating actor: Field Manager, with the relevant permissions.

Signing actor: Worker.

The worker does not require an application management account.

This process collects the worker's signature only. Any required owner signature is handled separately.

Preconditions

- The worker and document exist.
- The initiating user has permission to access the document and initiate signing.
- The selected version is available and eligible for signing.
- The worker is physically present.
- The worker has not already completed this signing process for the selected version.
- Offline signing requires the authorised document and relevant data to be available on the device.

Main Flow

1. The initiating user selects the worker and document version.
2. The system clearly identifies the intended worker and document.
3. The initiating user confirms that the selected worker is present.

4. The system presents the document for the worker to review.
5. The worker chooses to proceed.
6. The system displays a signature drawing area.
7. The worker draws a signature using a finger or compatible pen.
8. The worker reviews the drawing and its association with the document.
9. The worker explicitly confirms the signing action.
10. The system saves the signature result with the worker, version, initiating user and recorded date and time.
11. The system generates and retains a timestamped worker-specific PDF rendition linked to that version and signing result, preserving the source. Offline output is labelled pending until central acceptance; file-transfer status is tracked separately.
12. The system displays whether the result is pending synchronisation or accepted by the server.

Alternative Flows and Exceptions

A1 — The worker redraws the signature.
Before confirmation, the worker may clear the drawing and try again.

A2 — The worker cancels.
No completed signature is created by the cancelled interaction. The document is not marked as signed for that worker by this interaction.

A3 — No signature has been drawn.
The system requests a drawing before accepting confirmation.

A4 — The document cannot be reviewed.
Signing cannot proceed. The system explains the access or retrieval limitation.

A5 — A signing result already exists.
If the worker has already confirmed a signature for the selected version, the system displays its actual state.

A pending or uncertain result is checked or resumed rather than replaced by another signing interaction.

An accepted signature is not overwritten.

A6 — The document is shared.
Each worker completes an individual review and signing interaction. One worker's signature does not establish another worker's signature.

A7 — The device is offline.
The system saves the confirmed result locally as pending synchronisation. It does not claim server acceptance.

A8 — A newer version exists.
The signature remains linked to the version actually reviewed. It is not transferred to the newer version.

A9 — An older version is synchronised.
If the original version remains available and eligible, the server may accept the signature for that version. The system identifies that a newer version exists.

A10 — The original version is withdrawn or deleted.
The pending result is preserved for protected owner inspection or rejection with a reason. Owner approval cannot make a withdrawn/deleted source eligible; no automatic source resurrection or signature transfer is allowed. A replacement requires fresh worker review and signing under Q-DOC-04. Server cleanup protects pending signing operations it knows about; it cannot guarantee protection for signatures still exclusively on disconnected devices.

A withdrawn version does not automatically accept a pending signature.

A version deleted under the retention policy is not recreated merely because an offline signing operation reconnects.

No newer version is marked as signed by that pending operation.

A11 — The submission outcome is uncertain.
The system checks the existing operation before initiating another signing event. It does not automatically ask the worker to sign again.

A12 — Worker created in the field.
A worker created under UC-07 may participate offline if a suitable authorised document version is already available and its content genuinely applies to that worker. With document-association and signing permission, confirm the association as a pending dependency before recording the signature; do not change the source. Accept worker identity and document association before its signature. Failure or duplicate resolution preserves every linked proposal/file. Creating/uploading new source versions or generating a template still requires connectivity.

Postconditions **On local confirmation:** The signature result, identifying information and the generated local PDF rendition are saved on the device with separate file status. The result is pending until server acceptance where connectivity is unavailable.

On server acceptance: The signature is associated with the intended worker and reviewed version and is available for authorised consultation with its worker-specific PDF rendition. The signed source, signature evidence and rendition remain retained; the PDF neither overwrites the source nor displays another worker's application-captured signature.

On cancellation: No completed signature is created by the cancelled interaction.

On unresolved submission: The pending or rejected state is explicit. No other document version is marked as signed.

In all cases: Participation does not grant additional application permissions.

Related Requirements and Rules FR-020, FR-041, FR-042, FR-043, FR-044, FR-045, FR-063.

BR-023, BR-026, BR-027, BR-028, BR-029, BR-030.

Decision References Q-DOC-02, Q-DOC-04, Q-DOC-07.

4.18 UC-15 — Record Transport and Olive Deliveries

Purpose Record departure and arrival, source areas and transport details; correct or cancel an erroneous delivery record, or delete it when permitted. Weights, analyses and amounts may arrive later. The shared conditions in Section 4.2 apply.

Actors Farm Owner; Field Manager with the relevant scope and permissions.

Preconditions

- Campaign and source areas exist in scope; the driver is a registered business worker.
- Entry/arrival permission is required; confirmed-record corrections, cancellation/deletion or reassignment require separate authority.
- Offline entry uses prepared campaign/area context and prepared or newly created local workers. A pending explicit start may precede dependent trips. Campaign reopening and structural management require connectivity.

Main Flow

1. The user selects the campaign and one or more source properties or optional parcels.
2. The user records known source quantities, collection origin, variety and harvesting method where available.
3. The user selects the registered worker driver, vehicle and destination (existing or free text), and enters the actual departure time.
4. The user may record the transport-document reference and its declared origin separately from actual load sources.
5. The user reviews and confirms the entry. The system saves it as in transport with explicit local/server status.
6. After arrival, the user confirms delivery and enters an arrival time not before departure; the system validates chronology.
7. Arrival confirmation enables the delivery count and eligible weight, yield and monetary contributions under BR-038. An offline confirmation updates labelled local provisional indicators; central totals await server acceptance. Driver selection generates no attendance or hours.

Alternative Flows and Exceptions

A1 — Retrospective registration.
The user enters both event times; the system saves directly as delivered and keeps event and entry dates distinct.

A2 — Mixed or incomplete source information.
The user records actual origins and known quantities. The system leaves unknown quantities unallocated and does not assign the whole load to a declared/dominant parcel or accept percentage allocation.

A3 — Source information arrives separately.
The system saves the delivery without requiring weights/results/amounts; Delivered still counts without them. The user may enter at most one ticket and one analysis, and the ticket's amount, before or after arrival confirmation. Available pre-arrival data awaits receipt confirmation in detail and contributes nothing yet.

A4 — Ordinary correction.
The authorised user reviews the current driver, vehicle, destination or timestamps and reopens a closed campaign first. The system validates chronology and affected associations.

A5 — Delete an erroneous record.
The user requests deletion and confirms. The system checks deletion authority and absence of dependent ticket, analysis, amount or evidence. It removes composed transport/source-batch rows in the same validated operation, without deleting external results. Otherwise it refuses deletion and offers eligible cancellation.

A6 — Cancel an erroneous record.
The authorised user provides a reason, reopens a closed campaign first and confirms cancellation. The system retains the record and linked sources and excludes their indicator contributions. Physical refusal and interrupted trips remain outside this workflow.

A7 — Change campaign or source areas.
The user requests new campaign/source associations and reopens closed campaigns first. The system validates permissions, coverage, dates and dependencies in every affected context, preserves dependent evidence/results and blocks unresolved inconsistencies.

A8 — Offline correction or cancellation.
The system saves the authorised local proposal against the known revision. It remains pending until required reopening and dependency checks succeed; the server revalidates every affected context.

A9 — Unavailable driver or invalid input.
The system preserves recoverable input and requires a registered worker driver. The authorised user creates the worker through UC-07 online or offline before selection; the trip preserves its dependency on that worker.

A10 — Cancel input or submission fails.
The user may cancel unsaved input without changing accepted records. On uncertain submission, the system checks/retries the existing operation without duplicates and distinguishes this from cancelling a delivery.

Postconditions Departure/arrival state and synchronisation status remain distinct. Successful corrections retain current values. Accepted cancellation preserves dependencies and removes indicator contributions; eligible deletion removes only the confirmed dependency-free record.

Related Requirements and Rules FR-020, FR-046, FR-047, FR-048, FR-061.

BR-001, BR-012, BR-031, BR-035, BR-036, BR-038.

Decision References D-12, D-13, Q-DA-10.

4.19 UC-16 — Record Weighing Data and Adjustments

Purpose Manually record or correct the delivery's single ticket and cooperative-reported weights, adjustments, timing and optional evidence. The shared conditions in Section 4.2 apply.

Actors Farm Owner; Field Manager with the relevant scope and permissions.

Preconditions

- The delivery exists and is not cancelled; ticket entry is allowed In transport or after Delivered.

- The user has the requested entry or correction permission. Managers initially may consult complete values/tickets; structured monetary entry or correction requires the relevant additional financial permission.
- Offline use requires prepared delivery context and valid offline authorisation.

After confirmation, completing an unknown ticket/analysis field requires correction permission under Section 6.3.2; the closed-campaign exception does not waive it.

Main Flow

1. The user selects the delivery and creates its ticket, or opens the existing ticket.
2. The user records the available ticket reference, cooperative and weighing date/time.
3. The user enters reported clean net weight in kg and other source weights with explicit meanings/units, leaving unknown values missing.
4. The user may enter weighing start/end, excess duration/unit and the reported monetary penalty with the required financial permission.
5. The user records known adjustments and whether the source already includes them.
6. With the applicable entry/attachment permission, the user may attach PDF/JPG/PNG evidence within the initial 10 MB limit. Monetary content does not require a classification step or financial-entry permission for file upload.
7. The user reviews and confirms. The system saves structured values and shows separate attachment/local/server status.
8. For a Delivered, non-cancelled delivery, the system recalculates eligible indicators using current source values. In transport, it shows available data in delivery detail as awaiting receipt confirmation, without indicator contribution or a false missing-data label. It does not derive clean net weight or penalties from unconfirmed formulas.

Alternative Flows and Exceptions

A1 — Ticket already exists or concurrent creation.
The user opens the existing ticket and may correct it with separate authority; the server rejects concurrent creation of a second ticket. A reference alone is not globally unique.

A2 — Incomplete/ambiguous weight.
The system retains source facts as incomplete, without substituting zero, and excludes them from calculations requiring known semantics.

A3 — Penalty already included or treatment unknown.
The user records the known application status. The system neither calculates the penalty nor deducts it again from reported weight or amount.

A4 — Financial content.
The system displays complete ticket contents to authorised readers, including monetary data. Financial consultation is initially enabled for managers; structured monetary entry/correction still needs additional grants. The system neither classifies mandatory monetary content nor redacts printed values. Revoked consultation follows current access rules.

A5 — Offline or interrupted attachment transfer.
The system saves permitted structured proposals locally and tracks recoverable file transfer

separately. The user may retry the file; the system reports server availability only after upload acceptance.

A6 — Correction or late receipt.
The separately authorised user updates current fields only, including previously unknown fields after confirmation. The system permits existing-delivery ticket updates after closure without reopening.

A7 — Invalid timing or cancelled delivery.
The system identifies inconsistent known timestamps or invalid parent state and preserves input for resolution; it rejects source data against a cancelled delivery.

Postconditions One current ticket remains linked to the delivery. Unknowns remain visible. Financial/source and synchronisation classifications stay separate; rejected proposals do not become accepted totals.

Related Requirements and Rules FR-049, FR-050, FR-053, FR-054, FR-061.

BR-012, BR-032, BR-034, BR-036, BR-038, BR-044.

Decision References D-10, D-11, Q-DA-02, Q-DA-08, Q-AT-02.

4.20 UC-17 — Record Analysis Results

Purpose Record the delivery's single current cooperative-reported yield analysis and optional source evidence. The shared conditions in Section 4.2 apply.

Actors Farm Owner; Field Manager with the relevant scope and permissions.

Preconditions

- The delivery exists and is not cancelled; analysis entry is allowed In transport or after Delivered.
- The user has analysis-entry or separate correction permission.
- Prepared delivery/ticket context is required offline; the ticket may still lack its clean net weight.

After confirmation, completing an unknown ticket/analysis field requires correction permission under Section 6.3.2; the closed-campaign exception does not waive it.

Main Flow

1. The user selects the delivery and opens or creates its single analysis.
2. The user records the available analysis reference, entity and analysis date separately from entry time.
3. The user enters the reported yield percentage where known.
4. The system displays the clean net weight linked through the delivery ticket, without a separately editable analysis weight.

5. The user may attach PDF/JPG/PNG evidence within the initial 10 MB limit under the applicable entry/attachment permission. File contents are consulted as complete authorised evidence, without mandatory monetary-content classification.
6. The user reviews and confirms. The system saves the proposal and shows structured-record and attachment status.
7. The system updates campaign yield only for Delivered, non-cancelled deliveries with known yield and an eligible positive ticket clean net weight. Available In transport results await receipt confirmation in delivery detail and contribute nothing yet; awaiting arrival alone does not mean missing analysis.

Alternative Flows and Exceptions

A1 — Result not yet available.
The system leaves the analysis absent or incomplete without inserting a zero yield.

A2 — Clean net weight missing.
The system keeps the reported yield available and excludes it from the average until both arrival confirmation and a valid positive ticket weight exist. Disclose missing-weight coverage within the Delivered, non-cancelled set.

A3 — Analysis already exists.
The user opens the same record. With correction authority, the user may replace current values/evidence, including completing unknown fields after confirmation; the system keeps one analysis without field-change history.

A4 — Ticket weight corrected.
The system refreshes the reference weight and dependent indicators through the ticket association, without a duplicated editable weight.

A5 — Closed campaign.
The authorised user may add or correct the analysis for an existing eligible delivery without reopening; correction permission still applies to confirmed records.

A6 — Offline, failed upload or conflict.
The system preserves the local proposal and separate file status. On reconnection the server revalidates parent eligibility, authority and revision; the authorised user resolves conflicts without silently overwriting accepted data.

A7 — No eligible denominator.
The system displays the weighted average as unavailable and includes the number of contributing deliveries.

Postconditions At most one current analysis exists per delivery. Its entry does not alter official weights or delivery dates. The weighted indicator uses current eligible clean net weights and discloses missing data.

Related Requirements and Rules FR-051, FR-052, FR-054, FR-058, FR-061.

BR-012, BR-033, BR-036, BR-038, BR-044.

Decision References D-11, Q-DA-08, Q-AT-02.

4.21 UC-18 — Manage Cooperative-Reported Ticket Values

Purpose Consult, enter and correct the ticket's current reported monetary amount, online or offline; do not calculate prices or execute payments. The shared conditions in Section 4.2 apply.

Actors Farm Owner; Field Manager with the relevant scope and permissions.

Preconditions

- A ticket exists for a non-cancelled delivery; value entry is allowed In transport or after Delivered.
- The user has the requested financial consultation, entry or correction permission.
- Offline use requires authorised prepared ticket context and valid offline access.

Main Flow

1. The user opens the ticket directly or through its delivery.
2. The system displays the current amount or states that no amount has been reported.
3. For authorised entry/correction, the user provides the reported EUR amount, cooperative/reporting date and provisional/final classification.
4. The user may provide a source reference, observations and permitted PDF/JPG/PNG evidence within the initial 10 MB limit.
5. The user reviews and confirms. The system replaces the current amount without creating an additive historical revision.
6. For Delivered, non-cancelled deliveries, the system updates eligible financial indicators and identifies central acceptance or a local pending proposal. Available In transport values await receipt confirmation in delivery detail and contribute nothing yet; awaiting arrival alone does not mean a missing amount.

Alternative Flows and Exceptions

A1 — Final replaces provisional.
The system keeps one current value and distinguishes reported final/provisional classification from pending/accepted synchronisation status.

A2 — No source amount.
The system shows awaiting value and does not infer a payment, price or zero amount from weights/analysis.

A3 — Correction.
The system requires separate financial correction permission and retains only current values. Final-to-provisional reclassification remains pending.

A4 — Mixed load or reported penalty.
The system leaves mixed-load monetary totals unallocated by area and does not deduct the cooperative-reported penalty again.

A5 — Campaign closed.
The authorised user may enter/correct a late amount for an existing eligible ticket without reopening.

A6 — Offline.
The system supports authorised prepared consultation, entry and correction and marks proposals pending. Financial indicators/PDFs may include eligible values after a locally confirmed arrival, with freshness and pending labels. Central totals await server acceptance.

A7 — Evidence or synchronisation fails.
The system retains recoverable input and file status separately, checks uncertain outcomes and retries without duplicate monetary contributions.

A8 — Permission revoked, expired access or conflict.
The system blocks unauthorised access and new operations, preserves protected proposals for the agreed recovery workflow and rechecks current grants/revisions before acceptance.

Postconditions One current valuation is visible online or as an identified offline proposal. No ticket weight, analysis, payment or permanent value-change history is created by the operation.

Related Requirements and Rules FR-053, FR-054, FR-059, FR-060, FR-061, FR-062.

BR-012, BR-034, BR-035, BR-036, BR-040, BR-044.

Decision References D-15, Q-DA-09, Q-AT-02.

4.22 UC-19 — Consult Campaign Indicators

Purpose Allow an authorised user to understand the progress and results of a harvest campaign through operational, production and permitted monetary indicators.

The same consultation supports ongoing work and the final campaign review. Closing a campaign does not freeze its indicators.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Additional actor: Field Manager, with the relevant consultation permissions.

Monetary indicators require financial consultation permission, initially enabled for managers within assigned scope and revocable by the owner. Financial entry/correction remain separately granted.

Preconditions

- The campaign exists.
- The user has permission to consult the requested scope.
- Offline consultation requires a previously prepared operational summary and relevant authorised local information.

Operational activity is not a precondition: a campaign may have no recorded work or deliveries yet.

Main Flow

1. The user selects one harvest campaign.

2. The system identifies the campaign and the authorised consultation scope.
3. The user selects the campaign overview, a property summary or parcel detail.
4. The user may apply date, worker and task filters where relevant.
5. The system retrieves or calculates the eligible indicators using approved rules.
6. The system presents totals, tables and useful simple charts.
7. Each indicator identifies its meaning, unit, scope and relevant completeness.
8. The user may open an indicator to inspect its contributing records.
9. The user may export the selected summary as a PDF.

Indicator Groups Work indicators:

- Distinct workers with confirmed daily presence and an explicitly identified actual assignment within the selected campaign.
- Distinct start dates with confirmed actual work periods.
- Total confirmed actual working hours.
- Hours by worker.
- Hours by task type: machinery operation and general work.
- Actual campaign assignments whose hours or task information remain incomplete.

Production indicators: Only Delivered, non-cancelled deliveries contribute to delivery counts, weights, yield and monetary totals (D-30). Delivered counts do not require a ticket or analysis; each other result requires its own eligible source inputs.

- Confirmed deliveries.
- Known weight totals, separated by weight meaning.
- Available yield results and approved yield aggregates.
- Pending analyses and relevant result coverage.

Monetary indicators:

- Current cooperative-reported ticket amounts.
- Separate provisional and final totals.
- Tickets awaiting monetary values.

Monetary consultation is available online or from authorised prepared offline data, with financial permission.

Alternative Flows and Exceptions

A1 — No matching activity exists. The system explains that no records match. Zero is used only where it is a valid result for the defined indicator. An undefined average is not shown as zero.

A2 — Data are incomplete. The system identifies pending information. Meaningful partial results show their known coverage and exclusions within the Delivered, non-cancelled set. Available source data for In transport deliveries remains in detail as awaiting receipt confirmation, without indicator contribution or a false missing-data label.

A3 — The campaign is closed. The user may consult its current results. Late analysis results, ticket information and reported values update the relevant indicators when accepted.

A4 — An accepted correction exists. Indicators use the current accepted values under the applicable correction rules. Previous values are not added as additional contributions.

A5 — Loads have mixed provenance. Known totals may contribute to the campaign summary. Property or parcel allocation is shown for known source quantities. The current model stores no monetary amounts by origin, so mixed-load monetary totals remain unallocated by area; percentage allocation is outside this version. Unallocated information remains identifiable.

A6 — Yield calculation is unsupported. Use the reported clean net weight from the same delivery's ticket. Missing yield or eligible positive weight excludes the delivery; without a positive denominator the average is unavailable.

A7 — Access covers only part of the campaign. Results are limited to the authorised scope. A restricted view is not labelled as the unrestricted campaign total.

A8 — The device is offline. The system displays the prepared operational summary and its last successful update. Financial indicators are available from prepared data only with financial permission; missing local information remains explicit.

A9 — Local changes are pending. The operational summary combines the prepared server baseline with relevant local changes.

Creates are included once. Updates replace their corresponding baseline values. Locally accepted removals or cancellations remove the affected local contributions.

Affected indicators are identified as provisional until server acceptance. A locally confirmed arrival enables the delivery count and available eligible weights, yield and amounts in that provisional view. Central totals change only after server acceptance; pre-arrival source data remains excluded. The interface does not imply knowledge of unsynchronised changes from other devices.

A10 — A pending change is rejected. The system distinguishes the unresolved proposal from server-accepted information.

The accepted summary is restored or recalculated from accepted values. Any still-visible local proposal is separately identified and is not presented as part of the accepted total.

A11 — Requested detail is unavailable offline. The system explains that connectivity is required for that detail. It does not present unavailable records as an empty result.

A12 — Retrieval or calculation fails. The system identifies the affected indicator and offers a retry where appropriate. Failure is distinguishable from zero or a successful query with no records.

A13 — A PDF is requested. The system exports the selected authorised summary with its campaign, filters, generation date and time, completeness and synchronisation limitations. The export represents that moment, rather than a permanently frozen campaign. An offline PDF may include authorised prepared financial indicators. It identifies pending local contributions, last synchronisation and incomplete scope; financial permission is required.

Postconditions **On success:** The user can inspect the current indicators for the selected authorised scope and access available supporting records.

On offline consultation: The displayed operational summary identifies its synchronisation date and pending local changes.

On export: A PDF reflects the selected summary and its relevant limitations at generation time.

In all cases: Consultation does not modify source records, create payments or close the campaign.

Related Requirements and Rules FR-055, FR-056, FR-057, FR-058, FR-059, FR-060, FR-061, FR-062.

BR-014, BR-018, BR-019, BR-020, BR-035, BR-036, BR-037, BR-038, BR-039, BR-040.

Decision References Q-DA-01, Q-DA-05, Q-IND-01, D-15, D-30.

4.23 UC-20 — Generate Worker Documents

Purpose Generate a reviewed unsigned version from a template, then sign it through UC-14 and retain the worker-specific PDF. The shared conditions in Section 4.2 apply.

Actors Farm Owner; Field Manager with the relevant scope and permissions.

Preconditions

- The user has generation permission; managers have no such permission by default.
- The template and selected worker records belong to the authorised business context.
- Generation/acceptance of a source version requires connectivity. Concrete templates remain subject to selection and review with the client.

Main Flow

1. The user selects an available template and existing worker(s).
2. The system prefills known profile values and identifies missing fields.
3. The user completes required fields, reviews contents/worker associations and confirms.
4. The system accepts an unsigned version under UC-13 without changing worker profiles.
5. The user initiates UC-14 review/signing now or later; the worker explicitly confirms.
6. The system retains version-specific evidence and a timestamped worker-specific PDF.
7. The system exposes the result in the authorised document view with local/server and file status.

Alternative Flows and Exceptions

A1 — Missing information or template.
The system preserves draft input for correction and identifies missing fields/template without inventing values. The template inventory remains pending.

A2 — Several workers.
The system allows shared associations only for suitable templates; each worker signs separately and sees only their captured signature in their PDF.

A3 — Review cancelled or generation fails.
The system preserves accepted versions and creates no signature.

A4 — Document already exists.
The user reviews a new unsigned version; the system never overwrites signed contents or inherits signatures.

A5 — Prepared offline signing.
The user prepares the accepted source online, then initiates UC-14 offline. The system labels local signatures/PDFs pending until server acceptance; source generation remains online.

A6 — Identification or work-permit evidence.
The user uploads originals through UC-13. Verification forms neither issue identification/work permits nor establish legal approval.

A7 — Attendance log requested.
A dedicated actual-hours export remains pending and is not implied by generation.

Postconditions The system retains the distinct source, signature evidence and worker-specific PDF under signed-version retention. Worker profiles remain unchanged; unreviewed sources are never marked signed.

Related Requirements and Rules FR-038, FR-040, FR-041, FR-042, FR-043, FR-044, FR-045, FR-063.

BR-022, BR-023, BR-024, BR-026, BR-027, BR-028, BR-030, BR-041, BR-044.

Decision References D-14, Q-DOC-01, Q-DOC-02, Q-DOC-04.

5 Functional Requirements

The catalogue groups related behaviours into FR-001 to FR-064. Each ID groups a related capability; its lettered criteria are the independently referenced verification conditions. Groups organise the catalogue by affinity, while shared constraints are stated once. Catalogue inclusion does not establish MVP priority or stakeholder approval. Requirements retain stable identifiers across revisions. FR-063 and FR-064 specify document generation and account updates, respectively.

5.1 Requirements Catalogue

Common acceptance conditions. The shared conditions in Section 4.2 apply to every relevant requirement: business/operation authorisation, accurate local/server status, retry without duplicate effects, preservation of accepted data on cancellation, recoverable validation failures and explicit handling of unknown values. Connectivity remains operation-specific. The server revalidates offline proposals, including access and conflicts. Shared conditions remain mandatory without repetition in every row. A criterion referencing a business rule or section incorporates its applicable conditions; verification records the criterion and those conditions together.

Criterion references for verification. Verification checks and implementation tasks identify the requirement and the relevant lettered acceptance criterion. For example, FR-041(a) covers acceptance of a new current version, FR-041(b) covers retention and deletion, and FR-041(c) covers incorrect uploads and withdrawal.

5.1.1 Registration, Authentication and Access

Table 3: Functional requirements: Registration, Authentication and Access.

Reference	Requirement and acceptance criteria
FR-001	**The system shall register an owner and establish a business workspace.**
FR-001(a)	Require name, email, password and business name; reject duplicate emails under a consistent normalisation policy.
FR-001(b)	Verify email before access; allow replacement messages and reject invalid or expired links.
FR-001(c)	Activation creates one owner workspace without operational records; retries create no duplicate account or workspace.
FR-002	**The system shall authenticate account holders and enforce business access.**
FR-002(a)	Only activated accounts may log in with email and password; invalid credentials must not unnecessarily disclose account information.
FR-002(b)	Validate session, membership, scope and operation permission for protected records and files; expired sessions require authentication.
FR-003	**The system shall terminate sessions, isolate accounts and clean eligible local copies.**
FR-003(a)	Online or offline logout blocks local access. When connected, identify pending work and offer synchronisation first.
FR-003(b)	Preserve pending work in protected, account-isolated storage; other accounts cannot access or submit it. Explicit discard requires the originating authenticated user's confirmation of the identified losses.
FR-003(c)	Normal recovery requires the originating account and renewed authentication/authorisation. Deferred server-side logout completes on reconnection.
FR-003(d)	Remove eligible centrally accepted local copies after 30 days of account/business/device inactivity. Check on opening before resetting last use; authorised use resets it.
FR-003(e)	Never automatically delete pending work or recovery dependencies. Cleanup leaves central/signed retention unchanged, differs from offline expiry and requires removed data to be prepared online again.
FR-004	**The system shall enforce bounded offline continuity and revalidate access.**
FR-004(a)	Prepared offline access lasts 15 days from successful online permission validation; only another successful online validation renews it.
FR-004(b)	Expiry blocks consultation/new operations until connected authentication and validation; preserve pending work and dependencies. Enforce communicated revocation without claiming instant enforcement offline.

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Table 3: Functional requirements: Registration, Authentication and Access (continued).

Reference	Requirement and acceptance criteria
FR-004(c)	Reconnection checks uncertain outcomes and revalidates membership, scope, permissions, revisions and dependencies before safe synchronisation. Keep the separate 30-day cache policy in FR-003.
FR-004(d)	Support pending campaign starts and new workers, preserving dependent operations and conflicts without silent overwrite. Present routine status under Section 7.6.
FR-005	**The system shall allow users to recover their account access through a verified reset process.**
FR-005(a)	A recovery request gives a generic acknowledgement; eligible accounts receive a reset message.
FR-005(b)	Only valid, unexpired, unused authorisation permits a policy-compliant reset; consume it and invalidate online sessions on success.
FR-005(c)	Preserve ownership/permissions. Offline devices detect invalidation on reconnection or authorisation expiry; recovery without inbox access is not assumed.
FR-006	**The system shall invite managers and manage invitation acceptance.**
FR-006(a)	The owner sets recipient email, scope and permissions and may inspect, resend or cancel pending invitations.
FR-006(b)	Only the intended verified account may accept a valid invitation. Invitee registration does not automatically create an owner workspace.
FR-006(c)	Repeated acceptance/resending creates no duplicate membership; expired or cancelled invitations grant no access.
FR-007	**The system shall configure delegated scope and enforce the initial manager profile.**
FR-007(a)	Only the owner assigns properties and independent operation grants. Scope includes their parcels and current/future campaigns until revoked; no independent campaign/parcel scope grants apply. Managers cannot delegate authority.
FR-007(b)	Restrict multi-property campaign records/indicators to assigned properties. Expose relevant workers/insurance and owner-explicitly available workers, without revealing their other-property work or full mixed-property delivery/file totals outside the grant.
FR-007(c)	Apply owner-adjustable defaults and separately granted operations from Section 2.4. Evidence upload uses its entry/attachment grant, even when the source contains amounts.
FR-007(d)	Keep financial and complete-ticket consultation consistent as full-content access. Apply revocation to prepared data/files without redaction; enforce saved grants online and on reconnection.
FR-008	**The system shall revoke manager access and review existing pending proposals.**
FR-008(a)	Revocation blocks normal online business access without deleting accounts or records; offline devices enforce it on reconnection or expiry.
FR-008(b)	After origin-account authentication, submit preserved proposals through restricted recovery without normal business access. The owner accepts or rejects each; acceptance is a new explicit authorisation, subject to date, lifecycle, coverage, dependency and conflict checks.

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Table 3: Functional requirements: Registration, Authentication and Access (continued).

Reference	Requirement and acceptance criteria
FR-008(c)	Preserve originator and accepting owner separately. Do not infer prior creation or permission from device timestamps; unresolved proposals remain protected under BR-004. Signing-source eligibility cannot be overridden.
FR-064	**The system shall update verified email and password while preserving account identity.**
FR-064(a)	Connected authenticated users may change passwords after identity revalidation under the configured password/session policy.
FR-064(b)	Validate/verify the normalised new email before activation; retain the old address until then. Failed/expired links and retries change no ownership, membership or authorship.
FR-064(c)	Preserve account ID, ownership, memberships, signatures and authorship; retries create no replacement identity.

5.1.2 Properties and Campaigns

Table 4: Functional requirements: Properties and Campaigns.

Reference	Requirement and acceptance criteria
FR-009	**The system shall create properties with required and optional descriptive information.**
FR-009(a)	Require name and textual location; declared hectares and olive varieties are optional and may be added later.
FR-009(b)	Unknown area stays unknown; derive ownership from the workspace. No map is required.
FR-010	**The system shall create named parcels within a property.**
FR-010(a)	Require a name and authorised parent property; area and varieties are optional.
FR-010(b)	Parcels are optional. Campaign setup creates reusable property parcels under the same grant, not campaign-only divisions.
FR-011	**The system shall allow authorised users to consult and update properties and parcels.**
FR-011(a)	Preserve identity and historical associations; new parcels do not redistribute earlier property-level records.
FR-011(b)	Prepared details are consultable offline; creation/changes require connectivity. Missing local data does not establish non-existence.
FR-012	**The system shall archive and reactivate properties and parcels.**
FR-012(a)	Confirmed archival preserves history and blocks new operational associations through the property and its parcels.
FR-012(b)	Reactivation requires connected authority; parent reactivation does not undo independent parcel archival.

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Table 4: Functional requirements: Properties and Campaigns (continued).

Reference	Requirement and acceptance criteria
FR-013	**The system shall delete eligible properties and parcels.**
FR-013(a)	Delete only with connected authority, confirmation and no associated records.
FR-013(b)	Otherwise refuse and offer archival; cancellation preserves the record.
FR-014	**The system shall create Planned campaigns and select their coverage.**
FR-014(a)	Require name, planned start and eligible property/parcel coverage; allow multiple properties. Create Planned campaigns even where other campaigns are planned or executing.
FR-014(b)	Prevent duplicate parent/child coverage. Selection reserves no execution; apply FR-019 at start/reopening. New parcels remain reusable.
FR-015	**The system shall allow authorised users to consult and update campaigns and coverage.**
FR-015(a)	Prepared details are consultable offline; management changes require connectivity.
FR-015(b)	Validate added coverage. Removal requires reopening if closed and resolving dependencies through permitted correction/deletion; unresolved dependencies block removal, without cascade deletion.
FR-016	**The system shall explicitly start campaigns and maintain distinct planned and actual dates.**
FR-016(a)	Explicit start confirms a separate actual start and changes Planned to In progress; validate authority, chronology and exclusive coverage.
FR-016(b)	Allow prepared authorised offline start. Save a stable pending operation; dependent work/deliveries await ordered server acceptance.
FR-016(c)	Without an actual start, first actual assignment may propose its working date, but starting remains explicit. Planning, presence and deliveries never start campaigns silently; recording time proves neither actual start nor acceptance.
FR-016(d)	Atomically validate current permissions and competing execution on reconnection. Preserve conflicting starts/fieldwork for resolution; never accept both or silently change actual dates.
FR-017	**The system shall close and reopen campaigns with execution validation.**
FR-017(a)	Connected authorised closure requires an In progress campaign, actual end and activity validation; retain records and FR-035 and FR-054 completion exceptions.
FR-017(b)	Reopening returns Closed to In progress only after authority, actual-date and exclusive-coverage checks; it bypasses no correction or execution constraint.
FR-017(c)	Confirmed-work correction/removal requires reopening without a grace period. Planned overlap is allowed; conflicting In progress execution blocks reopening.
FR-018	**The system shall archive, unarchive or delete eligible campaigns.**
FR-018(a)	Archival changes visibility, not lifecycle or execution coverage; unarchiving does not reopen.

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Table 4: Functional requirements: Properties and Campaigns (continued).

Reference	Requirement and acceptance criteria
FR-018(b)	Delete only without operational records and with connected authority and confirmation; otherwise offer archival.
FR-019	**The system shall prevent conflicting campaign execution without blocking preparation.**
FR-019(a)	Allow overlapping Planned campaigns; starting, reopening or extending In progress coverage requires exclusive execution in the affected area.
FR-019(b)	Whole-property execution conflicts with every parcel; distinct parcels may run concurrently. Archival never releases In progress coverage.
FR-019(c)	Validate actual chronology and historical overlap, including retrospective changes. Central validation accepts at most one conflicting execution and preserves rejected proposals.

5.1.3 Workers, Planning and Actual Work

Table 5: Functional requirements: Workers, Planning and Actual Work.

Reference	Requirement and acceptance criteria
FR-020	**The system shall create identified worker records online or offline and maintain existing profiles.**
FR-020(a)	Require name, telephone and official identification type/country/number; generate a stable internal ID. Creation provides no account, identity verification or scanned-copy requirement.
FR-020(b)	Record machinery capability and optional notes separately from tasks; apply same-business duplicate warnings and reviewed distinct records under BR-014, without cross-business merging.
FR-020(c)	Allow authorised same-day offline creation with identical required fields and a stable pending ID. Permit immediate local attendance, work, driving and eligible document/signing use under BR-013.
FR-020(d)	Accept worker identity before dependants; recheck duplicates/authority and preserve work/files during resolution. Retries reuse the same worker and operation IDs.
FR-020(e)	Existing-profile administration remains online, preserving associations, current values and modification time. Identification types and country-specific validation remain under Q-ID-01.
FR-021	**The system shall inactivate and reactivate workers.**
FR-021(a)	Preserve work/documents, exclude inactive workers from new-work selection and flag future plans without changing them.
FR-021(b)	Reactivation reuses identity; daily non-selection or absence is not inactivation.
FR-022	**The system shall delete eligible worker records.**
FR-022(a)	Delete only without associated records and with connected authority and confirmation.

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Table 5: Functional requirements: Workers, Planning and Actual Work (continued).

Reference	Requirement and acceptance criteria
FR-022(b)	Otherwise refuse and offer inactivation; cancellation preserves the worker.
FR-023	**The system shall allow authorised users to consult prepared worker profiles and locally created worker records.**
FR-023(a)	Consult prepared authorised worker profiles and locally created worker records offline, showing pending/availability limits. Missing local data does not prove central non-existence.
FR-023(b)	Profile access alone grants no document rights. FR-020 permits offline creation; existing-profile administration remains online.
FR-024	**The system shall prepare independent daily work plans with group and individual values.**
FR-024(a)	Planning is optional. Select campaign, eligible area, workers and dates; create an independent plan for each date.
FR-024(b)	Hours/tasks are optional for one area per worker/date; multiple areas require periods for every relevant assignment.
FR-024(c)	Apply group periods, tasks and optional counted/non-counted breaks with individual overrides. Plans confirm neither attendance nor work.
FR-025	**The system shall reuse a plan or actual work as a reviewed planning proposal.**
FR-025(a)	Copy workers, areas, periods, tasks and applicable breaks to reviewed target-date proposals; preserve source records and do not copy attendance/actual confirmation.
FR-025(b)	Validate each target without silent overwrite/duplication; review inactive workers for eligible replacement or reactivation.
FR-026	**The system shall validate planned periods and scheduling conflicts.**
FR-026(a)	Defined periods must end after their start; reject same-worker planned overlaps across properties, campaigns and adjacent dates.
FR-026(b)	Allow equal consecutive boundaries; use complete overnight timestamps and start-date association. Missing hours cannot establish conflict-free scheduling.
FR-026(c)	Validate planned and actual interval sets separately; a matching plan does not duplicate actual work.
FR-027	**The system shall adjust eligible plans while preserving their role as an expectation.**
FR-027(a)	Allow worker removal/replacement before actual work exists in the affected daily-area context.
FR-027(b)	Thereafter preserve that plan as a reference; record differences as actual work/corrections. Other target-date plans remain independently editable.
FR-028	**The system shall allow authorised users to consult schedules through daily, worker and weekly views, including offline preparation.**
FR-028(a)	Use the same plans in all views with campaign/property/parcel/worker/task filters; identify missing hours/tasks, breaks, inactive workers and overnight ends.
FR-028(b)	Prepare the current and following week; prepare extra dates explicitly. Show freshness and pending changes.

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Table 5: Functional requirements: Workers, Planning and Actual Work (continued).

Reference	Requirement and acceptance criteria
FR-028(c)	Distinguish empty results, failures and unavailable local data; consultation confirms no attendance or work.
FR-029	**The system shall create and adjust plans offline using prepared reference data.**
FR-029(a)	Save supported plan changes locally as pending; revalidate authority and business rules on reconnection.
FR-029(b)	Rejected/conflicting changes remain unaccepted; preserve source and target-plan integrity during recovery.
FR-030	**The system shall distinguish daily attendance from campaign participation and unrelated workers.**
FR-030(a)	Keep one attendance decision per worker/start date/business. Selected/expected workers await explicit Present/Absent confirmation; unrelated unselected workers generate neither absence nor alerts.
FR-030(b)	Permit direct actual entry without planning, including newly created offline workers. Confirm campaign-area participation separately; presence may precede hours.
FR-030(c)	Absence cannot coexist with actual assignments/periods. Areas share daily presence; missing-hours alerts require presence and incomplete campaign work. Driver selection creates no presence.
FR-031	**The system shall explicitly confirm actual periods with their task and area.**
FR-031(a)	Review a plan or enter actual work directly; permit individual changes in group entry without silently confirming attendance or rewriting plans.
FR-031(b)	A confirmed complete period requires task, area and valid start/end instants. Split task/area changes; missing details remain incomplete assignments, not unclassified counted hours.
FR-031(c)	Use the local start date for overnight work. Keep presence, assignment, complete period and synchronisation distinct; first work may explicitly propose start under FR-016.
FR-032	**The system shall record breaks and calculate counted working time.**
FR-032(a)	Record each break's worker, start/end and counting treatment; deduct only non-counted time intersecting that worker's actual periods.
FR-032(b)	Do not deduct counted breaks or gaps outside periods; deduct overlapping break time once and attribute it to affected tasks.
FR-032(c)	Apply Section 6.4.2; no payroll calculation is implied.
FR-033	**The system shall validate actual periods and prevent worker-time overlap.**
FR-033(a)	Actual periods must end after their start; reject same-worker overlaps across areas, campaigns and adjacent dates.
FR-033(b)	Allow equal consecutive boundaries; reject zero duration and any positive overlap.
FR-033(c)	Use complete overnight timestamps; validate plans separately without counting them as actual work.

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Table 5: Functional requirements: Workers, Planning and Actual Work (continued).

Reference	Requirement and acceptance criteria
FR-034	**The system shall record attendance and actual work offline with its dependencies.**
FR-034(a)	Use prepared campaign/area context and prepared workers or workers created under FR-020; support pending FR-016 starts and retain parent/dependency IDs.
FR-034(b)	Apply the same task, chronology, attendance and overlap checks locally; label provisional totals and revalidate centrally. Preserve work at expiry under FR-004.
FR-035	**The system shall allow authorised users to enter retrospective work and complete missing information.**
FR-035(a)	Preserve actual working date separately from entry timestamp.
FR-035(b)	After closure, complete missing details/periods within actual dates without changing confirmed values/associations; validate attendance, coverage and overlap under BR-011.
FR-035(c)	Confirmed-work changes/removals require reopening; work outside actual campaign dates also requires date review. Entry never silently changes campaign dates.
FR-036	**The system shall correct attendance, actual work and valid record associations.**
FR-036(a)	Require correction authority and valid dates, coverage, overlaps and open campaign state in every affected context; no age-based deadline applies.
FR-036(b)	Allow worker, date, campaign, area, period, task and break corrections without changing plans or unrelated work.
FR-036(c)	Changing presence to absence requires confirmed removal of all contradictory assignments, periods and breaks for that business date, including assignments without hours; cancellation preserves them.
FR-036(d)	Moving a period preserves other work and attendance consistency; retain current values without permanent field-change history.
FR-037	**The system shall support offline corrections and authorised conflict resolution.**
FR-037(a)	Save prepared corrections as pending; compare changed accepted values with the proposal for authorised manual review.
FR-037(b)	Keep accepted values or resubmit a reviewed correction for validation; temporary comparison creates no permanent correction history.

5.1.4 Worker Documents and Signing

Table 6: Functional requirements: Worker Documents and Signing.

Reference	Requirement and acceptance criteria
FR-038	**The system shall allow authorised users to consult and download authorised worker documents and retained versions.**
FR-038(a)	List contract/insurance/other documents with title, version and worker-specific signing status; open/download available PDF/JPG/PNG files.
FR-038(b)	Apply business-wide owner access and manager worker/document grants under BR-024; profile access alone is insufficient.

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Table 6: Functional requirements: Worker Documents and Signing (continued).

Reference	Requirement and acceptance criteria
FR-038(c)	Apply BR-026 retention and BR-025 expiry, including unknown, expired and withdrawn states. Current does not imply valid/signed; unavailable retrieval is not an empty list.
FR-039	**The system shall prepare insurance documents for field consultation.**
FR-039(a)	Prepare current/previous retained insurance for workers planned for the current/following week and additional existing workers selected online.
FR-039(b)	Show preparation failures, missing files and freshness; apply received access/deletion changes. Preparation coverage does not determine offline expiry.
FR-040	**The system shall upload worker documents and their metadata.**
FR-040(a)	Online upload requires modification permission, one or more same-business workers, category, title and file; notes are optional, and version/upload time are recorded automatically.
FR-040(b)	Apply BR-044 formats/initial size limit, subject to representative-file validation; unknown insurance expiry remains unconfirmed.
FR-040(c)	Allow shared files; external-signature declarations follow BR-027 and create no worker application signatures.
FR-040(d)	Validate/review before acceptance; incomplete files cannot become current, and retries create no duplicate versions.
FR-041	**The system shall manage versions, signed retention, withdrawal and eligible deletion.**
FR-041(a)	An accepted replacement becomes current without inherited signatures/declarations; failure, cancellation and retry trigger no replacement or extra cleanup.
FR-041(b)	Apply BR-026: retain all contracts/signed versions and current/previous insurance/other versions; clean only older unsigned unprotected versions. Protect server-known pending signing and preserve local signatures for Q-DOC-04 resolution.
FR-041(c)	Delete an incorrect unsigned upload only with authorisation, confirmation and no dependencies. Withdraw signed errors while preserving source/evidence/renditions; a withdrawn current version remains current until replacement, with no fallback.
FR-041(d)	Replacement creates a new version; management is online and accepted deletions propagate to local copies on synchronisation. Signed-retention periods remain pending.
FR-042	**The system shall conduct in-person document review and explicit worker signing.**
FR-042(a)	An authorised initiator identifies the present worker and eligible version; no worker account or owner signature is required.
FR-042(b)	The worker reviews the document and draws a signature, may redraw or cancel, then explicitly confirms the non-empty signature and reviewed version.
FR-042(c)	Record worker, version, initiator and timestamp; never replace/transfer confirmed drawings. Upload/generation creates no signing event.
FR-042(d)	For uploaded/generated sources, retain a timestamped worker-specific signed PDF linked to version/evidence, preserving the source and excluding others' captured signatures. Offline output remains pending with separate file-transfer status.

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Table 6: Functional requirements: Worker Documents and Signing (continued).

Reference	Requirement and acceptance criteria
FR-043	**The system shall prevent duplicate signing and recover uncertain submissions.**
FR-043(a)	Do not repeat completed worker/version signing; show accepted results and check/resume pending or uncertain operations before requesting another signature.
FR-043(b)	Distinguish no confirmed signature, local pending, accepted and resolution-required outcomes; preserve recoverable results without duplicate events.
FR-044	**The system shall maintain individual signing results for shared documents.**
FR-044(a)	Track signing per worker/version; one worker's signature never signs for another.
FR-044(b)	Show only the selected worker's captured signature. Do not imply redaction of source content/external signatures; enforce source-sharing permissions.
FR-045	**The system shall prepare and synchronise offline signing for the reviewed version.**
FR-045(a)	Allow explicit preparation/signing of authorised documents, including contracts, without bulk contract download. For new local workers, separately authorise association/signing and queue worker/association before signature; preserve the reviewed source.
FR-045(b)	Accept pending signatures only against the original eligible version; disclose newer versions where applicable.
FR-045(c)	Preserve withdrawn/deleted-source proposals for protected owner inspection/rejection; no owner approval can validate an ineligible source. Never transfer signatures or resurrect deleted versions. A valid replacement needs fresh worker review/signing.
FR-063	**The system shall generate reviewed worker documents from available templates.**
FR-063(a)	With online generation authority, select template/workers and prefill known fields; identify gaps and permit completion without editing source profiles.
FR-063(b)	Review content/associations before accepting an unsigned version under normal document rules. Failure/cancellation changes no accepted version; select templates/categories with the client. Verification forms issue no identity documents.
FR-063(c)	Use UC-14 signing, including prepared offline signing, and FR-042(d) signed PDFs; generation alone creates no signature.

5.1.5 Deliveries, Weighing, Analyses and Reported Values

Table 7: Functional requirements: Deliveries, Weighing, Analyses and Reported Values.

Reference	Requirement and acceptance criteria
FR-046	**The system shall register, correct or cancel deliveries and their provenance, and delete delivery records when permitted.**
FR-046(a)	Require campaign, eligible origins, registered worker driver, vehicle, destination and departure. Record known source quantities separately from declared transport origin/reference; infer no allocation or driver attendance/hours.

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Table 7: Functional requirements: Deliveries, Weighing, Analyses and Reported Values (continued).

Reference	Requirement and acceptance criteria
FR-046(b)	Delivery changes require separate correction authority and valid chronology/contexts. Reopen closed affected campaigns for corrections, cancellation, deletion or reassignment; FR-054 still covers source-result updates.
FR-046(c)	Confirmed deletion requires no ticket, analysis, valuation or evidence; remove composed transport/batches atomically. Alternatively cancel with a reason, preserving dependencies and excluding indicator contributions under BR-031.
FR-046(d)	Reassignment validates all affected permissions, dates, coverage and dependencies without cascade deletion. Block unresolved inconsistencies; cancellation implies no reversal/reinstatement workflow.
FR-047	**The system shall confirm a delivery or retrospectively register the complete transport.**
FR-047(a)	Save departure as In transport; authorised arrival confirmation changes it to Delivered with arrival not before departure.
FR-047(b)	Allow retrospective entry as Delivered with both event times, distinct from entry times; one associated transport per delivery remains the proposal.
FR-047(c)	Arrival enables eligible counts/results under BR-038. Offline confirmation updates labelled local provisional indicators; central totals await acceptance.
FR-048	**The system shall record departures and delivery confirmations offline.**
FR-048(a)	Use prepared campaign, area and worker data; allow a new destination name.
FR-048(b)	Keep physical and synchronisation states separate; validate current authority and campaign context centrally.
FR-049	**The system shall record one weighing ticket per delivery and its reported source values.**
FR-049(a)	Allow one ticket per non-cancelled delivery, including In transport. Record available reference, cooperative, weighing time, clean net kg and other identified weights/units; pre-arrival values await receipt under BR-038.
FR-049(b)	Record optional weighing times, excess duration/unit and reported monetary penalty. Record whether adjustments are applied; derive neither clean weight nor penalties and never deduct adjustments twice.
FR-049(c)	Expose incomplete inputs and exclude dependent calculations. References are not globally unique; concurrent creation accepts at most one ticket.
FR-049(d)	Apply BR-044 evidence limits, Section 2.4 grants and Section 9.1.7 transfer recovery. Full-ticket access includes printed amounts without redaction/classification; structured financial writes require separate grants. Prepared authorised data/files work offline.
FR-050	**The system shall correct the current ticket and adjustment information.**
FR-050(a)	Separate correction authority updates the existing ticket, including post-confirmation completion; retain current values without a second ticket or permanent field history.
FR-050(b)	Refresh indicators under FR-061; preserve uniqueness and use the common conflict workflow.

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Table 7: Functional requirements: Deliveries, Weighing, Analyses and Reported Values (continued).

Reference	Requirement and acceptance criteria
FR-051	**The system shall record one analysis per delivery and its reported yield.**
FR-051(a)	Allow one analysis per non-cancelled delivery, including In transport. Record available yield percentage, entity, reference and analysis date separately from entry time; pre-arrival results await receipt under BR-038.
FR-051(b)	Weight yield by the delivery ticket's clean net weight in kg, never an editable analysis copy; exclude missing yield/weight and ineligible or non-positive weight, disclosing incomplete coverage.
FR-051(c)	Allow prepared offline entry and BR-044 evidence; track record/file states separately. No automatic extraction occurs.
FR-052	**The system shall correct the current analysis without creating another result for the delivery.**
FR-052(a)	Separate correction authority updates the existing analysis, including post-confirmation completion; retain one current result without permanent field history.
FR-052(b)	Refresh indicators without changing ticket weights or delivery dates; ticket correction updates the derived reference weight.
FR-053	**The system shall allow authorised users to consult and manage the current reported ticket amount online and offline.**
FR-053(a)	Require operation-specific financial authority and an existing ticket/non-cancelled delivery, including In transport. Store the EUR amount to two decimal places, the cooperative/date and provisional/final status; reference, notes and PDF/JPG/PNG evidence are optional. Apply BR-038 pre-arrival exclusion.
FR-053(b)	Final replaces provisional; corrections retain one current value. Missing differs from zero; final-to-provisional reclassification remains pending.
FR-053(c)	Do not calculate prices or payments, reapply reported penalties or infer area allocations. Source files retain separate transfer/retention status.
FR-053(d)	Allow prepared authorised offline consultation/entry/correction. Apply Section 2.4 financial/full-file grants; label freshness/pending proposals in indicators/PDFs. No monetary-content classification is required.
FR-054	**The system shall complete and correct external information after campaign closure.**
FR-054(a)	Authorised ticket, analysis and reported-value additions/corrections for existing deliveries/tickets need no reopening.
FR-054(b)	This does not permit confirmed-work correction, new activity outside actual dates, delivery reversal or campaign reassignment.

5.1.6 Campaign Indicators and Reports

Table 8: Functional requirements: Campaign Indicators and Reports.

Reference	Requirement and acceptance criteria
FR-055	**The system shall allow authorised users to consult campaign indicators and property/parcel breakdowns.**
FR-055(a)	Consult one campaign during/after activity, with overview, property summaries and available parcel detail.
FR-055(b)	Allow campaigns without activity; preserve identity and label partial authorised scope. Cross-campaign comparison remains excluded.
FR-056	**The system shall filter indicators and inspect their supporting records.**
FR-056(a)	Filter by date/property/parcel/worker/task only where source relationships exist; label indicator scope and date basis.
FR-056(b)	Enforce permission/local availability for contributing records. Worker/task filters allocate no deliveries/money; unavailable detail is not empty.
FR-057	**The system shall present actual participants, working days and counted hours.**
FR-057(a)	Count distinct present workers with actual campaign assignment, including incomplete hours, and distinct eligible period-start dates.
FR-057(b)	Show counted hours overall/by worker/task using confirmed complete actual periods and worker breaks, never plans or machinery capability.
FR-057(c)	Recalculate distinct workers/dates across areas; disclose unknown periods and incomplete assignments.
FR-058	**The system shall present delivery, weight and clean-net-weighted yield indicators.**
FR-058(a)	Only Delivered, non-cancelled deliveries contribute. Count them without tickets/analyses; sum each eligible compatible reported weight once.
FR-058(b)	For that set, weight known yield by eligible positive ticket clean net kg. Disclose exclusions, contributing count and incomplete coverage; no eligible denominator means unavailable, not zero.
FR-058(c)	Count mixed loads once; allocate only known source quantities. Under BR-035, mixed-load money remains unallocated by area and percentage allocation is excluded.
FR-058(d)	Show pre-arrival ticket/analysis data as awaiting receipt; exclude it from indicators without falsely labelling it missing.
FR-059	**The system shall present current reported monetary indicators with financial authorisation.**
FR-059(a)	For Delivered, non-cancelled deliveries, show current provisional/final totals separately and tickets awaiting amounts, online or from authorised prepared data.
FR-059(b)	Count each eligible current valuation once; show pre-arrival amounts as awaiting receipt without contribution. Keep financial and sync states distinct; infer no payment.
FR-060	**The system shall present authorised prepared indicators offline.**
FR-060(a)	Show missing/unallocated inputs, available scope, freshness and pending contributions; imply no knowledge of later remote changes.

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Table 8: Functional requirements: Campaign Indicators and Reports (continued).

Reference	Requirement and acceptance criteria
FR-060(b)	Combine prepared accepted data with local proposals: create once, replace updates and suppress removals. Local arrival enables eligible provisional contributions; only server acceptance changes central totals.
FR-060(c)	Distinguish unavailable data, failures and zero; financial indicators require their grant and prepared data.
FR-061	**The system shall refresh derived totals after source changes without duplicate contributions.**
FR-061(a)	Recalculate affected origin/destination totals after accepted corrections or late data, including after closure; indicators are read-only.
FR-061(b)	Local proposals affect only labelled provisional results; rejected/unresolved proposals never appear as centrally accepted. Retain unresolved proposals separately.
FR-061(c)	Closure does not freeze results; previous exports remain snapshots.
FR-062	**The system shall export an authorised campaign snapshot as PDF online or offline.**
FR-062(a)	Include campaign, authorised scope, filters/date basis, generation time, completeness and synchronisation limits.
FR-062(b)	Offline PDFs identify freshness/pending contributions; include money only with permission and prepared data, keeping financial and sync states distinct.
FR-062(c)	Export changes no source records. Detailed Excel and dedicated printing are not included in this PDF capability; a dedicated work-attendance export remains pending under Q-DOC-01.

5.2 Mapping to Use Cases

This reverse index records the main and supporting functional requirements explicitly linked by each use case. Supporting links identify dependencies, not additional ownership of the same capability. Each use case also links to its requirements and business rules, while each NFR identifies the functional capabilities it constrains. These links support impact analysis and criterion-level verification; they do not establish implementation or exhaustive test coverage.

Table 9: Use-case index to the functional catalogue.

Use case	Main and related functional requirements
UC-01	FR-001
UC-02	FR-002, FR-003, FR-004
UC-03	FR-005, FR-064
UC-04	FR-004, FR-006, FR-007, FR-008
UC-05	FR-009, FR-010, FR-011, FR-012, FR-013
UC-06	FR-004, FR-010, FR-014, FR-015, FR-016, FR-017, FR-018, FR-019
UC-07	FR-004, FR-020, FR-021, FR-022, FR-023

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Use case	Main and related functional requirements
UC-08	FR-024, FR-025, FR-026, FR-027, FR-029
UC-09	FR-028
UC-10	FR-004, FR-016, FR-020, FR-030, FR-031, FR-032, FR-033, FR-034, FR-035
UC-11	FR-033, FR-036, FR-037, FR-061
UC-12	FR-038, FR-039, FR-041, FR-044
UC-13	FR-040, FR-041, FR-063
UC-14	FR-020, FR-041, FR-042, FR-043, FR-044, FR-045, FR-063
UC-15	FR-020, FR-046, FR-047, FR-048, FR-061
UC-16	FR-049, FR-050, FR-053, FR-054, FR-061
UC-17	FR-051, FR-052, FR-054, FR-058, FR-061
UC-18	FR-053, FR-054, FR-059, FR-060, FR-061, FR-062
UC-19	FR-055, FR-056, FR-057, FR-058, FR-059, FR-060, FR-061, FR-062
UC-20	FR-038, FR-040, FR-041, FR-042, FR-043, FR-044, FR-045, FR-063

6 Business Rules and Record States

The forty-four consolidated rules define shared constraints once. States, correction principles and formulas below describe their application. Pending questions and reviewed decisions are centralised in Section 11.

6.1 Business Rules

Table 10: Consolidated business rules.

ID	Business rule
BR-001	Self-registration by an owner requires name, email, password and business name. One consistently normalised email identity identifies one account. Verification precedes active access and creates one initially empty owner workspace that may later contain several properties and campaigns. Account, business membership and worker record are distinct; workers have no management accounts and invited managers do not automatically receive an owner workspace.

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ID	Business rule
BR-002	Offline access requires prior authentication and authorisation valid for 15 days after the last successful online validation of permissions. Only a successful online validation renews that period; opening cached data or changing the device clock does not. At expiry, block consultation and new operations until renewed online authentication and permission validation, preserving protected pending records and recovery dependencies. Initial login, activation, recovery and access administration require connectivity. Revocation is enforced online or at reconnection; disconnected access is independently bounded by expiry. Logout isolates retained data. The separate 30-day inactivity cleanup removes only eligible centrally accepted local copies, never automatically pending work. Reset tokens remain single-use/time-limited and do not change ownership or permissions.
BR-003	Only the owner invites managers and assigns properties plus independent operation permissions. Property scope includes its parcels and current/future campaigns until revoked; separate campaign/parcel grants are excluded. Multiple managers may share a property. Multi-property campaign reads and indicators disclose only authorised property records; never unrestricted totals. Relevant worker profiles/insurance and explicitly owner-made-available workers are accessible without exposing other-property work. Section 2.4 defines adjustable defaults, including planning, attendance/work entry, deliveries, weighing/analysis entry, and financial/full-ticket consultation. Structured monetary entry/correction, confirmed-record corrections, worker/property/campaign management and document upload/generation/signing require their separate grants. Consultation of insurance is mandatory for workers under the manager's responsibility: it is a permission guarantee, not a prerequisite review before each operation. Contracts and other documents require an additional consultation grant. Only the intended verified account may accept the invitation; managers cannot extend their own powers.
BR-004	Revocation blocks normal business actions when communicated. The originating authenticated account may submit protected pending proposals through restricted recovery without regaining business access. The owner accepts or rejects each; acceptance supplies new explicit authority and must pass normal lifecycle, date, coverage, revision, dependency and integrity checks. Preserve originator and accepting owner separately. Device timestamps do not prove prior creation or permission; automatic proof of old versus fabricated work is not claimed. Preserve unresolved proposals. Owner approval cannot make a withdrawn/deleted signing source eligible or transfer a signature.
BR-005	A property may have no parcels; each parcel belongs to one property. Declared hectares and varieties are optional. Declared area is the reference when known; map estimates are separate future information. Property-level quantities are not redistributed among parcels and parcel-area totals are not assumed to equal the whole property.
BR-006	Property/parcel deletion requires no associated records; otherwise use archival. Campaign deletion requires no operational records. Archival preserves history. An archived property blocks new operational use through itself and its parcels; reactivating the parent does not reactivate independently archived parcels.
BR-007	A campaign covers at least one whole property or eligible parcel and may cover several properties. Parent and child areas must not duplicate coverage. Parcels created during setup remain reusable property records. New divisions do not silently redistribute historical records.

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ID	Business rule
BR-008	Campaign lifecycle is Planned, In progress or Closed, with independent archival. Several Planned campaigns may cover the same area, including an area currently in use. A Planned campaign does not reserve execution. Starting or reopening a campaign into In progress requires exclusive execution coverage: a whole property conflicts with all its parcels; distinct parcels may operate concurrently. Validate actual dates and relevant historical activity as well as current execution. In-progress coverage changes undergo the same checks. Archival never releases coverage of an In progress campaign. Offline starts are provisional until central validation; two devices cannot establish two accepted simultaneous executions on the same area.
BR-009	A campaign is created Planned with a required planned start. An authorised user explicitly starts it and confirms its separate actual start date, online or offline on a prepared device. Without a recorded start, the first actual campaign-area assignment may propose its working date, but still requires explicit start confirmation. Planning, presence and delivery entry do not silently start the campaign. An offline start and dependent field records remain pending; the server validates permissions, dates and coverage before acceptance, preserving conflicting proposals. Device recording time is not proof of the actual event or acceptance. Actual end is recorded at closure; existing actual dates are not silently changed.
BR-010	Remove campaign coverage only after resolving dependent records through permitted correction or deletion workflows. Reopen a closed campaign before changing coverage or operational associations. Validate affected contexts and do not cascade-delete records; unresolved or undefined corrections block removal.
BR-011	Closed campaigns permit missing work details or previously unrecorded periods within valid actual dates, without replacing/removing confirmed values or changing confirmed associations. Attendance, coverage and overlap must remain consistent. Correcting/removing confirmed work requires reopening, with no age-based deadline. Work outside valid actual dates requires reopening and date review.
BR-012	Ticket information, analyses and reported amounts for existing deliveries/tickets may be added or corrected after closure without reopening. This exception does not authorise new operational activity outside campaign dates or exceptional delivery changes.
BR-013	Property/parcel structure, campaign creation/coverage/closure/reopening, existing worker-profile edits/inactivation/deletion, document source-version management, template generation and access administration require connectivity. Explicit campaign start and basic same-day worker creation are supported offline, in addition to authorised prepared consultation, planning, attendance/work, delivery/results, signing and financial operations. New local workers have stable internal identifiers and may be used immediately in permitted local records; dependent operations remain pending until identity and permissions are validated. Signing requires a suitable available document version and authorised worker association; creating a worker does not certify documents or grant signing powers. Local confirmations are distinct from central acceptance.

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ID	Business rule
BR-014	A worker belongs to one business and is reused across its properties/campaigns. Creation requires full name, telephone and an official identification type, issuing country and identification number, including offline creation. Keep a stable generated internal identifier; the official number is not the primary key. Identification values are text, not numeric quantities. Matching name, telephone or the normalised identification tuple within that business produces a duplicate warning, not mandatory rejection or automatic merging. Cross-business identities are not merged. Entering identification neither verifies the document nor requires uploading its copy. Accepted document types and jurisdiction-specific validation remain pending. Keep current profile values and last-modified time; preserve all dependent records during reconciliation.
BR-015	Machinery capability is profile information and does not establish the work actually performed or prove certification. A capable operator may perform general work. Every confirmed actual work period requires its actual task; task/area changes use separate periods. Attendance or an actual assignment may be confirmed before periods are complete, but an unknown task cannot be accepted as a completed period. Profile edits do not relabel historical work.
BR-016	Worker inactivation preserves history and flags existing future plans for review rather than deleting them. New assignments require an eligible worker or authorised reactivation. Non-selection or absence on a date is not inactivation.
BR-017	Advance planning is optional: actual work may be recorded directly without a prior plan. Multi-date plans remain independent. Copying planned or actual work creates a reviewed proposal, never attendance or confirmed actual work. Planned hours/tasks are optional for one area per worker/date; multiple areas require explicit periods for all relevant assignments. Actual work freezes the corresponding daily-area plan as an expectation. Defined planned periods must be positive and non-overlapping; consecutive boundaries are allowed. Validate planned and actual sets separately.
BR-018	Attendance is one decision per worker and start-based working date within the business, separate from campaign/area participation. A person neither selected/expected nor assigned that day has no participation expectation and generates no missing-attendance alert; lack of selection is not absence. Expected/selected workers without a decision await confirmation. Present and absent are explicit decisions; absence cannot coexist with contradictory actual assignments or periods. A worker may participate in several areas under the same daily attendance. Missing-hours alerts concern confirmed presence plus campaign participation, not every active worker in the business. Plans and breaks never become actual work without explicit confirmation.
BR-019	Actual area/task changes require distinguishable periods; actual intervals cannot overlap for the same worker across contexts. Record work at minute precision using complete timestamps and the business timezone, including midnight/daylight-saving transitions. The working date is the local start date, distinct from entry date. Sum durations without rounding each period; display totals in hours and minutes.
BR-020	Only explicitly non-counted breaks belonging to the same worker reduce intersecting actual periods. Use the union of such intervals to deduct time once. Counted breaks and gaps already outside recorded periods are not deducted. Attribute each deduction to the affected task/area period; no wage calculation follows.

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ID	Business rule
BR-021	Specific authority is required for corrections in every affected context. Retain current values without permanent field-change history. Conflicting accepted values require authorised comparison and renewed validation, not silent overwrite. Local proposals remain provisional and rejected proposals do not contribute to accepted central totals.
BR-022	Worker documents are contract, insurance or other. Upload requires existing same-business workers, category, title and PDF/JPG/PNG. The initial per-file limit is 10 MB, subject to representative-file validation; the byte convention remains to be fixed. Observations are optional and version/time metadata automatic. Generation follows BR-041.
BR-023	A shared document has one source/version history and several same-business worker associations. Application signatures remain per worker/version and each worker-specific view shows that worker's captured signature. Source content may contain other information; individual-signature display does not redact it.
BR-024	Profile consultation does not grant all document access. Document consultation, modification and signing initiation are separate permissions. Relevant insurance consultation is mandatory within a manager's worker responsibility; contracts/other documents require additional authority. Financial access is separately controlled.
BR-025	Insurance validity reflects recorded expiry, not upload order: missing date is unconfirmed, passed expiry is expired and an unpassed expiry has valid recorded expiry status. It is not independent coverage verification. Expired retained files remain consultable and withdrawal is a separate restriction.
BR-026	A new accepted version becomes current without inherited signatures or external declarations. Retain all contract versions and every signed version, including externally declared signed uploads. Insurance/other documents additionally retain current and immediate previous versions; only older unsigned unprotected versions are eligible for rolling cleanup after accepted replacement. Protect pending signing operations known to the server; preserve signatures still exclusively local for Q-DOC-04 resolution if the source becomes unavailable. Signed source/evidence/renditions are not automatically deleted; retention periods remain pending. Unsigned dependency-free errors may be deleted after confirmation; signed errors are withdrawn. A withdrawn current stays current until replacement, without fallback. Failed/cancelled/retried replacements trigger no extra deletion.
BR-027	An external-signature declaration describes the uploaded version, not a verified application signing event or signatures for all associated workers.
BR-028	Signing is in person on an authorised initiator's device. The initiator identifies the worker; the worker reviews document and drawing and explicitly confirms a non-empty signature. Redraw/cancel is allowed only before confirmation. Preserve worker, version, initiator and timestamp; do not replace or repeat a confirmed drawing for that pair. No worker account or owner signature is required.
BR-029	Prepare current/previous retained insurance for workers planned in the current/next week, with additional existing workers selected while connected. Other authorised versions may be explicitly prepared for offline signing; this does not automatically download all contracts. Preparation coverage does not define offline-session duration.

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ID	Business rule
BR-030	A pending signature remains tied to the reviewed version. A newer version never inherits it. If the original remains eligible it may be accepted with a newer-version notice; withdrawal/deletion requires resolution and must not recreate a deliberately deleted version. Shared deletion reaches prepared copies on synchronisation, not instantaneously while disconnected.
BR-031	One associated transport belongs to each delivery in one campaign. Require eligible source property/parcel entries, a registered worker driver, vehicle, destination and departure; arrival is not earlier than departure. Driver selection creates no attendance/hours. Delivery corrections require separate authority and valid affected contexts; reopen closed campaigns before delivery edits, cancellation, deletion or reassignment. Deletion requires no dependent ticket, analysis, valuation or evidence. Remove the delivery's composed transport and source-batch rows in the same validated operation; they do not independently prevent eligible delivery deletion. Cancellation requires a reason, retains dependencies and excludes linked results from indicators. Campaign/source moves validate all dependencies without cascading deletion; unresolved inconsistencies block the move. Physical refusal/interrupted trips and reinstatement are not defined by this cancellation rule.
BR-032	Each delivery has at most one ticket, possibly added later. Preserve reported clean net weight in kg and other weights/adjustments with their meanings and units. Missing is not zero. Optional weighing start/end, excess duration/unit and monetary penalty reproduce cooperative source data. Do not calculate final clean weight or penalties, or deduct reported adjustments twice. Correction updates the existing current ticket; source reference is not globally unique.
BR-033	Each delivery has at most one analysis, possibly added later; corrections retain only current values. The reported yield percentage is weighted by the clean net weight from the same delivery's ticket. Do not store a competing editable analysis reference weight. Exclude missing yield or missing/non-positive eligible weight, disclose partial coverage and show unavailable if no positive denominator exists.
BR-034	A ticket has at most one current cooperative-reported EUR amount, provisional or final. Final replaces provisional and corrections retain only current values. Do not add revisions, invent missing amounts, derive prices, reapply deductions or infer payment. Final-to-provisional reclassification is not assumed. Monetary source attachments are not worker documents.
BR-035	Mixed loads count once in eligible campaign totals. Keep actual source areas/known quantities separate from the declared origin in the transport document. Distribute weight or money only using known source quantities/amounts, not dominant-parcel assumptions or invented percentages. The current schema represents source quantities but no source monetary amounts; mixed-load monetary totals remain unallocated by area. Unknown source allocation is identified as unallocated, not zero. Manual percentage allocation is outside this version; parcels remain optional.

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ID	Business rule
BR-036	Supported prepared operational and financial actions, explicit campaign starts and basic worker creation may be confirmed locally under valid authority. Local creates count once in labelled provisional views; updates replace and cancellations/removals suppress contributions. Keep accepted central values distinct from pending proposals. Automatically attempt synchronisation on reconnection and expose a simple status; reveal resolution only for a genuine problem, never require routine technical queue management. Revalidate identity dependencies, permission, revision, campaign execution and business rules before acceptance. Financial consultation/full-ticket access follow the default/revocable grant; structured financial writes need additional authority. The 15-day authorisation gate follows BR-002; preserve pending work at expiry. Separate 30-day cache inactivity cleanup is not that authorisation period. Approved recovery targets are specified by the NFRs; concrete retry timing and recovery mechanisms still require validation.
BR-037	Review one campaign at a time. Participants are distinct workers with confirmed daily presence and an actual campaign-area assignment, including assignments whose hours are still incomplete. Work days derive from eligible actual-period start dates. Counted hours use confirmed periods with required actual tasks and same-worker breaks, never plans or profile capability. Show incomplete assignments separately rather than inventing zero hours. Distinct people and dates are not additive across area breakdowns. No alert is inferred merely from an unselected business worker.
BR-038	Only Delivered, non-cancelled deliveries contribute to counts, weights, yield and amounts. Count deliveries even without a ticket/analysis. Sum eligible compatible weights once; weight known yield by positive clean net kg; separate current provisional/final amounts without implying payment. Source data may be entered In transport but awaits receipt confirmation in delivery detail, without contribution or a false missing-data label. A local offline arrival updates only labelled provisional indicators until server acceptance. Disclose missing/ambiguous eligible inputs and partial coverage. Theoretical oil is not oil actually received.
BR-039	Filters apply only to indicators with a defined source relationship and labelled date basis. Worker/task filters do not allocate deliveries or money. Underlying-record access remains separately authorised; partial scope is identified without revealing restricted information through detail or differences between totals.
BR-040	Campaign closure does not freeze indicators: accepted late data and authorised source corrections update totals. PDFs are timestamped snapshots with filters, scope, completeness and synchronisation limitations. Authorised prepared financial content may appear offline, with freshness and pending status; financial permission is required. Detailed spreadsheets and cross-campaign comparisons are not included in this version. A dedicated attendance export remains pending under Q-DOC-01.
BR-041	Generation selects a provided template and workers, prefills known fields, permits completion and requires review before an unsigned version is accepted. It does not silently edit source profiles or issue identity/permit documents. Signing follows the reviewed version; timestamped worker-specific PDF renditions preserve source and evidence. Exact templates, supplied by the team/client, remain pending.
BR-042	Verified email/password changes preserve account identity, workspace ownership, memberships and authorship. A new email becomes active only after uniqueness checks and verification; the old address remains active until then. Account/business deletion remains pending and ownership transfer is outside this version.

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ID	Business rule
BR-043	The business has a configured timezone. Work entry/display uses minutes and complete date/time instants; the working date follows local start. Durations use elapsed time across midnight and clock changes, without per-period rounding; totals display hours and minutes. Client-recorded, event and server-acceptance times remain distinct.
BR-044	Uploaded worker documents and source attachments accept PDF/JPG/PNG up to the initial 10 MB per-file limit. Failed uploads remain retryable without duplicates. Structured operational records may be saved before evidence transfer succeeds. A document version cannot become current before its source is fully available; offline viewing/signing requires prepared files.

6.2 Record States and Transitions

6.2.1 Accounts, Invitations and Access

An account moves from *Pending verification* to *Active* after valid email verification. An expired link does not itself delete the pending registration. Active identity is separate from business membership.

Invitations move from *Pending* to *Accepted*, *Cancelled* or *Expired*. Only acceptance by the intended verified account activates membership. A manager membership moves from *Active* to *Revoked*; restoration requires a new explicit owner authorisation through invitation, not duplicate operational records.

A session is authenticated, expired or terminated. Offline access additionally requires available authorised context and authorisation valid for 15 days since the last successful online permissions validation. Expiry blocks access/new operations but preserves pending dependencies; validation online renews the period. Account suspension/deletion and ownership transfer are not defined by these states.

6.2.2 Properties, Parcels and Workers

Properties/parcels move between *Active* and *Archived*; workers between *Active* and *Inactive*. Deletion is terminal and subject to dependency checks, not a reversible availability state. A parcel's availability also depends on its parent; parent reactivation does not undo independent parcel archival. Worker inactivation preserves future plans with warnings.

6.2.3 Campaign Lifecycle and Archival

Creation produces *Planned*. Explicit accepted start transitions to *In progress*; closure records actual end and produces *Closed*. Authorised reopening returns Closed to In progress after coverage/date validation; "Reopened" is not a fourth permanent state. Only In progress reserves execution. Planned campaigns may overlap coverage and actual dates remain independently relevant to historical records. Archival is a separate visibility attribute and does not release In progress coverage. A local offline start has pending sync state, not another accepted lifecycle state or a central reservation guarantee. Deletion requires no operational records or blocking pending dependencies.

6.2.4 Attendance and Completeness

For selected/expected workers, attendance is *Awaiting confirmation*, *Present* or *Absent*, once per worker and working date in the business. Initial confirmation establishes presence/absence; later changes follow correction rules. Unselected unrelated workers have no participation expectation and need no DailyAttendance row; this is not a fourth confirmed absence/presence state. Actual campaign-area assignment is separate. A late arrival uses actual times, not another attendance state. Work completeness is derived for the record/calculation; a confirmed period may support a labelled partial total. No mandatory daily-close action is introduced.

6.2.5 Synchronisation

Operations use *Pending*, *Synchronising*, *Accepted*, *Transmission failed*, *Rejected*, *Conflicted* or *Outcome uncertain*. Transmission failed means delivery could not be completed without evidence of acceptance; Rejected is an explicit server refusal. Conflicted requires comparison with accepted revisions. Outcome uncertain means the server may have committed without an acknowledgement; check the existing operation before retry. These states do not use a generic Failed label. Local confirmation is not server acceptance. An offline delivery may be delivered physically while its confirmation is pending.

6.2.6 Documents and Signatures

Version currency (current, previous or older retained version), withdrawal (available/withdrawn), recorded insurance expiry (valid/expired/unconfirmed) and signature information are independent. Online and offline, recorded expiry is compared with the current business date. Missing expiry is unconfirmed; an unpassed recorded expiry is not independent verification of coverage. Expired and withdrawn retained files remain historically consultable, but withdrawal prevents operational use. A withdrawn current version remains current until replacement, without fallback.

An external-signature declaration is not an application event and does not establish individual signers. Application signing is tracked per worker/version: no confirmed signature, locally confirmed/pending, accepted, or resolution required. Transport failure is a synchronisation condition, not a signature state. Retention, withdrawal and cleanup follow BR-026; the version example is in Section 6.3.5.

6.2.7 Deliveries and Source Information

Departure establishes *In transport*; arrival confirmation establishes *Delivered*. Retrospective registration may create Delivered directly with both timestamps. Ordinary timestamp correction does not reverse delivery; cancellation of an erroneous entry changes it to *Cancelled*, preserves its linked records and excludes their indicator contributions. Require reason, authority and reopening of closed affected campaigns. Deletion checks and composed-row removal follow BR-031. Reinstatement or physical interrupted/refused delivery scenarios remain outside this defined flow.

Weighing/analysis completeness is independent: *Not recorded*, *Incomplete* or *Available*. A supplied attachment does not itself establish complete or calculation-eligible structured data; field thresholds remain under Q-DA-08.

Tickets, analyses and amounts may be recorded In transport. Available data then awaits receipt confirmation in delivery detail; it is not missing and contributes to no indicator. Delivered, non-cancelled eligibility follows BR-038, including locally confirmed arrivals in provisional views.

Reported amounts progress from *Not reported* to *Provisional* or directly to *Final*; final replaces provisional. Missing differs from reported zero. Final-to-provisional reclassification remains under Q-DA-09, not an assumed normal transition.

6.2.8 Indicators and Report Snapshots

Indicators are derived, not editable records. A result may be available, partial, unavailable, provisional due to pending local changes and/or based on last synchronisation. These conditions may coexist. Financial provisional classification is distinct from local pending status. Zero is valid only when supported by the available scope and evidence. A PDF is a timestamped snapshot; source corrections require a new export. Offline reports may include authorised prepared financial information and identify freshness, pending changes and incomplete scope.

Generation and signed output. Template input is a draft until reviewed and accepted as an unsigned document version. Signing does not change its source identity. A worker-specific PDF rendition is a derived output linked to that version and worker, not another source version or a signature for other workers.

6.3 Corrections and Change History

6.3.1 Identity, Dependencies and Current Values

Descriptive edits preserve record identities and associations; parcel creation does not distribute past property-level records. Ordinary profile/work/source-entry corrections retain current values rather than permanent field-change history. A delivery has a single current analysis. Document versions, signatures and signed PDF renditions remain domain records subject to their own retention. Security-event retention is a separate NFR decision; secrets must not be placed in event logs.

Coverage correction follows BR-010: reopen if closed, resolve dependent records, then remove the area. No cascade deletion or undefined exceptional reassignment is introduced. Campaign archival does not substitute for reopening.

6.3.2 Completion and Correction

Completion supplies previously unrecorded information without replacing or removing confirmed values or changing confirmed associations. It includes missing hours/tasks and previously unrecorded periods within valid actual dates. In a closed campaign it still requires attendance, coverage and overlap consistency. An absent worker cannot acquire actual work through completion; attendance must first be corrected. Confirmed work changes/removals require reopening, with no additional age-based deadline. Work outside valid dates requires reopening/date review. Existing-delivery tickets, analyses and amounts follow the separate BR-012 exception. For tickets and analyses, entry permission creates the initial record. After confirmation, completing a previously unknown field in that same record is an update requiring the separate correction permission (and the corresponding financial permission for monetary content). The work-completion exception does not grant additional source-record rights. Campaign closure does not waive these permission checks.

6.3.3 Work Corrections

Require correction permission and valid origin/destination campaign contexts. Worker, date, task, periods, breaks and area may be corrected without rewriting the plan. Moving a period preserves unrelated work and daily-attendance consistency. Presence-to-absence requires explicit removal confirmation for all contradictory actual assignments, including incomplete ones, periods and breaks on that date; all affected permissions and campaign reopening rules apply.

6.3.4 Access Revocation and Offline Conflicts

Pending proposals keep their origin account/business. Normal acceptance requires current authorisation. Existing revoked-manager proposals may undergo owner review under BR-004 without restoring access. Acceptance is new explicit owner authorisation, not proof of pre-revocation creation. Validate normal domain rules and record originator and accepting owner; device timestamps prove no prior authority. Owner approval cannot bypass business rules or signing-source eligibility. Logout protects retained pending information from other accounts.

Concurrent accepted values require authorised comparison against the pending proposal; retain server values or submit a reviewed correction and validate again. Temporary conflict data is not permanent correction history. Pending totals remain provisional and rejected proposals do not enter accepted totals.

6.3.5 Documents and Signing

Metadata correction preserves signed contents. Content replacement creates a new version without inherited signatures or external declarations. Shared updates affect the same version across worker associations; hiding other application signatures does not redact the source.

Retention, deletion and withdrawal follow BR-026. For example, after accepted v3, contracts retain v1/v2/v3. Insurance/other documents retain v2/v3 and also v1 if signed or declared signed externally; unsigned v1 is cleanup-eligible only when unprotected and dependency-free. Failed/cancelled v3 leaves v1/v2 unchanged. Shared worker associations do not create separate retained copies. Propagate accepted deletions to local copies and align storage/backups with the retention policy.

Server-known pending signing operations protect their source from cleanup. Signatures still exclusively offline may require Q-DOC-04 resolution. Preserve those proposals without transferring the signature or recreating deleted content. Uncertain submissions check/resume the existing operation, not another drawing.

6.3.6 Delivery-Related Corrections and Indicators

Delivery correction, cancellation, deletion or campaign/source reassignment requires specific authority in all affected contexts. Reopen closed campaigns first; validate chronology, coverage and dependent records. Delete only under BR-031, including its composed-row treatment. Cancellation requires a reason, preserves ticket/analysis/amount/evidence records and removes their indicator contributions. No cascading deletion or automatic reinstatement is introduced. Unresolved dependencies block reassignment. Ticket and analysis corrections update their single current records; the late-source exception remains. Amount correction requires financial permission online or offline, and retains one current valuation. Source evidence is distinct from worker-document retention and field-change history.

Indicators cannot be corrected directly. Recalculate affected origin/destination contributions after source changes; local updates replace their baseline rather than adding it again. Closure does not freeze current results and existing PDFs remain unchanged snapshots. Changing a viewing filter does not change source data.

6.4 Calculation Rules

6.4.1 Areas, Dates and Eligibility

Declared hectares are optional; unknown is not zero. Parcel totals need not equal property area and no quantity allocation follows automatically from hectares. Future map estimates remain separate from the declared reference. Only In progress campaigns reserve exclusive execution under BR-008; Planned coverage does not. Dates on Closed campaigns remain relevant. Working-date filters use period start dates; delivery counts use arrival dates. Ticket/analysis/financial date bases remain under Q-IND-01.

6.4.2 Counted Working Time

A valid period has complete start and end instants, with end later than start. An earlier clock time alone does not establish the end date; validate full intervals across adjacent dates. All durations below are elapsed hours. Let W_i be an eligible actual period and $w(i)$ its worker. Let B_w be the union of applicable non-counted break intervals for that same worker. Then

$$h_i = \text{duration}(W_i) - \text{duration}(W_i \cap B_{w(i)}), \qquad H = \sum_i h_i.$$

Durations in this expression are in hours. A break affects only intersections with that worker's periods; the union prevents duplicate deduction. Counted breaks and already excluded gaps do not reduce hours. Task/area attribution uses the intersecting share of each period. A complete confirmed period requires its actual task. Planned or missing periods do not establish actual hours or zero. No wages are calculated. Enter/display work at minute precision and use the configured business timezone for date association. Compute elapsed durations without rounding each period, then display totals as hours and minutes. Clock changes use complete time instants. Detailed break-entry validation remains under Q-WF-03.

Acceptance examples.

Table 11: Counted working-time examples.

Input	Result
08:00–18:00 with no non-counted break.	10 hours.
08:00–18:00 with a counted 12:00–13:00 break.	10 hours.
08:00–18:00 with a non-counted 12:00–13:00 break.	9 hours.
08:00–12:00 and 13:00–18:00.	9 hours, without another deduction for the excluded gap.
08:00–11:00 machinery and 11:00–13:00 general work.	3 machinery hours and 2 general-work hours.

Continued on next page

Input	Result
22:00–02:00 the next day, on a night without a clock change.	4 hours, associated with the start date.
A and B work 08:00–12:00; only A takes a non-counted 30-minute break.	A: 3 hours 30 minutes; B: 4 hours; total: 7 hours 30 minutes.
A draft two-hour interval without its actual task.	Incomplete; choose the task before confirming it as counted work.

6.4.3 Campaign Work and Delivery Indicators

Let P_C be distinct workers with confirmed presence and actual assignment to campaign C; let E_C be eligible actual periods, with start date d_i. Let D_C be distinct deliveries in scope for campaign C whose arrival is confirmed (*Delivered*) and which are not cancelled. This is the common delivery set for counts, weights, yield and monetary totals, under D-30. A delivery belongs to D_C even without a ticket or analysis. Then

$$N_{\text{workers}} = |P_C|, \qquad N_{\text{days}} = |\{d_i : i \in E_C\}|, \qquad N_{\text{deliveries}} = |D_C|.$$

Participation can be known without hours; a no-period assignment does not add a measured working day. Driver selection alone is not participation.

$$H_C = \sum_{i \in E_C} h_i, \qquad H_{C,w} = \sum_{\substack{i \in E_C \\ w(i)=w}} h_i, \qquad H_{C,f} = \sum_{\substack{i \in E_C \\ \text{task}(i)=f}} h_i.$$

Task totals use the required actual task of each confirmed complete period. Distinct workers and days cannot be summed across area breakdowns. Incomplete intervals support no invented duration; partial known totals disclose missing coverage.

6.4.4 Weights and Yield

Each delivery has at most one ticket and one analysis. For each weight meaning/unit m, let $E_w^{(m)} \subseteq D_C$ contain deliveries whose ticket reports a known, eligible weight $q_i^{(m)}$. The corresponding total is

$$Q_C^{(m)} = \sum_{i \in E_w^{(m)}} q_i^{(m)}.$$

Count each reported weight once and disclose missing or incompatible data within D_C. Clean net weight is recorded in kg as supplied by the cooperative; do not derive it from other weights or apply deductions until their source treatment is confirmed under Q-DA-02. Optional excess time and monetary penalties are reported facts, not application calculations, and are not deducted again from a reported ticket value.

Let $E_a \subseteq D_C$ contain deliveries with a known reported yield r_i and an eligible positive reported clean net weight w_i from their ticket. Then

$$\bar{r} = \frac{\sum_{i \in E_a} w_i r_i}{\sum_{i \in E_a} w_i}.$$

The percentage remains a reported source result; its reference weight is reached through the ticket, not an independently editable analysis copy. A missing yield or clean net weight excludes that

delivery and is disclosed, not inserted as zero. No eligible positive denominator means unavailable. A reported zero yield with a valid positive weight is not missing. Show contributing deliveries and incomplete coverage. For 1,000 kg at 20% and 500 kg at 18%, the result is approximately 19.33%; percentage display precision remains a team setting. Theoretical oil is not oil actually received; its inclusion is not assumed.

6.4.5 Current Monetary Totals and Allocation

Let T_C be the tickets belonging to deliveries in D_C. Let $E_f \subseteq T_C$ and $E_p \subseteq T_C$ contain tickets with a known current EUR amount a_t, classified as final and provisional respectively. Then

$$V_{\text{final}} = \sum_{t \in E_f} a_t, \qquad V_{\text{provisional}} = \sum_{t \in E_p} a_t.$$

Use one current EUR amount per ticket and decimal monetary precision; $E_f \cap E_p = \emptyset$. Neither total establishes a payment. Replaced values contribute nothing; missing amounts are disclosed, not zero. Do not reapply reported penalties or deductions. Mixed-load totals contribute once to eligible campaign results; source-area quantity allocation requires known source quantities. The current model has no source-amount fields, so mixed-load monetary totals remain unallocated by area. Percentage allocation is outside this version; a transport document's declared/dominant origin does not distribute physical or monetary totals. Otherwise show unallocated information, not duplicate totals or invented zeros.

Ticket, analysis and amount entry is permitted while a delivery is *In transport*, subject to the relevant grants. Available source data remains visible in delivery detail as awaiting receipt confirmation, rather than missing, and contributes nothing until arrival is confirmed. Missing-data coverage concerns eligible D_C records, not deliveries still awaiting receipt.

For central totals, D_C uses server-accepted arrival confirmations and source values. An authorised offline arrival confirmation may put a delivery into the local D_C and update local provisional indicators using available eligible data. Identify pending synchronisation; central totals change only after server acceptance.

Indicator refresh and PDF snapshots follow FR-061 and FR-062, including authorised offline output and freshness labels. Undefined results, unavailable local data and retrieval failures must not be displayed as zero.

7 Non-Functional Requirements

This section specifies verifiable performance, usability, accessibility, security, privacy, reliability and offline-continuity properties for the functional baseline.

Status and verification. The working-specification status in Section 1.4 applies. Existing policies are 15-day offline authorisation, separate 30-day cache inactivity cleanup and the initial 10 MB upload limit. Authentication, recovery, availability, performance and usability targets have now been approved for this specification. Concrete mechanisms, deployment resources and operational language still require the residual validations in Section 11.3. Verification requires configuration and recorded evidence.

Use PERF-01–PERF-08, USA-01–USA-09, SEC-01–SEC-11, PRV-01–PRV-07, REL-01–REL-08 and AOC-01–AOC-06 with lettered criteria, for example SEC-03(d). Each requirement's *Related* links identify its functional applicability; Section 11.1 provides planned verification scenarios.

Record approved configuration, stimulus, environment, expected response and observed measure when executing a quality scenario.

7.1 Performance

All workloads, sampling plans and timing limits in this subsection are approved working-specification acceptance targets under Q-NFR-07; they are not measured capacity, executed results or an infrastructure procurement commitment. Correct results and accurate save states remain required: speed does not compensate for lost work, duplicate effects or incorrect totals.

7.1.1 Reference Conditions

Use synthetic data in one owner workspace: 20 properties, 40 optional parcels, 10 campaigns, 100 workers, 180 working dates, 5,000 daily attendance decisions, 10,000 actual periods, 2,000 deliveries, 2,000 tickets and 2,000 analyses. These are test fixtures, not capacity limits. Validate campaign coverage and dates, unique business/day attendance, separate area assignments and at-most-one ticket/analysis. Use client-validated weight/yield meanings; do not settle open source questions through invented fixture calculations. The central history is not assumed to be entirely prepared offline: record each offline scenario's authorised local subset.

Exercise five simultaneous authenticated sessions. Use a representative mid-range phone with at least 4 GB RAM, a representative tablet and any claimed desktop browser. Record exact hardware, operating system, browser, server resources, build and dataset versions. The reference online profile is at least 10 Mbps download, 2 Mbps upload and at most 150 ms round-trip latency. Document cache state and separate cached/uncached runs; initial installation, login and preparation are online prerequisites. Disable the network only after successful scoped preparation.

For PERF-01–PERF-07, collect at least 30 independent samples per applicable device, operation and mode; at least 95% must meet the approved limit. Measure from the initiating action to usable content, completed calculation or accurate save state, including applicable rendering and network time. Record errors as failures. PERF-08 instead uses ten complete reconnect cycles; its benchmark is defined in PERF-08.

The initial upload policy remains BR-044. Reference file sizes do not reduce that limit. Worker-document generation under FR-063 is included, while its template inventory and separate performance fixture remain subject to Q-DOC-01 and Q-NFR-07.

7.1.2 Performance Requirements

PERF-01 — Application and property access

Authorised home and property views shall become usable promptly online and within prepared offline scope.

Rationale: Field navigation must not wait for unavailable central services.

Related: FR-002, FR-011, FR-015, FR-023.

(a) Target: usable authorised home and property list within 3 seconds under the reference profile; apply the same limit to prepared offline reopening.

(b) Measure online opening and prepared reopening separately, distinguishing cached and uncached runs. Expose unavailable local detail without blocking unrelated navigation.

PERF-02 — Worker and schedule consultation

Worker searches, prepared profiles and schedule changes shall provide timely usable results.

Rationale: Daily consultation should remain practical with the reference workforce and history.

Related: FR-023, FR-028.

(a) Target: each worker search, prepared-profile opening and daily/worker/weekly view change completes within 2 seconds.

(b) Test filters, matching and empty results, and unavailable prepared dates. Offline timing covers only authorised available scope; a retrieval failure is not a successful empty query.

PERF-03 — Working-time previews and validation feedback

Supported local work calculations and validation feedback shall respond promptly to input changes.

Rationale: Group entry needs immediate feedback before explicit confirmation.

Related: FR-026, FR-031, FR-032, FR-033.

(a) Target: update supported previews and validation feedback within 1 second for a 25-worker group with up to 100 periods.

(b) Use the counted-time rules in Section 6.4.2, including overnight work and worker-specific overlapping breaks. Missing tasks/hours remain incomplete, not invented durations or counted unclassified periods.

(c) Compare local checks with central validation on the same fixtures. Local feedback neither guarantees acceptance nor settles pending source-weight or yield interpretations.

PERF-04 — Local saving and connected acknowledgement

Work submissions shall provide timely and distinct local-save and central-acceptance feedback.

Rationale: A quick interface response must reflect the actual persistence outcome.

Related: FR-029, FR-034, FR-037.

(a) For 25 workers with up to 100 periods and 25 breaks, the local-save target is 3 seconds from submission to durable local confirmation.

(b) For a valid online submission under the reference profile, the server-acknowledgement target is 3 seconds from submission. Record it separately from local persistence.

(c) Interrupt transmission before and after central commit/acknowledgement. Report the actual or uncertain outcome; a timeout proves neither rejection nor acceptance.

PERF-05 — Campaign indicators and source refresh

Eligible indicators and their supported filters shall load and refresh promptly after relevant changes.

Rationale: Campaign review depends on current, correctly scoped source contributions.

Related: FR-055, FR-056, FR-058, FR-060, FR-061.

(a) Target: campaign indicators and applicable property/parcel filters become usable within 4 seconds.

(b) Refresh affected visible results within a 4 seconds after that client receives an accepted change or durably saves a supported local proposal; do not imply instant delivery to disconnected devices.

(c) Verify restricted scope, late source information after closure, mixed-load allocation and Delivered-only eligibility. Keep accepted and local provisional contributions distinct.

PERF-06 — Prepared and online file opening

Authorised prepared and online documents shall open within a measured reference-file budget.

Rationale: Field consultation depends on both file readiness and usable viewing.

Related: FR-038, FR-039, FR-040, FR-045, FR-049, FR-051.

(a) For a prepared authorised PDF/JPG/PNG reference file up to 2 MB, opening limits are 2 seconds offline and 10 seconds online.

(b) Test current/previous insurance and explicitly prepared signing documents; record larger-file performance separately. The reference size is not an upload limit: BR-044 still applies to worker documents and source evidence.

(c) Missing local files, denied access and failed transfers produce truthful guidance rather than a successful-open result.

PERF-07 — Campaign PDF snapshot export

A campaign indicator snapshot shall be produced promptly from available authorised information.

Rationale: Export must preserve the meaning and limitations of the visible campaign results.

Related: FR-059, FR-060, FR-062.

(a) Target: a snapshot of up to ten pages is ready to open or download within 10 seconds under the applicable reference mode.

(b) Inspect online and offline scope, generation time, completeness, freshness and pending labels. Include prepared monetary content only with its financial grant; offline finance is not categorically excluded.

(c) This benchmark covers campaign snapshots, not template-based worker-document generation. FR-063 remains included; its representative templates and performance fixture require separate review.

PERF-08 — Reconnection throughput

A valid prepared structured-operation batch shall synchronise within an agreed reconnection budget.

Rationale: Reconnection must recover field work without duplicate effects or hidden failures.

Related: FR-004, FR-029, FR-034, FR-037, FR-048.

(a) Accept 100 valid structured operations totalling at most 1 MB within 60 seconds, excluding files. Start timing after connectivity and current authority validation complete; record access-validation time separately.

(b) All ten reference reconnect cycles must meet the budget, preserving dependency order and one business effect per operation. Count only valid centrally accepted operations as complete.

(c) Exclude file transfers from this benchmark; define the operation unit, dependency mix and payload profile before testing. Test denied authority, conflicts, business-rule failures and attachment recovery separately.

7.2 Usability and Accessibility

Use an approved sample of five representative participants, including an owner representative, after at most 15 minutes of introductory guidance. Exercise the included manager paths within their grants and worker participation without an account. Record exact devices, tasks and facilitation under Q-NFR-08; record task completion, timing, assistance and errors. This small sample supports a prototype review, not a population-wide claim.

The numerical success, timing and interface targets below are approved acceptance targets, requiring observed validation evidence. Local confirmation, central acceptance, financial provisional/final classification and unavailable data retain their meanings in Section 6.2.5 and the business rules. Routine offline interaction follows Section 7.6.

USA-01 — Role-appropriate navigation

The owner and included managers shall find their permitted operational workflows through consistent navigation.

Rationale: The owner is an operational user as well as a business reviewer.

Related: FR-007, FR-024, FR-028, FR-030, FR-038, FR-046, FR-055.

(a) Target: at least four of five participants find each applicable planning, work, document, delivery and indicator task within 30 seconds without facilitator help.

(b) Exercise owner and authorised manager paths. Expose only permitted tasks; do not confine the owner to an executive-only dashboard.

USA-02 — Efficient group and retrospective entry

Group defaults and individual adjustments shall support practical end-of-day entry without continuous phone use.

Rationale: Recorded actual work may differ from planning and need retrospective confirmation.

Related: FR-024, FR-031, FR-032, FR-035, FR-047, FR-049.

(a) Target: at least four of five participants complete the reference task in less than 10 minutes without facilitator help, from opening the workflow to local confirmation.

(b) Use 25 prepared workers, two eligible work areas, machinery/general tasks, one worker-specific break, one individual adjustment and two completed deliveries with supplied ticket data. Confirm valid campaign-start context before entry.

(c) Check required tasks, attendance, times and individual differences. Measure synchronisation separately; successful local entry is not central acceptance.

USA-03 — Planning, attendance and actual-work clarity

Users shall distinguish planned work, daily presence, campaign-area participation, confirmed periods and synchronisation.

Rationale: Profile capability and incomplete records must not be mistaken for actual counted work.

Related: FR-020, FR-028, FR-030, FR-031, FR-033.

(a) Show missing tasks/hours and awaiting attendance explicitly. Machinery capability does not establish the actual task; overnight periods show both dates and their start-based working date.

(b) Target: at least four of five participants correctly interpret each reference example, including a worker in two areas and an incomplete assignment without a task. An unclassified period is not a valid confirmed complete period.

USA-04 — Insurance and version clarity

Document views shall distinguish version currency, withdrawal, recorded expiry and worker-specific signing information.

Rationale: A current file is not necessarily valid, signed or prepared on the device.

Related: FR-038, FR-039, FR-041, FR-044.

(a) Present current/previous or older retained status separately from expiry, withdrawal and signatures; missing expiry is unconfirmed, not verified coverage.

(b) Ask participants to locate prepared insurance and interpret expired, unconfirmed, withdrawn and unsigned examples. Explain missing local files/preparation failures without implying missing business records.

USA-05 — Reviewable signing interaction

Signing shall identify the worker and exact version and require deliberate review and confirmation.

Rationale: Signing speed must not limit document reading or trigger duplicate signatures.

Related: FR-042, FR-043, FR-044, FR-045.

(a) Show worker/version, permit drawing, clearing/redrawing or cancellation before explicit confirmation, and reject empty drawings.

(b) Target: at least four of five participants complete the interface interaction within 2 minutes after document review. Record reading time separately without a reading-time limit.

(c) Test prepared online/offline and shared-document paths. Check an uncertain prior submission before requesting another drawing; preserve worker/version-specific results.

USA-06 — Terminology, sources and summary meaning

Operational labels and summaries shall make their source, scope, units and uncertainty understandable.

Rationale: Readable totals must not imply unsupported prices, payments or allocations.

Related: FR-049, FR-051, FR-053, FR-055, FR-056, FR-058, FR-059, FR-060.

(a) Use English for the academic report/prototype. Validate operational language, locale and client terminology under Q-NFR-08. Use explicit units and unambiguous dates; distinguish property/parcel provenance, tree/ground/mixed collection and actual task.

(b) Explain incomplete, unallocated and unavailable results separately from valid zero. Keep reported financial provisional/final status distinct from local pending acceptance.

(c) Use unknown yield, mixed provenance and partial-authority examples. Participants identify scope/source meaning without assuming price calculation, payment or automatic area allocation.

USA-07 — Outdoor touch interaction

Core field controls shall support readable and deliberate touch interaction on the supported devices.

Rationale: Outdoor conditions can make small controls and colour-only feedback difficult to use.

Related: FR-028, FR-030, FR-031, FR-038, FR-042, FR-046, FR-049.

(a) Targets: primary touch controls at least 44 by 44 CSS pixels and text contrast at least 4.5:1. Space controls to reduce accidental activation and never encode meaning by colour alone.

(b) Use suitable mobile numeric/time inputs. Measure controls/contrast and test representative field devices outdoors, recording sunlight and glove conditions rather than claiming universal hardware compatibility.

USA-08 — Responsive forms and accessible controls

Core forms, actions and feedback shall remain usable across the agreed viewport, zoom and input modes.

Rationale: Phone, tablet and keyboard users need access to the same permitted actions.

Related: FR-024, FR-028, FR-030, FR-036, FR-038, FR-042, FR-055.

(a) Targets: core forms and primary actions usable at 360 CSS pixels without horizontal scrolling, and controls/messages accessible at 200% zoom. Document viewers may pan source pages while retaining usable controls.

(b) Provide associated labels, keyboard operation and visible focus for web controls, including those around document review and drawing. Validate on the agreed phone/tablet and supported desktop browser matrix.

(c) Record which screens and interactions were checked; these selected criteria alone do not establish full accessibility conformance.

USA-09 — Recoverable errors and informed corrections

Errors and confirmations shall preserve recoverable input and explain the consequences of the requested operation.

Rationale: A failed or cancelled action must not destroy unrelated work.

Related: FR-035, FR-036, FR-037, FR-041, FR-043.

(a) Identify invalid fields and required corrections while preserving unaffected input. Cancellation before confirmation preserves accepted records; destructive actions identify consequences and require confirmation.

(b) Test interrupted saves/uploads and presence-to-absence removal. Distinguish permitted completion after closure from confirmed-work correction requiring reopening; never silently rewrite plans or unrelated records.

7.3 Security

The API revalidates current account, workspace, scope and operation authority. Prepared local data may be stale or affected by later revocation; a device timestamp does not establish prior permission or acceptance. Figure 1 summarises the proposed control boundaries, not an implemented security result.

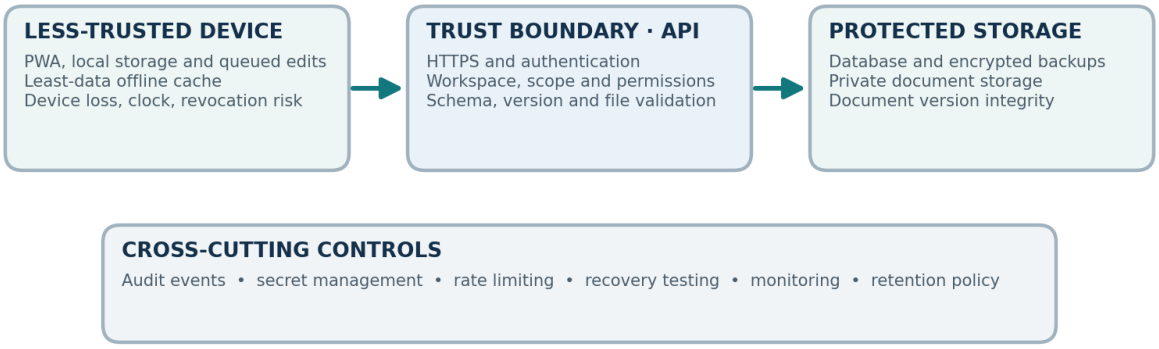


Figure 1: Proposed security boundaries and cross-cutting controls, adapted from the team's NFR contribution.

SEC-01 — Account authentication and password protection

The backend shall authenticate every application account using a unique, verified email identity and a password.

Rationale: Protects the owner and included manager accounts against account takeover and enumeration.

Related: FR-001, FR-002, FR-005, FR-064.

(a) Only accounts with a verified email can access a business workspace; unverified accounts are refused.

(b) Application-managed authentication uses established backend libraries and salted Argon2id password hashes; benchmark and record concrete cost parameters before deployment. Plaintext passwords are never stored or logged. A managed identity service is not the selected baseline.

(c) Enforce at least 15 characters in the password-only flow and accept passphrases of at least 64 characters without silent truncation. Reject common/compromised passwords on registration, reset and change.

(d) After five failed logins within 15 minutes, apply progressive throttling using account/source signals, without permanent attacker-triggered account lockout. Exact delay progression and recovery configuration require implementation review and tests.

(e) Authentication failure responses do not disclose whether an email address is registered.

SEC-02 — Online session lifecycle

Protected online requests shall use short-lived, revocable credentials.

Rationale: Limits the exposure of stolen credentials and supports timely revocation.

Related: FR-002, FR-003, FR-004, FR-005, FR-008, FR-064.

(a) Online access credentials expire after 15 minutes; the server refuses expired credentials. Renewal is permitted only while the originating online session remains valid.

(b) Renewable online sessions expire at most seven days after login, requiring authentication again; renewal does not extend that absolute limit. Implement revocation and rotation/replay protection. Cookie-based credentials use Secure, HttpOnly and an appropriate SameSite policy.

(c) Logout revokes the applicable online session. Successful password reset or change invalidates existing online sessions. Membership revocation immediately denies that business online without deleting unrelated memberships.

(d) An expired online session requires online renewal or authentication under its policy. It does not delete pending work or change the separate 15-day offline grant in AOC-02.

SEC-03 — Workspace and operation-level authorisation

The server shall check authorisation on every protected read, write, export and file operation.

Rationale: Preserves business isolation and independently granted consultation, entry and correction rights.

Related: FR-002, FR-004, FR-007, FR-008, FR-023, FR-038, FR-053; BR-024.

(a) Each request evaluates account status, current business-workspace membership, responsibility scope and operation-specific permission.

(b) A user-supplied identifier never establishes authority by itself.

(c) Distinguish consultation, planning, entry, corrections, document association/modification, signing initiation and financial grants. Apply the included Field Manager defaults and mandatory scoped insurance access in Section 2.4.

(d) Supported financial actions may operate offline with the corresponding grants and prepared context; central acceptance revalidates them. Consultation/full-ticket access starts enabled for managers within assigned scope and is revocable; structured financial writes require additional grants.

(e) Denied requests return a generic response that contains no protected data.

SEC-04 — Encryption and private storage

Protected business data shall be encrypted in transit and in central storage; service credentials shall be protected and stored files private by default. Password storage follows SEC-01 rather than reversible password encryption.

Rationale: Protects employment, insurance and signing information across the device, API and storage boundaries.

Related: FR-002, FR-038, FR-039, FR-040, FR-042.

(a) Production requests use HTTPS and a reviewed TLS/HSTS configuration. Plain HTTP is redirected or refused before protected content or credentials are served. The minimum supported protocol and deployment settings are reviewed under Q-NFR-01.

(b) Encrypt the central database, backups and private file storage at rest, with controlled key access and rotation. The hosting/key-management implementation is selected and verified during architecture review; this does not settle local-device protection under Q-DOC-07.

(c) Document storage is private by default; protected files are served only after API authorisation, using short-lived scoped retrieval links where applicable.

(d) The central database is not directly reachable from the public internet.

SEC-05 — Secure API, PWA and file validation

The backend shall validate every input and uploaded file independently of the PWA.

Rationale: Prevents malicious requests, script injection and unsafe file handling; client-side validation does not replace server-side validation.

Related: FR-001, FR-038, FR-040, FR-045, FR-049, FR-051, FR-053, FR-063, FR-064.

(a) Types, identifiers, relationships and file sizes are validated on the server, including references across business workspaces.

(b) Use parameterised database access, context-appropriate output encoding, restrictive CORS and a deployable Content Security Policy. Cookie-authenticated state changes also require effective CSRF protection; CORS alone is not an authorisation or CSRF control.

(c) Worker uploads accept PDF, JPG and PNG within the initial 10 MB per-file limit in BR-044. Validate content, size and safe file handling server-side, not only the extension. Representative-file and exact-byte validation remains under Q-DOC-06.

(d) Ticket, analysis and valuation evidence uses the same initial formats and 10 MB limit. Complete source files may be prepared offline with their consultation grants; monetary content neither forces classification/redaction nor prohibits upload under the applicable entry grant. Source retention/upload recovery remains under Q-AT-02.

(e) Scan application and bundled dependencies for known vulnerabilities in the delivery pipeline, triage findings and record remediation or explicit risk acceptance before release. A successful scan is not a claim of absence of vulnerabilities.

SEC-06 — Secrets and deployment identities

Production credentials shall be kept out of source control and PWA assets, and every pipeline and service shall use least-privilege identities.

Rationale: Prevents compromise through repository leakage or excessive pipeline privileges.

Related: FR-001, FR-005, FR-040.

(a) Do not commit production service credentials or bundle database/email/storage secrets into the PWA. Device-local protection material, if used, follows the separate reviewed design under Q-DOC-07.

(b) Runtime services and delivery-pipeline jobs use separate identities with least privilege.

(c) Keep production secrets in an access-controlled secrets mechanism with documented rotation and access review. Provider/tool selection remains an implementation decision.

SEC-07 — Security-event audit logging

The server shall record security-relevant events.

Rationale: Supports investigation and accountability while preserving the rule that ordinary business updates retain current values only (Section 6.3.1).

Related: FR-003, FR-005, FR-007, FR-008, FR-036, FR-037, FR-042.

(a) Each event records server timestamp, initiating account when known, workspace where applicable, action, minimal target identifier and outcome.

(b) Mandatory server-observable categories: successful/failed authentication, logout, account recovery, membership and permission changes, restricted-document access requests handled by the API or controlled file service, signing acceptance, denied operations, revoked-access recovery decisions and security-relevant synchronisation conflicts. For document access, record the authorised retrieval/link-issuance decision or refusal; this does not prove that the user viewed the file.

(c) Logs exclude passwords, raw tokens/full lifecycle links, document bodies, signature images, full official identification and raw monetary values; use minimal authorised identifiers.

(d) Retain security events for 90 days from the server event, accessible only to authorised operations/investigation personnel. Purge thereafter, except justified incident preservation with a documented purpose and scope; test the configured schedule.

(e) This requirement does not impose a permanent field-by-field history on ordinary corrections.

(f) Opening a prepared file entirely offline is not logged in this version. Server preparation/retrieval/link issuance does not prove viewing. This coverage choice does not weaken file-access authorisation.

SEC-08 — Document and signature integrity

Each accepted document version and each confirmed signature shall be tamper-evident and bound to the exact version signed.

Rationale: Preserves integrity and provenance while respecting category-specific retention and withdrawal.

Related: FR-038, FR-041, FR-042, FR-043, FR-044, FR-045; BR-026.

(a) Each accepted document source has an immutable file reference and a recorded cryptographic digest. Verify the source file against its recorded digest. Before accepting or exposing a signed rendition, validate its association with the exact source version and worker-specific signature. The derived PDF has different content and is not checked against the source digest. Select the digest algorithm and integrity-metadata protection during architecture review; a hash alone is not proof of signer identity.

(b) A confirmed drawn signature is linked to the identified worker, the initiating authorised account, the exact version and the confirmation timestamp.

(c) A signature is never reused for a newer version or for another worker; signed file content is never replaced in place.

(d) Rolling cleanup cannot delete signed or externally declared signed versions. Preserve source, evidence and renditions under BR-026; signed erasure requires the separately validated retention/rights procedure in Q-DOC-03 and PRV-04. Protect server-known pending signing sources and preserve exclusively local proposals for Q-DOC-04 resolution.

(e) The server records its own acceptance time alongside the device-captured confirmation time.

(f) A drawn mark alone is not represented as a qualified electronic signature.

SEC-09 — Verification, recovery and invitation tokens

Account-lifecycle links shall be single-use, short-lived and unpredictable.

Rationale: Protects registration, recovery and invitation flows from replay, enumeration and unauthorised business membership.

Related: FR-001, FR-005, FR-006, FR-064.

(a) Email-verification links, including replacement-email verification, expire after 24 hours and are single-use. An unverified replacement cannot supersede the active email; a resend invalidates the preceding link.

(b) Password-reset links expire after 30 minutes and are single-use. Resending invalidates the preceding link; invalid/expired/consumed links cannot change credentials.

(c) Invitations expire after seven days and are single-use, bound to the intended verified email and business. Resending/cancelling invalidates superseded links without duplicate membership.

(d) Generate token values unpredictably and store only a verification hash or equivalent protected verifier when persisted. Consume/invalidate lifecycle authorisations transactionally. Retries may report an already completed action without repeating activation, membership creation or credential changes.

(e) A successful reset revokes existing online sessions without changing business ownership.

SEC-10 — Abuse protection and rate limiting

The API shall limit request rates to protect availability and prevent automated abuse.

Rationale: SEC-01 throttles login attempts only; recovery, invitation, upload and email-triggering endpoints can also be abused.

Related: FR-001, FR-005, FR-006, FR-040.

(a) Use account/source/recipient-aware limits as applicable: unauthenticated operations cannot depend on an authenticated account key. Apply stricter reviewed controls to authentication, recovery, invitations, uploads and email-triggering requests.

(b) Exceeded limits return a generic, retryable rejection and do not affect other accounts.

(c) Resending verification, reset and invitation messages is limited per recipient address.

(d) The approved login threshold is five failures within 15 minutes under SEC-01. Endpoint-specific recovery, resend, upload and invitation limits and progressive delays still require configuration review; verify recipient/account isolation before release.

SEC-11 — Email delivery security

Verification, recovery and invitation messages shall be sent through an authenticated and monitored email delivery service.

Rationale: Protects account activation, recovery, verified-email changes and invitations that depend on email delivery.

Related: FR-001, FR-005, FR-006, FR-064.

(a) The sending domain is authenticated with SPF, DKIM and DMARC.

(b) Messages contain the intended lifecycle link and minimal account context, never passwords, worker data or protected documents.

(c) A send failure preserves the pending account/invitation state without duplicate identities or memberships. Offer rate-limited resend, invalidating the preceding lifecycle link. Recovery acknowledgements remain generic regardless of registration status.

(d) Record the provider, its processing role/terms, sending-domain configuration and any transfers in the processing inventory under PRV-05; provider selection remains under Q-NFR-05.

Email delivery operation. Only the backend calls the transactional email service using protected credentials. Record technical send outcomes and provider failures without full links. Provider acceptance does not prove inbox delivery or reading. Development/demo sending uses controlled recipients or a test environment. Provider, authorised sender domain and concrete configuration remain under Q-NFR-05.

Technical references. Password and session controls are informed by the OWASP Authentication, Password Storage and Session Management guidance. These references do not approve project-specific numeric targets.

7.4 Privacy

Protect account data and worker records, including required official identification, insurance, employment documents and drawn signatures. The processing basis and retention duties must be validated for the actual context; technical controls alone do not establish GDPR compliance. PRV-04 defines governance procedures without silently adding account deletion or worker self-service to the functional scope.

PRV-01 — Personal-data minimisation and purpose limitation

The application shall collect and expose only the personal data needed for documented purposes.

Rationale: Workers have no management accounts; owners and authorised managers process their records within a business workspace.

Related: FR-020 to FR-023, FR-038, FR-039, FR-053.

(a) Document the purpose and necessity of account/worker attributes, including the official type, country and number required by FR-020. Required entry does not itself establish a lawful basis or require collection of a scanned copy.

(b) Each cached dataset and exported report is limited to the authenticated user's authorised responsibilities.

(c) Full document contents and unrelated personal identifiers are not copied into dashboards or technical logs.

PRV-02 — Lawful processing and transparency

The responsible business shall document the lawful basis for worker data before real data are processed.

Rationale: Consent may be inappropriate or insufficient in an employment relationship; compliance depends on purpose and legal context, not on hosting location.

Related: FR-020 to FR-023, FR-038 to FR-045.

(a) Before real worker data are used, the purposes, applicable lawful bases, privacy notice and processor responsibilities are documented for worker records, insurance, employment documents and signatures.

(b) Do not assume that a generic GDPR consent signature authorises all employment processing. Validate the applicable basis by purpose, including any sensitive information in supplied documents; unresolved questions remain under Q-NFR-03.

(c) Academic demonstration and test environments use synthetic or appropriately anonymised data unless real-data processing has been authorised.

PRV-03 — Protected local data and account isolation

Data held on the device shall be limited, isolated by account and protected after logout or revocation.

Rationale: Prepared insurance and signing documents must be available offline while unsent work is safeguarded on logout.

Related: FR-003, FR-004, FR-008, FR-039, FR-045.

(a) Prepared local datasets contain only authorised required context, with pending local records/dependencies created through supported offline operations stored separately from the accepted baseline.

(b) Private data and pending operations are logically isolated by originating account and business workspace.

(c) Logout immediately blocks workspace access and isolates retained information. Disposable accepted copies may be removed under the local policy; the 30-day inactivity cleanup in FR-003 removes only eligible accepted copies and never pending recovery dependencies.

(d) Normal recovery requires the originating account and renewed authority. The separate revoked-manager owner-review procedure in FR-008/REL-08 preserves original authorship without restoring normal access or exposing pending work to another account.

(e) On reconnection, refresh authority before normal protected access/acceptance and apply authorised access/deletion changes. Preserve quarantined pending proposals and required recovery files under the separate recovery policy.

(f) The system does not claim that a disconnected device can be remotely wiped immediately.

(g) Encrypt device-held personal data, protected files and pending operations, isolated by origin account and business. Logout locks protected access while preserving recovery dependencies. Another account cannot read or submit them. Concrete encryption/key lifecycle and trusted-time mechanisms require a technical proof under Q-DOC-07; browser-origin isolation or screen lock alone is insufficient.

PRV-04 — Rights, retention and permitted erasure

Personal-data requests shall follow a documented authorised procedure that respects applicable rights, dependencies and validated retention duties.

Rationale: Distinguishes rights handling from ordinary record deletion and avoids inventing unapproved self-service workflows.

Related: FR-003, FR-021 to FR-023, FR-038, FR-041; BR-026.

(a) Define how authorised staff retrieve information and handle access, correction, export, restriction and erasure requests through existing capabilities or controlled procedures. Any

additional application feature requires scope approval; no new account/worker self-service is implied.

(b) Assess dependencies and the validated retention/legal basis before restricting or erasing records. Archival/inactivation preserves history but is not itself erasure or a complete response to an erasure request.

(c) While retained, signed source contents and evidence remain immutable. Only an approved retention/erasure procedure may remove signed versions or derivatives; ordinary older-unsigned cleanup cannot do so. Preserve unresolved local signatures under Q-DOC-04.

(d) Retention periods for employment and insurance documents are taken from the validated legal and business schedule rather than from an assumed common expiry.

(e) Backup expiry and restore suppression follow the approved schedule and REL-04. Do not claim immediate erasure from every backup; restored deleted/restricted data must not become ordinarily accessible again.

(f) Account/business deletion remains pending under Q-AA-07. Verified email changes are already included in FR-064 and preserve account identity; they are not an undecided scope item.

PRV-05 — Processing location and breach response

Processing locations, providers and transfers shall be reviewed, and an operational personal-data incident procedure shall exist.

Rationale: Hosting location alone does not establish GDPR compliance; organisational responsibilities and transfer arrangements also matter.

Related: FR-001 to FR-008, FR-038 to FR-045.

(a) Approve primary, backup, telemetry and support-processing locations before real-data deployment. EU-region hosting is the submitted preference for review under Q-NFR-03, not a settled provider/deployment decision or a substitute for transfer assessment.

(b) Processing terms and any international transfers are verified for infrastructure, support, telemetry and email-delivery providers.

(c) An incident-response procedure identifies the controller contact, escalation route, containment and assessment steps.

(d) Where GDPR Article 33 applies, the controller assesses notification without undue delay and, where feasible, within 72 hours after awareness, unless the breach is unlikely to risk individuals' rights and freedoms. Record the assessment and any reasons for delay; applicability and responsibility are validated under Q-NFR-03.

PRV-06 — Privacy-preserving telemetry

Logs, analytics and diagnostics shall not become an uncontrolled second copy of personal or financial data.

Rationale: Observability must not expose protected documents or monetary values.

Related: FR-038, FR-045, FR-053 to FR-062.

(a) Logs, analytics, error reports, queue diagnostics and deployment artefacts minimise personal data and exclude full official identification, document bodies and captured drawings. Operational identifiers are limited to the authorised diagnostic purpose.

(b) Diagnostic records use record identifiers and minimal contextual metadata, with access control and a retention schedule.

(c) Do not copy monetary values or protected document contents into technical diagnostics. This does not prohibit authorised financial business data, prepared offline consultation or PDF indicators under FR-053, FR-060 and FR-062.

PRV-07 — Offline document retention and deletion propagation

Offline document preparation shall be selective, and deletions made centrally shall reach local copies.

Rationale: Offline insurance and signing (FR-039, FR-045) must not expand personal-data exposure unnecessarily.

Related: FR-038, FR-039, FR-041, FR-042, FR-045.

(a) Preparation covers current and previous insurance versions for the prepared worker window and explicitly selected workers, and only specifically prepared versions for offline signing.

(b) Contracts are never bulk-downloaded automatically.

(c) Local file/reference metadata identifies the originating account and workspace, associated worker and version or authorised business target, access context and freshness without implying that a shared source contains only one worker's information.

(d) Apply received access, withdrawal and deletion changes to prepared copies. Remove eligible accepted retention-deleted copies while protecting pending signatures/proposals and necessary recovery evidence. Withdrawn retained sources remain historically consultable but not eligible for operational signing; exclusive local conflicts follow Q-DOC-04.

(e) Queued retries never recreate a deleted version.

Regulatory reference. Validate the applicable duties against Regulation (EU) 2016/679, including purpose/minimisation, lawful basis, rights, processors, security, notification and transfers. The incident-notification time in PRV-05 is a conditional legal requirement, not a proposed service-level target or a certification of compliance.

7.5 Reliability and Recovery

These requirements protect accepted data and recoverable proposals during retries, concurrent edits, interrupted transfers, upgrades and central recovery. The protocol in Section 8.4 supplies the interaction detail; the synchronisation state definitions remain in Section 6.2.5. Backup and service targets are approved specification targets, not measured outcomes or provider guarantees.

REL-01 — Idempotent server processing

Every mutating operation shall have the same business effect however many times it is submitted.

Rationale: Retries after disconnection must not duplicate business records or signatures.

Related: FR-001, FR-003, FR-006, FR-029, FR-041, FR-043, FR-048.

(a) Mutations carry stable operation_id, origin, payload fingerprint and dependencies. Persist minimal outcomes for 90 days after processing. Identical retries have one effect; reused IDs with different content are refused. Expired-context submissions require reviewed recovery, never blind replay that could duplicate effects or recreate deleted records.

(b) Creation of users, memberships, attendance, deliveries, document versions and signatures is protected against duplicate retries.

(c) Related database changes and the idempotency outcome commit atomically. Transfers to object storage have a recoverable readiness/cleanup protocol; this is not a claim of one database/object-store transaction.

REL-02 — Optimistic concurrency and conflict resolution

Concurrent changes to shared records shall never be silently overwritten.

Rationale: Supports informed user review of competing corrections.

Related: FR-008, FR-036, FR-037, FR-043.

(a) Every mutable shared record has a server-managed revision, and each update carries the client's base revision.

(b) A version mismatch never overwrites accepted server state; the pending proposal is preserved.

(c) The user sees accepted and proposed values side by side for authorised manual review.

(d) Resubmission after review revalidates permissions, campaign state, dependencies and business rules.

(e) Automatic last-write-wins is not used for any persisted business record.

REL-03 — Cross-entity integrity and consistent calculations

The API and the database shall enforce integrity rules, and server-side validation shall be authoritative even for data entered offline.

Rationale: Ensures dashboard correctness and prevents cross-workspace or cross-campaign corruption.

Related: FR-026 to FR-037, FR-046 to FR-062.

(a) Foreign-key, uniqueness, business-scope and status-transition constraints are enforced.

(b) Actual work periods are not double counted across assignments, campaigns or adjacent dates.

(c) Delivery-based totals require Delivered, non-cancelled parents under BR-038. Pre-arrival source data awaits receipt confirmation. Ticket deductions are not applied twice; missing analyses are not treated as zero.

(d) For one ticket, replacing a provisional amount with final retains one current valuation, never additive revisions. Separate provisional/final totals across different eligible tickets remain valid and do not imply payments.

(e) Server-side validation prevails over local validation for data that originated offline.

REL-04 — Central backup and tested recovery

Central data shall be backed up and recoverable within stated targets.

Rationale: Makes central persistence recoverable; measurable objectives depend on an agreed deployment and a rehearsed restore procedure.

Related: FR-003, FR-038 to FR-045, FR-046 to FR-062; BR-026.

(a) Automate encrypted central database and compatible file/evidence backups at least daily. Preserve database/file dependencies together, restrict access and alert on failed scheduled backups.

(b) Retain central backups for at least 14 days. This is a recovery policy, not an employment or signed-document retention period or proof of immediate erasure from all backups.

(c) Rehearse RPO at most 24 hours and RTO at most four hours against the agreed deployment/data restore scope. The accepted RPO permits up to 24 hours of central-data loss. These targets do not recover unsent records on a lost device.

(d) Before restoring ordinary service, reapply retained authorised deletion/restriction and access decisions that postdate the backup. Retain recovery metadata independently long enough to enforce them; if it is insufficient, quarantine affected data rather than exposing erased/restricted records.

(e) At least one representative restore rehearsal is performed before operational deployment, validating relationships and accessible retained document versions.

REL-05 — Monitoring and incident response

Failures shall be detected and reach a named team contact during the harvest period.

Rationale: Field work may continue locally during cloud incidents, but failures must be detected for recovery and eventual synchronisation.

Related: FR-004, FR-037, FR-043, FR-048.

(a) Monitoring covers API reachability, elevated error rates, failed scheduled backups, storage capacity and abnormal synchronisation failures.

(b) Alerts reach an assigned operational contact under a documented response arrangement and produce an incident record. The response coverage and thresholds must be agreed; no unapproved round-the-clock service guarantee is implied.

(c) Monitoring complies with PRV-06.

REL-06 — Validated deployments and rollback

Releases shall be tested, reversible and safe for devices holding queued work.

Rationale: Protects campaign continuity from software releases made while clients are disconnected.

Related: FR-003, FR-004, FR-037, FR-045.

(a) Changes pass automated tests and schema-compatibility checks in a staging environment before release.

(b) A deployable previous version and a documented rollback procedure, including migration compatibility, are preserved.

(c) A release never discards locally queued changes or requires a destructive upgrade.

(d) Staging and test environments contain no production personal data (PRV-02).

REL-07 — Older PWA client compatibility

Devices running an earlier PWA release shall still be able to synchronise or be told how to recover.

Rationale: Prepared clients may run old releases; an expired offline grant still leaves pending data requiring safe recovery.

Related: FR-003, FR-004, FR-037, FR-043.

(a) Support the immediately preceding client release and required payload formats for 30 days after a new release. Preserve queued records/files from unsupported or stale contexts for reviewed recovery; refresh prepared data without blind replay. The server-issued context, not device time alone, establishes eligibility.

(b) Schema transformations never silently discard fields or reinterpret identifiers.

REL-08 — Revoked-access and rejected-operation recovery

Pending operations associated with a revoked business membership, and rejected operations, shall remain protected and recoverable under the applicable review procedure. Revoking membership does not revoke the account in other businesses.

Rationale: Implements the FR-008 recovery flow and offline signing safety.

Related: FR-003, FR-004, FR-008, FR-037, FR-041, FR-043, FR-045.

(a) Revoking a manager membership immediately denies normal online actions in that business. Disconnected enforcement follows BR-002 and AOC-02; the manager role is included.

(b) After origin-account authentication, allow restricted recovery submission without normal business read/write access. The owner accepts or rejects each proposal. Acceptance is new explicit authorisation, subject to normal lifecycle, date, coverage, revision, dependency and integrity checks. Client timestamps do not prove pre-revocation creation; automatic distinction from fabricated prior work is not promised. Concrete protected transport/quarantine/expiry remain implementation validation tasks.

(c) The originating account and the accepting owner are recorded as distinct in the recovery event.

(d) Rejected/conflicting/uncertain outcomes remain distinct and recoverable. A newer version permits acceptance against the exact reviewed older source only while it remains present and eligible, with a newer-version notice. Withdrawn/deleted sources permit inspection/rejection with a reason, not owner-authorised signature validity or automatic resurrection. A replacement needs fresh worker review and signing; rejected-proposal retention remains pending.

Synchronisation metadata lifetime. Retain authorised change-feed/deletion markers for 90 days. Beyond the supported window, require a scoped refresh and reviewed recovery of older proposals; never infer that a missing operation outcome means an unprocessed operation. Validate server-context expiry and anti-resurrection safeguards in the technical slice. These metadata are not a permanent business correction history and do not replace security-event or document retention.

7.6 Availability and Offline Continuity

Offline continuity is operation-specific and follows BR-013. Online availability and prepared field use are separate properties. The 15-day authorisation already agreed in D-22 remains unchanged; additional approved availability/performance targets are coordinated through Q-NFR-02 and Q-NFR-07.

Simple routine interaction. Use the ordinary field forms, clear locally-saved/pending status and automatic sync attempts with a manual retry when connected. Do not require routine outbox management. Genuine conflicts or changed grants need an explicit recoverable exception workflow. The demonstration's simplified interface does not establish production persistence, authorisation enforcement or recovery.

AOC-01 — Prepared offline operational continuity

After online authentication and preparation, the PWA shall support the operations that the use cases designate as available offline.

Rationale: Maintains field continuity with the same forms and operation-specific grants rather than an unrestricted offline application.

Related: FR-004, FR-011, FR-015, FR-016, FR-020, FR-023, FR-028, FR-029, FR-034, FR-037, FR-039, FR-045, FR-048–FR-054, FR-060, FR-062.

(a) Support prepared consultation, optional planning, attendance/work and authorised corrections, deliveries/results, selected-version signing and financial consultation, entry, correction and PDF output under their FR grants. Explicit campaign start and identified same-day worker creation also work offline with stable pending dependencies.

(b) Initial authentication, account/access administration, property/parcel structure, campaign creation/coverage/closure/reopening, existing-worker administration, source-version management and template generation require connectivity. Do not confuse these with the supported worker-create/campaign-start exceptions.

(c) Offline operation is not presented as access to all historical data; unavailable local data is distinguished from an empty result.

(d) Prepared financial values, complete tickets and source files follow the financial/full-content grants in Section 2.4. Do not exclude monetary data from offline views or require content classification/redaction. Structured records and file-transfer readiness remain separate.

(e) Expose readiness for schedules in the current and following week and retained current and previous insurance, including explicitly selected existing workers. Other permitted signing documents require explicit preparation; apply BR-029 without automatically downloading all contracts.

(f) Distinguish metadata and file readiness. Interrupted preparation, unavailable files and insufficient storage must not produce a false ready state; coordinate storage guidance with AOC-05.

AOC-02 — Bounded offline authorisation

A device shall operate offline only for a limited time after its latest successful online permission validation.

Rationale: Balances seasonal field connectivity against the risk of lost devices and delayed revocation.

Related: FR-002, FR-003, FR-004, FR-005, FR-008.

(a) Prepared offline authorisation expires 15 days after the latest successful online validation of permissions, as already defined in FR-004 and D-22. Only another successful online validation renews it.

(b) The 15-day grant is separate from online credential/session expiry, the current/following-week preparation window and the 30-day accepted-cache inactivity cleanup.

(c) On expiry, protected offline access is locked until the device is revalidated online.

(d) Expiry never deletes pending operations: they stay protected in the local queue and are submitted after revalidation, subject to server-side authorisation (including REL-08 for revoked managers).

(e) Protect the validation/expiry context against clock rollback and fail closed when its trusted-time context cannot be established. Keeping server time or detecting rollback is a proposed safeguard, not proof against arbitrary clock manipulation; the final method and boundary tests remain under Q-DOC-07.

(f) Reconnection revalidates current access and permissions before any pending operation is accepted.

AOC-03 — Persistent operation queue and visible states

Every offline change shall be stored durably, and its synchronisation state shall be visible.

Rationale: Prevents a locally saved entry from being mistaken for a server-accepted record and makes connectivity failures recoverable.

Related: FR-003, FR-004, FR-029, FR-034, FR-037, FR-043.

(a) Persist the business proposal and outbox entry atomically before local confirmation, with stable record/operation identities, origin account/workspace, base revision, dependencies, payload and recorded time. File-transfer readiness is tracked separately.

(b) After successful durable local saving, pending records and confirmed signing proposals survive normal PWA closure/reopening, temporary network loss and device restart while the supported browser storage remains intact. This does not guarantee survival after eviction, deliberate clearing or device loss.

(c) Use the states in Section 6.2.5: Pending, Synchronising, Accepted, Transmission failed, Rejected, Conflicted and Outcome uncertain. Routine forms show a simple local/pending/synced presentation and freshness; actionable detail appears for genuine problems, without routine technical queue management.

(d) Browser-storage eviction or device loss before synchronisation remains an explicit residual risk and is not covered by central backups.

AOC-04 — Service availability and synchronisation performance

The online service shall meet an agreed availability target while reconnection throughput follows the shared performance benchmark.

Rationale: Provides concrete quality gates without conflating online uptime with field continuity.

Related: FR-004, FR-029, FR-034, FR-037, FR-048.

(a) Target at least 99.5% availability per complete calendar month with continuous scheduled monitoring, including planned user-impacting maintenance as unavailable. Prepared offline continuity is evaluated separately.

(b) Use PERF-08 and the single reference profile in Section 7.1; do not maintain a competing reconnect benchmark. File-transfer performance remains separate.

(c) Unavailability of the online API does not suppress locally supported offline operations.

(d) Use one external probe per minute to check PWA delivery plus an authenticated database-backed synthetic read within 10 seconds. Availability is successful probes divided by all scheduled probes times 100; report monitoring gaps separately and never as success. Report file-service failures separately if other features remain available.

(e) Retain timestamped probe evidence and coverage. A short trial validates the procedure, not an achieved monthly target. Assign an alert recipient, communication channel and operating responsibility before operational use.

AOC-05 — Local storage capacity and persistence

The PWA shall check, request and report local storage so that offline work is not lost silently.

Rationale: The solution depends on sufficient local storage for pending operations and prepared documents (Section 3.5.3).

Related: FR-004, FR-028, FR-034, FR-039.

(a) Request persistent browser storage where supported and record whether it is granted. This request does not guarantee persistence against user clearing, device loss or browser policy.

(b) Use available browser storage estimates to assess preparation needs and warn on insufficient space. Estimates are advisory; actual writes and quota failures remain authoritative.

(c) If a change cannot be stored locally, the user is told before the interface reports local confirmation.

(d) While pending operations exist, the interface states that unsynchronised data are held only on this device.

AOC-06 — Outage guidance and maintenance communication

Unavailable online operations shall provide timely recovery guidance without blocking unrelated prepared field work.

Rationale: Users need a truthful outcome and a practical next step during outages.

Related: FR-004, FR-034, FR-037, FR-038, FR-043, FR-048, FR-062.

(a) Proposed target: within 30 seconds without observable request progress, replace indefinite loading with truthful failure or uncertain-outcome guidance and retry/prepared alternatives. Measure from submission or last progress; an active file transfer may continue with progress feedback.

(b) Preserve input and pending operations. Timeout neither proves rejection nor permits blind duplicate submission; apply the recorded-outcome/retry workflow.

(c) Announce planned user-impacting maintenance at least 24 hours in advance through the agreed channel; communicate emergency incidents as soon as practicable. Named responsibility/channel and response coverage remain to be assigned.

(d) Exercise API and file-storage outages separately. Central recovery objectives remain under REL-04; communication targets do not promise a restoration time.

8 Solution Architecture

8.1 System Context

OliveTrack is proposed as a progressive web application (PWA), supported by a central business API, relational persistence and private file storage. Phones and tablets are the primary field devices; desktop access uses the same responsive application where supported. The design aims to support quick field entry, prepared offline continuity, explicit central validation and business isolation.

The Farm Owner and invited Field Managers use individual accounts within the owner's business workspace. Workers have business-specific records and may review and sign documents on an authorised user's device; they do not need accounts. The cooperative and laboratory supply

information entered or attached by users. Direct integration with those organisations is outside the current scope. Email delivery is an external dependency for verification, invitations and recovery.

The approved working stack is React with TypeScript, Vite and Tailwind CSS for an installable PWA, IndexedDB for protected prepared data/outbox, one modular Node.js/NestJS backend and PostgreSQL. Shared TypeScript contracts and pure rules support previews; the server remains authoritative. The end-to-end technical validation under Q-ARCH-01 is still required. This description does not establish that a production system, deployment or synchronisation mechanism has been implemented or verified. The intermediate prototype follows Section 3.3; its status and presentation date are recorded in Section 1.4.

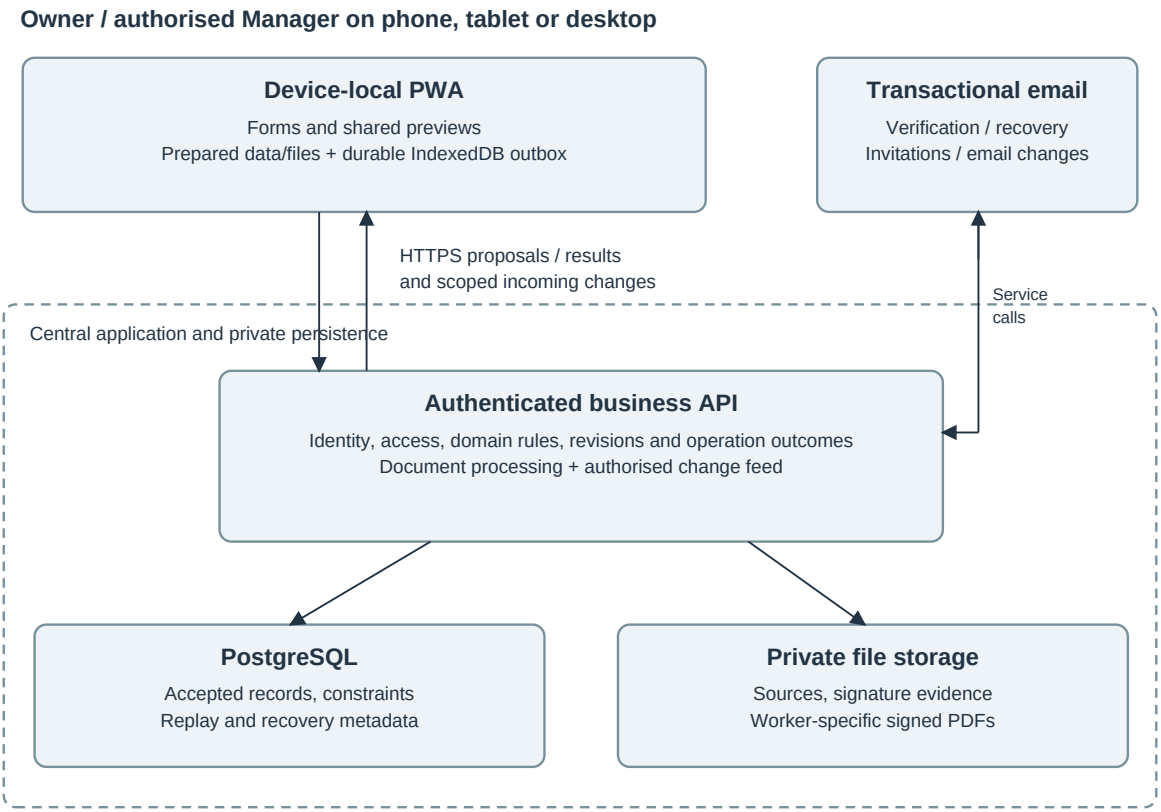


Figure 2: Proposed component responsibilities and authenticated interfaces. Device-local proposals and private central storage remain separate trust boundaries; no browser has direct database access.

8.2 Components and Responsibilities

8.2.1 Client Application and Local Storage

The React PWA uses Vite for builds and Tailwind CSS for responsive presentation. A service worker prepares the application shell; IndexedDB holds the authorised local dataset, file availability metadata and durable outbox. Browser caching alone is not a business-operation queue. Save each supported proposal and its outbox entry atomically before reporting local success.

Prepare only the required scope and permitted source versions/files. Include insurance consultation and selected-version signing under their grants, rather than excluding all contracts or insurance from offline use. Prepared financial consultation, entry, correction and PDF output also follow their grants. Record file readiness separately from structured-record acceptance. Preserve pending

dependencies through normal app closure and restart while browser storage remains intact; clearing storage or losing a device can destroy exclusively local work. Storage checks and warnings follow AOC-05.

The interface uses the business and synchronisation states in Section 6.2.5. A generic failure label must not obscure the difference between transmission failure, rejection, conflict and an uncertain server outcome. Normal field interaction remains simple, with recovery detail shown when action is needed, as defined in Section 7.6. Browser-background execution is not assumed to be continuously available: retry when execution and connectivity permit, and provide an explicit retry/review path.

8.2.2 Identity and Delegated Access

User identifies the account; WorkspaceMembership carries the contextual role and AccessGrant carries permitted operations and owner-assigned responsibility scope. Each owner has one business workspace. A real person working for different owners may have separate Worker records; no global worker directory or cross-business identity merge is introduced. The Owner role does not prove legal property title.

Every protected API read, write, export and file request validates account, membership, operation and scope. Never trust a workspace identifier or role sent by the browser. Preserve the reviewed manager defaults and mandatory scoped insurance consultation in Section 2.4; other document and financial operations retain their independent grants. Property-based scope includes parcels and current/future campaigns, permits multiple managers and filters multi-property views to authorised contributions. Explicit worker availability does not grant visibility of other-property work; see Q-AA-02.

Prepared offline authorisation expires 15 days after the latest successful online permission validation. It is distinct from online-session expiry and 30-day accepted-cache cleanup. Logout blocks local access while protecting recoverable pending proposals; it does not indiscriminately erase the outbox. Offline devices cannot receive immediate revocation or remote wiping. Central access checks, reconnection and the local expiry gate enforce the applicable boundary. The trusted-time, local protection and restricted owner-review mechanisms remain subject to the existing security decisions.

8.2.3 Business API and Domain Modules

The Node.js/NestJS API is organised into cohesive modules in one backend application. This modular structure does not require independently deployed microservices. Module responsibilities follow the current data model:

Table 12: Proposed backend modules.

Module	Responsibility
Access and workspaces	Accounts, verification/recovery, invitations, memberships and operation-specific grants.
Properties and campaigns	Properties, optional parcels, coverage, explicit execution start, closure, reopening and archival rules.
Workforce and work	Identified workers, capabilities, optional plans, daily attendance, actual assignments, periods, tasks and breaks.

Continued on next page

Module	Responsibility
Worker documents	Uploaded/generated source versions, worker associations, templates, signatures, renditions and retention.
Harvest logistics	Delivery-associated transport, actual load origins, driver/vehicle references and weighing tickets.
Analyses and indicators	Delivery analyses, current ticket-reported amounts, clean-net-weighted yield and scoped indicators/PDF summaries.
Synchronisation	Operation outcomes, revisions, ordered dependencies, authorised change feed and reviewed conflict/recovery paths.

8.2.4 Shared Rules and Central Persistence

A shared TypeScript package provides types and pure validation/calculation functions used by client and server. Local results are previews while acceptance is pending. Shared code does not replace server authorisation, complete-data checks or database integrity. The server recalculates and validates accepted changes.

PostgreSQL stores the model in Section 9. Use foreign keys, uniqueness, checks, transactions and suitable cross-record enforcement. Same-workspace references, actual-period non-overlap and exclusive accepted campaign execution need more than ordinary independent row checks. Keep technical idempotency and deletion metadata separate from permanent business field history.

Each delivery has at most one ticket and one analysis, and each ticket has one current reported monetary value. A reported final amount replaces its provisional predecessor; ordinary corrections retain current values. Store reported values and units without deriving clean net weight or applying deductions twice. Calculate aggregates from eligible sources under Section 6.4. Weights, yields and monetary totals include only Delivered, non-cancelled deliveries with the required inputs. Local arrival can update provisional local indicators; central totals require accepted arrival and source records.

8.2.5 Files, Generation and Signing

Files are required architecture components. Keep them private in an object store or equivalent protected service, with authorised access through the API or short-lived scoped links. A private S3 bucket is a deployment candidate; document workflows do not depend on selecting that provider. Validate each file request and keep secrets and reusable privileged credentials out of the browser bundle.

Maintain immutable source identities/content hashes, document versions, per-worker signature evidence and worker-specific derived signed PDFs. Capture the worker's review and confirmation for the selected version; the operator's account identifies the facilitator and does not replace the worker as signer. A later source version needs its own signing event. Template generation under UC-20 and FR-063 produces reviewed unsigned source versions; available templates remain under Q-DOC-01. The workflow does not issue official identity or work-permit evidence.

Retention follows BR-026 and Section 9.5: retain all contracts and protected signed/declared-signed versions; insurance/other cleanup does not remove protected older versions or pending dependencies. The initial 10 MB PDF/JPG/PNG policy also applies to source evidence. Manual entry may be supported by attached ticket/analysis evidence; excluding automatic extraction does not prohibit attaching images. A captured mark and content hash do not establish independently

certified identity or legal validity. Operational suitability, retention duration remains pending. Pending-signature behaviour follows the approved exact-source/conflict policy; protected recovery mechanisms still need verification.

8.3 Main Interaction Flows

1. **Connected preparation.** Authenticate and validate permissions, then prepare selected reference data, schedules and permitted files. Account/access, property structure, campaign creation/coverage/closure/reopening, existing-worker administration, source-version management and generation use connected workflows.
2. **Field entry.** Review and confirm a supported operation, durably save its local proposal and dependencies, and show its actual save state. Prepared explicit campaign starts and identified same-day worker creation are supported offline exceptions; they do not make all administration available offline.
3. **Central acceptance.** Authenticate again as needed, check current authority, operation identity, revision, dependencies and business rules, and commit the outcome consistently. Reconcile the result without duplicate business effects.
4. **Review and completion.** Consult authorised incoming changes and indicators, resolve conflicts explicitly and complete late source information. Campaign closure/correction exceptions follow the use cases and rules, not an append-only restriction on all business records.

8.4 Offline Operation and Synchronisation

The proposed protocol separates durable local proposals from centrally accepted records. Prepared datasets and local/central storage responsibilities are described in Section 9.6. Authentication limits, recovery mechanisms require technical validation even where their behaviour and selected stack are approved.

8.4.1 Local Proposals and Ordered Dependencies

Each proposed operation is identified by a stable `operation_id`, origin User, Workspace, entity type and ID, and operation kind. For validation, it also carries the payload, base revision for updates, dependency IDs and client recording time. The local outbox uses the states in Section 6.2.5, with separate retry detail. Client timestamps do not prove central acceptance or prior authorisation. Use stable client-generated record identifiers for offline creation; the selected UUID generation method must be consistent across devices and server.

The selected local store must persist each business proposal and its outbox entry atomically before the application reports a successful local save. Show local confirmation separately from acceptance. Submit dependencies first: a delivery before its ticket, a version before signing, and accepted metadata/file readiness before dependent references. Basic worker creation and explicit campaign start are supported exceptions. Queue a worker before attendance, assignments, driver references and document associations/signatures that depend on it; queue campaign start before dependent execution records. Administration of existing profiles and property creation remain online.

Use one stable client-generated Worker UUID and operation UUID through local creation and acceptance. Pending references must not be replaced by an unrelated identity if a duplicate warning appears. Resolve an existing/new identity only through an explicit reviewed procedure

that preserves work, document associations and files; no automatic merging or cross-business lookup is introduced.

An explicit start proposes Planned to In progress. The server validates authority, actual dates and competing execution atomically before accepting dependent records. Concurrent offline starts on one area cannot both become accepted; keep the rejected or conflicting start and its dependencies recoverable. A device timestamp is not an authoritative proof of an exclusive reservation.

8.4.2 Central Acceptance, Conflicts and Incoming Changes

For each request: authenticate the account, determine the operation workspace, validate current authority, check operation identity and compare its fingerprint, then validate base revision and domain invariants in a transaction. A reused ID with different content is not the same retry. Commit the business change and its idempotency outcome consistently, and return affected identities/revisions. An outcome ledger stores minimal replay status, not a permanent full copy of every old business row. Retain minimal operation outcomes for 90 days after processing and change/deletion-feed markers for 90 days. Support the preceding client/payload version for 30 days after a release. Server-issued validity context must route stale requests to refresh/reviewed recovery; missing expired metadata never proves that a request is new. Validate safeguards against duplication and resurrection.

An uncertain transmission triggers operation-status checking or safe retry. Never assume that timeout means no central change. A stale base revision produces a conflict, not Last Write Wins. Authorised users compare accepted values and the proposal, then keep accepted data or resubmit a reviewed correction for validation.

The server supplies only changes the account may receive, using a proposed durable cursor. Apply business changes, deletion markers and cursor advancement consistently. A deletion marker is a technical synchronisation fact, not a field-change history. After access reduction, the client removes inaccessible prepared data without submitting it through another account. Beyond the valid change-retention window, refresh authorised data rather than silently missing old deletions.

Track attachment upload separately from structured-record acceptance. Keep failed uploads recoverable and do not expose an unavailable object as stored. A replacement version becomes current only once its source is available. Offline retention cleanup must not revive deleted versions when a pending signature reconnects.

8.4.3 Access Expiry, Logout and Recovery

Logout blocks local access but preserves recoverable pending work in account-isolated protected storage. Renewed authentication is required for normal recovery. Revocation is enforced online and on reconnection or offline expiry; no instantaneous remote wipe of disconnected devices is asserted.

Existing pending manager proposals may enter a restricted Owner-review workflow without restoring manager workspace access. Preserve origin and reviewer, and run all campaign, date, revision and dependency checks on acceptance. Owner acceptance supplies new explicit authority; client timestamps cannot establish pre-revocation creation. The origin account authenticates only for restricted submission, without normal workspace access. Concrete protected transport/quarantine/expiry remains a verification task. Signing-source eligibility cannot be overridden by approval.

Enforce BR-002 through a local gate storing the last successful online permission-validation instant and expiry. Cached use and cleanup cannot renew it. Protect this context against

simple clock rollback; the trusted-time mechanism remains a security design decision. Expiry blocks consultation/new operations but preserves protected recoverable proposals until online revalidation. It does not deliver server changes; the authorised change feed does that.

The routine presentation follows Section 7.6. A simple status does not replace durable outbox storage, dependency checks or idempotency. Local protection and expiry follow PRV-03/AOC-02; the final mechanism remains under Q-DOC-07. Warning/retry timing and concrete local security mechanisms remain pending; approved session/backup/recovery/availability targets are specified by their owning NFRs. REL-08 and Q-NFR-06 govern restricted owner-review recovery, not a normal submission bypass.

8.4.4 Local Indicators and Operation Boundaries

Use the last accepted available baseline plus identifiable local proposals: create once, replace on update and remove the previous contribution on local removal. Label affected values provisional and disclose missing/partial scope and last sync. Apply the same Delivered, non-cancelled eligibility to all delivery-based totals. A local arrival confirmation enables eligible contributions in the provisional view; pre-arrival source data remains in detail awaiting receipt confirmation. Central totals use only server-accepted arrival confirmations and source values. Rejected proposals do not remain part of accepted totals. Never imply knowledge of other-device changes since last synchronisation. Offline PDF may include authorised prepared financial information and records generation time, scope, freshness, missing information and pending data. Financial source provisional/final status is separate from synchronisation status.

The outbox accepts only the operations supported in Section 7; connected administration and source generation use online endpoints. Prepared datasets/file manifests enforce the operation and complete-file restrictions in Section 9.1.7. Coordinate source, evidence and rendition cleanup with protected signing operations; exclusively local signatures may require Q-DOC-04 resolution. Cancellation remains a domain update in the change feed, rather than a deletion marker, preserving dependencies while removing indicator contributions.

Local personal data, protected files and proposals are encrypted and isolated by origin account/workspace. Logout locks access while preserving pending work; normal recovery requires origin-account authentication and current authority. Suspected clock rollback or an untrustworthy expiry context blocks protected access until online revalidation, preserving proposals. No instantaneous lost-device wipe or proof of browser cryptographic feasibility is asserted.

A pending signature remains bound to its exact reviewed source. A newer current version does not invalidate an older present/eligible source automatically; disclose the newer version. Withdrawal/deletion requires protected inspection/rejection, never automatic acceptance, resurrection or signature transfer. Owner approval cannot substitute for the worker's fresh review/signing of a replacement.

8.5 Deployment and Operation

8.5.1 Proposed Hosting Arrangement

Separate static PWA delivery, the application API, PostgreSQL and private file storage. The contribution suggests AWS in an EU region: S3/CloudFront for the application, an evaluated container service for the API, RDS for PostgreSQL, private S3 files, Secrets Manager and CloudWatch. This is an example topology under Q-ARCH-02; it is not a procurement decision, a free-tier guarantee or evidence of operational availability. Evaluate an academic/managed

alternative if necessary without weakening access, integrity or document controls. Application-managed authentication is selected; choose the transactional email provider separately.

The EU-region preference does not itself demonstrate privacy compliance. Review processing locations, provider roles, transfers and costs under Q-NFR-03 and Q-ARCH-02. Use synthetic or appropriately anonymised demonstration data until the processing basis for real personal data has been established. Configure encrypted transport/storage, least-privilege service access and secret management against Section 7.3.

8.5.2 Code Organisation and Delivery

The proposed monorepo separates client, server and shared contracts/rules. Pin dependencies and review generated or contributed code under the same standards. GitHub Actions can run build, lint, unit/integration checks and migration validation before merge/deployment. Docker is proposed for reproducible server packaging; choose one infrastructure-as-code approach, such as Terraform or AWS CDK, only if it fits the team's capacity. These engineering mechanisms remain planned.

Changes to local schemas, service workers and API payloads must preserve pending work and account for older cached clients. Store the client/payload version with sync requests and reject unsupported formats recoverably. Do not force an update that clears queued work. Use reviewed migration and rollback procedures; an old application release is usable only with a compatible schema and payload contract. The approved 30-day preceding-client compatibility policy is specified in REL-07 and Q-NFR-02.

8.5.3 Operational Verification

Assign monitoring, incident, backup and restore responsibilities before operational use. Managed services do not remove application-level configuration or restore testing. Approved availability, daily/14-day backup, RPO/RTO and performance targets are specified in the owning NFRs; do not create independent architecture targets or imply they have been measured. Central backups cannot recover changes that never left a lost device. Security audit events have a separate purpose and retention policy from business correction history.

Measure rendering, durable local saving, central acknowledgement, structured replay and file transfer separately. Synthetic probes must not expose worker data or disable supported offline entry. Validate phone/tablet usability and recovery feedback under Section 7.2 and AOC-06. Test retries, concurrent edits, offline starts/new workers, storage interruption and source/signature retention before claiming the corresponding behaviour works. Section 10 records the project risks affecting delivery.

8.6 Architecture Decisions and Remaining Validation

The reviewed specification already separates planned and actual work, local confirmation and server acceptance, source files and signed renditions, and business corrections and technical replay metadata. Stable identities, idempotent effects and optimistic revision checks with explicit conflict review are retained. Last Write Wins is not an alternative conflict policy in this baseline.

Q-ARCH-01 records the selected stack and the outstanding tested vertical slice; Q-ARCH-02 covers hosting, service selection, costs and operating ownership. These do not reopen the included manager, prepared document/signing flows, authorised offline finance or one-ticket/one-analysis rules. Property scope, owner-authorised recovery and metadata lifetimes are decided.

Concrete local encryption/key handling, trusted-time enforcement, safe expired-context recovery and legal/operational signing suitability retain the specific validations in Section 11.3. The implementation MVP remains subject to Q-SCOPE-01.

Rationale and trade-offs. The following decisions explain the proposal's main quality drivers. They record design reasoning, not comparative benchmark results or final procurement approval.

Table 13: Architecture rationale, trade-offs and validation needs.

Proposed choice	Reason / requirement driver	Trade-off / validation
Responsive PWA with IndexedDB outbox	One phone/tablet interface and prepared field continuity; atomic local proposals support FR-034 and AOC-03.	Browser storage and execution limits require device tests; protection and trusted time remain under Q-DOC-07.
Modular TypeScript API and shared pure rules	Modules follow domain responsibilities; shared previews reduce divergent calculations, while the server enforces REL-03.	Shared code does not supply authority or concurrency guarantees. Validate the selected stack through the technical slice under Q-ARCH-01.
PostgreSQL transactional acceptance	Relationships, uniqueness and serialised validation support single tickets/analyses, attendance and exclusive execution.	Cross-record checks need a tested transaction/locking design, not only independent row checks.
Stable operations and optimistic revisions	Safe retries and explicit conflict review preserve REL-01/REL-02 without silent last-write-wins.	Adds dependency and recovery work; expired-context safeguards and owner-authorised recovery remain verification tasks.
Private object storage plus relational metadata	Keeps source files, signature evidence and derived PDFs distinct under SEC-08 and BR-026.	Database/file readiness and cleanup need recoverable coordination; provider/cost and retention are separate decisions.
Managed hosting candidate	Separates application delivery, API, database and files while supporting configurable backup/monitoring.	Compare an affordable deployment against the agreed NFRs; AWS is an option under Q-ARCH-02, not a free or approved service guarantee.

Required technical slice. Authenticate and prepare an authorised campaign/worker; durably save field work offline; close/reopen the PWA and verify persistence; reconnect and accept centrally; retry the same operation without duplicate effects; then create a competing revision and verify explicit conflict recovery. Include stale-context recovery, local encryption/key lifecycle and fail-closed expiry boundary checks before claiming the corresponding architecture is feasible on the supported devices.

Transactional email. The backend sends verification, reset, invitation and verified-email-change links through a transactional service. Keep credentials server-side, authenticate the authorised sender domain with SPF/DKIM/DMARC, log technical outcomes without links, preserve pending state on send failure and provide controlled resend. Provider acceptance is not

proof of inbox delivery. Select provider/domain/configuration under Q-NFR-05; use controlled test recipients during development.

9 Data Architecture

This section describes the proposed entities, relationships, integrity constraints and storage representations. Sections 4–6 define behaviour; this model must implement those constraints. Types, indexes and technology choices remain design proposals where not yet agreed.

The companion field reference is compiled from `data-architecture-reference.tex`, using `technical/data-architecture/data-dictionary.tex`. It supplies field-level detail omitted from this summary and must be versioned with it. Any conflict with the requirements requires an explicit correction or decision, not silent precedence.

9.1 Conceptual Data Model

The proposed model contains 34 entities/associations, including the new WorkerAvailability access association. The complementary ER views show one model, with contextual entities in grey and endpoint cardinalities explicitly labelled. Each delivery may have at most one weighing ticket and one analysis, both of which may be recorded later. ManagerInvitation and AccessGrant are included. DocumentTemplate and DocumentRendition support the agreed UC-20 and signed worker-specific PDF output. Authentication challenges, outbox operations and recovery ledgers are technical persistence, not additional business actors.

9.1.1 Accounts and Business Access

User is the application identity; Worker is a business-specific person with no required login. Workspace separates one owner's properties, campaigns and records. WorkspaceMembership carries the contextual Owner/Manager role; each active workspace has one Owner and an account owns at most one workspace. ManagerInvitation records recipient and reviewed initial grants; acceptance requires the intended verified account. AccessGrant records actual operations and scope, including the mandatory insurance constraint. Revocation preserves author/initiator references. Email verification challenges do not change the User id or ownership. Workspace stores its business timezone. AccessGrant is keyed by membership, operation and required property; scopes include parcels and current/future campaigns. WorkerAvailability links explicitly available workers to manager membership before scoped work exists; other relevant workers follow authorised scoped records. It confers no other-property work visibility. Account/business deletion procedures remain pending; ownership transfer is excluded. Figure 3 summarises the access relationships.

9.1.2 Properties, Parcels and Campaign Coverage

Property and HarvestCampaign belong independently to a Workspace. Parcel is an optional reusable division of a Property. CampaignCoverage links a campaign to one or more whole properties or selected parcels, allowing a campaign to span several properties without duplicating their records. Figure 4 shows the coverage relationships.

A coverage entry always identifies a property and optionally one of its parcels; an absent parcel denotes whole-property coverage. Operational area references use the same representation. This

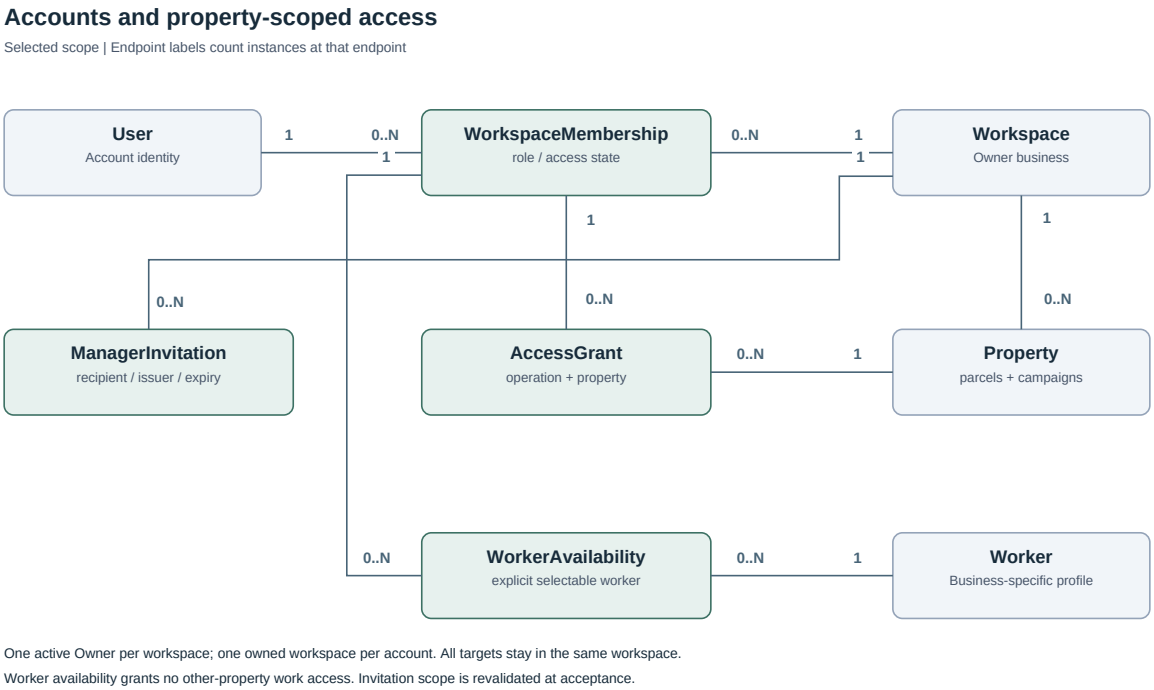


Figure 3: Included account, membership, invitation and owner-configured access relationships.

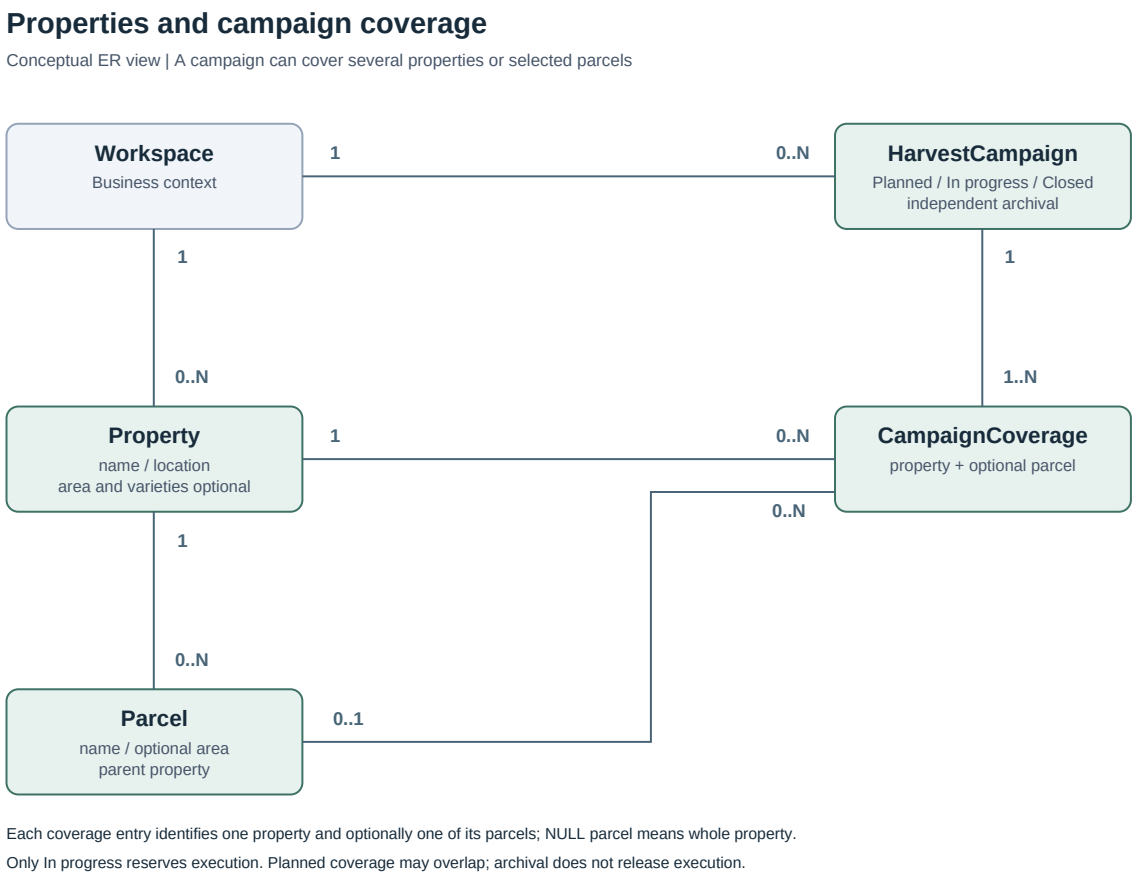


Figure 4: Properties, optional parcels and campaign coverage. A coverage entry always identifies a property and optionally one of its parcels.

preserves the difference between property-level and parcel-level information when divisions are added later.

HarvestCampaign stores planned start, actual start and actual end separately. Lifecycle and archival are separate fields because execution/closure govern dates, while archival controls normal-list visibility. CampaignCoverage is the set against which start/reopening and In progress coverage changes enforce BR-008. This cross-record constraint needs transactional validation; the ER relation alone cannot enforce exclusive execution.

9.1.3 Business Worker Records

Worker belongs to a Workspace and retains a stable internal UUID across planning, attendance, work, driving, document associations and signatures. Mandatory fields are full name, telephone, official identification type, issuing country and number; keep the number as text, separately from the UUID and from optional fiscal data. The type/country/number tuple supports non-unique same-business duplicate lookup; it is neither a primary key nor a global uniqueness constraint. Official document types and validation remain under Q-ID-01. Offline proposals and their dependants retain the same UUID through acceptance, as described in Section 8.4. Profile updates preserve identity and links; machinery capability is stored separately from actual tasks.

9.1.4 Work Planning, Attendance and Actual Work

The model separates expectations, daily presence and actual campaign participation. WorkPlan groups one campaign's assignments for a working date. PlannedAssignment identifies a worker and area, with optional PlannedWorkPeriod and PlannedBreak records. These tables retain the proposal independently of subsequent actual work. Figures 5 and 6 show planning and actual work, respectively.

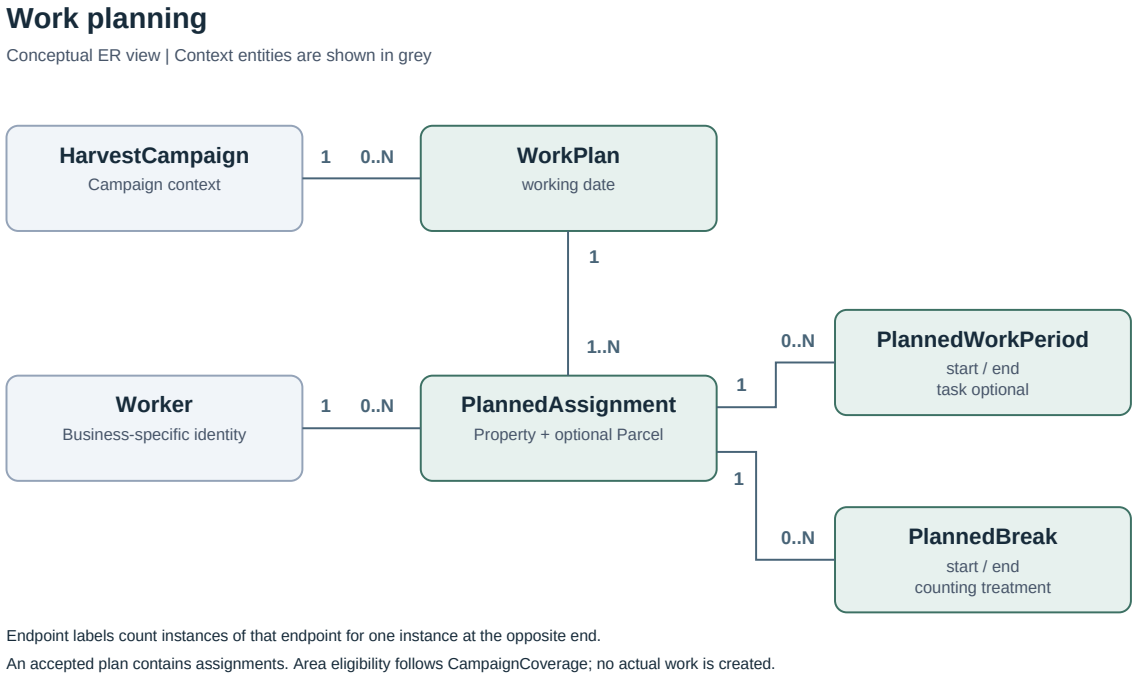


Figure 5: Conceptual relationships for work planning. Context entities are shown in grey; area references are summarised.

Daily presence and campaign participation. DailyAttendance identifies one worker and working date within a workspace, without a campaign reference. ActualWorkAssignment supplies the campaign and area context. This supports one daily attendance decision across several campaigns and records participation before hours or tasks are complete. An actual assignment may refer to a source PlannedAssignment, but no mandatory one-to-one correspondence is imposed.

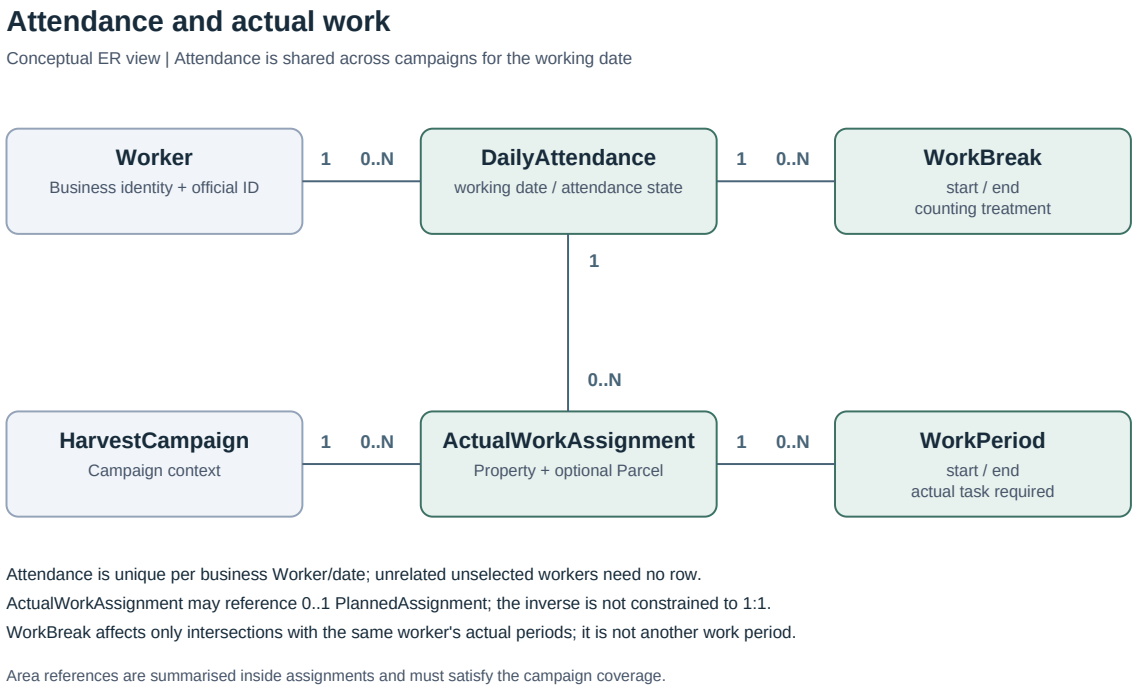


Figure 6: Conceptual relationships for daily attendance and actual work. The optional reference to a planned assignment does not impose a one-to-one correspondence.

Intervals and breaks. WorkPeriod belongs to an actual assignment; WorkBreak belongs to daily attendance. The latter placement lets a worker's break intersect consecutive task or campaign periods without duplicating it. Complete start/end timestamps support intervals crossing midnight and cross-campaign overlap checks. Missing hours are represented by the absence of complete periods; each confirmed complete period requires its actual task. An incomplete assignment is not converted into zero hours.

Planning and confirmation conditions are defined in BR-017–BR-021; the counted-time formula is in Section 6.4.2. Completion and correction use Sections 6.3.2 and 6.3.3, without a separate entity for every historical field value.

Expected versus unrelated workers. DailyAttendance is sparse: it does not require a row for every worker/date. Derive expectations from selected/planned or actual assignments under BR-018. This avoids recording non-selection as absence. Pending records reference locally created Worker UUIDs without creating a second daily attendance after acceptance.

9.1.5 Deliveries, Loads and Cooperative Results

Delivery belongs to one campaign and has one associated TransportRecord, with a registered Worker driver. OliveBatch stores one actual source property and optional parcel, with known

source quantities separate from official ticket weights. TransportRecord may retain a transport-document reference and declared origin; these do not allocate a mixed load or make parcels mandatory. Figure 7 shows the delivery and source-result relationships. WeighingTicket has

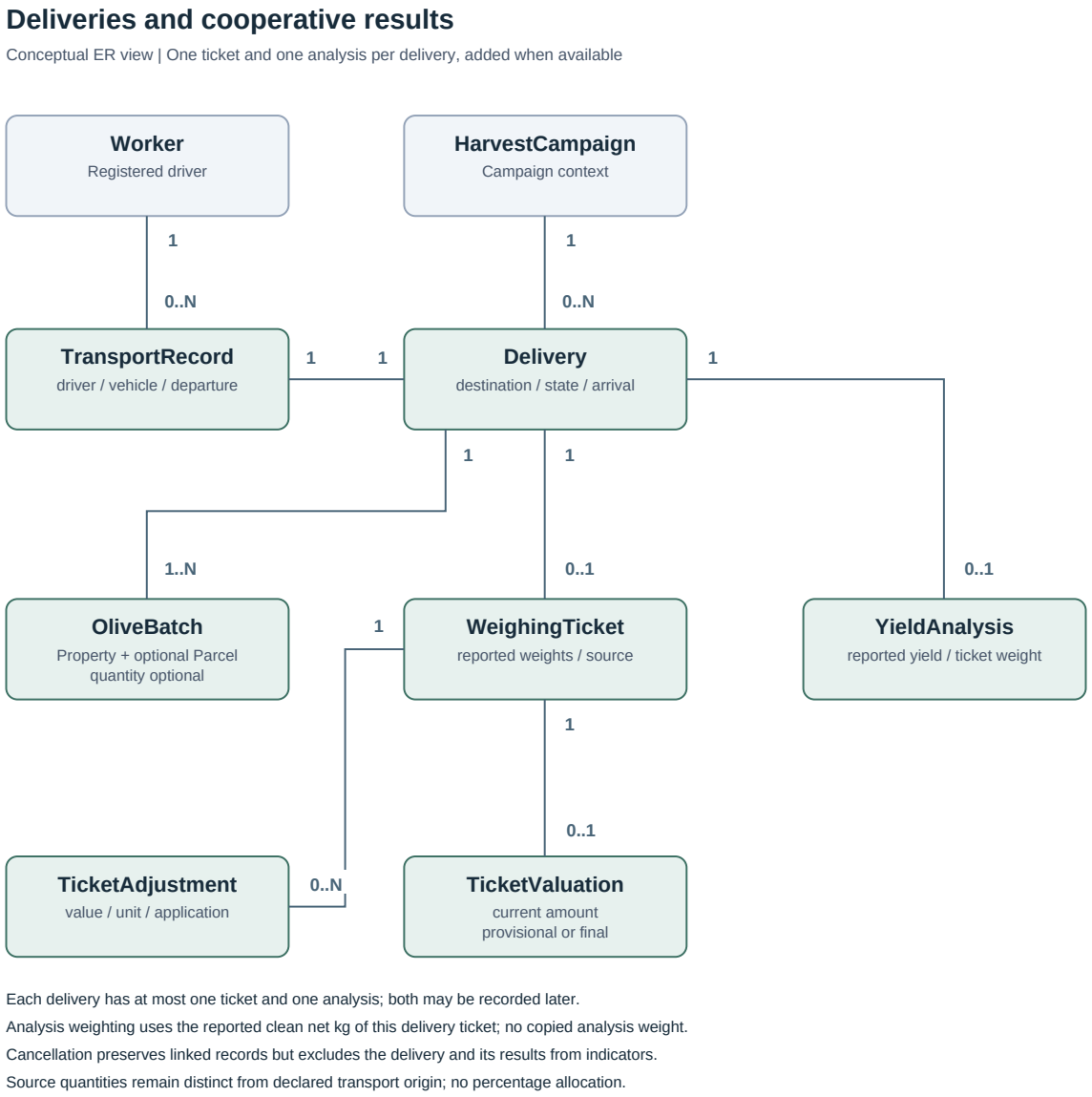


Figure 7: Each delivery has at most one ticket and one analysis. Source batches remain distinct from official weights.

a unique delivery reference and an optional canonical reported clean-net weight in kg. Other weights retain source meanings/units. Optional weighing start/end and reported excess duration are descriptive source facts. TicketAdjustment also represents reported monetary penalties, protected by financial grants; the application does not calculate or reapply them. YieldAnalysis has a unique delivery reference and the reported yield percentage. Its weighting reference is reached through the delivery's ticket; no independently editable reference-weight copy is stored. TicketValuation is one current reported amount per ticket. SourceAttachment supplies separate evidence links. Source records may exist while Delivery is In transport. Indicator queries use the Delivered, non-cancelled parent predicate in BR-038; source existence alone does not confirm receipt. The current model records known quantities in OliveBatch and one total amount in TicketValuation; it has no field for monetary amounts by source. Mixed-load amounts therefore remain unallocated by property/parcel, even when source quantities are known. Supporting

reported source amounts would require an explicit model extension and review. No monetary allocation proportional to weight is inferred. Delivery cancellation stores reason, time and actor while preserving dependent records; lifecycle and correction behaviour follow BR-031. Eligible hard deletion removes composed TransportRecord and OliveBatch rows in the same validated operation, only when no ticket, analysis, valuation or evidence blocks it.

9.1.6 Worker Documents, Generation and Signatures

Document identifies the shared category/title; DocumentWorker identifies covered workers; DocumentVersion holds accepted source contents and version-specific metadata. WorkerSignature binds one associated worker to one reviewed version and identifies the initiating User separately. The worker signs, not the initiating account. Figure 8 shows the shared sources and individual signing results. DocumentVersion separates source identity, currency, withdrawal, recorded expiry

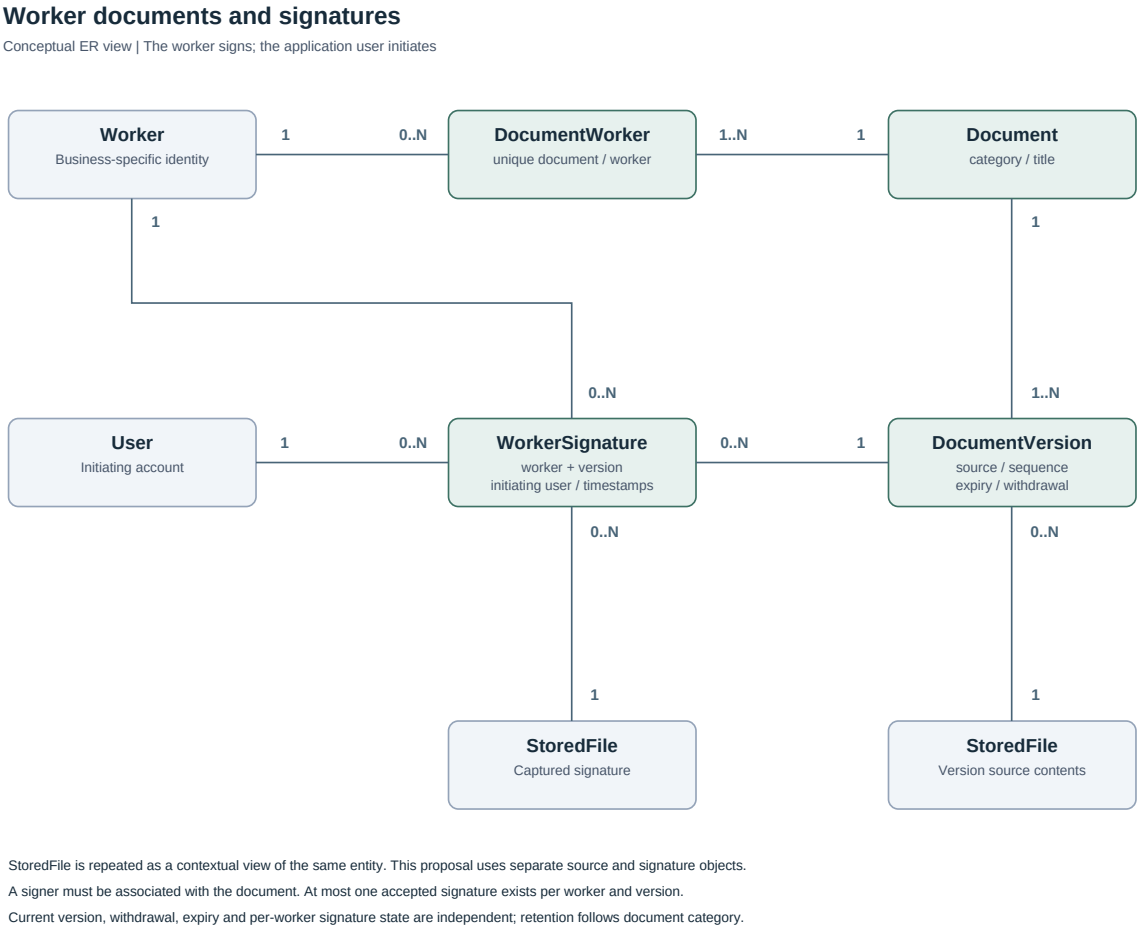


Figure 8: Shared source versions and individual worker/version signing results. StoredFile is one entity shown in contextual views.

and external-signature declaration. WorkerSignature records individual signing; retention/cleanup follows BR-026, including protection of server-known pending operations and preservation for Q-DOC-04 resolution.

DocumentTemplate supplies an available model, category, revision and field schema. Generated DocumentVersion records the template reference/revision and reviewed input snapshot. Signing preserves that source and adds worker-specific evidence and a derived signed PDF under FR-042(d), for uploaded and generated versions alike. DocumentRendition links a derived PDF to that

version and worker. The worker-specific rendition contains only that worker's captured signature; it does not redact a shared source or replace source contents. Regeneration is not another signing event. Figure 9 shows how generation and renditions relate to the source. The template inventory

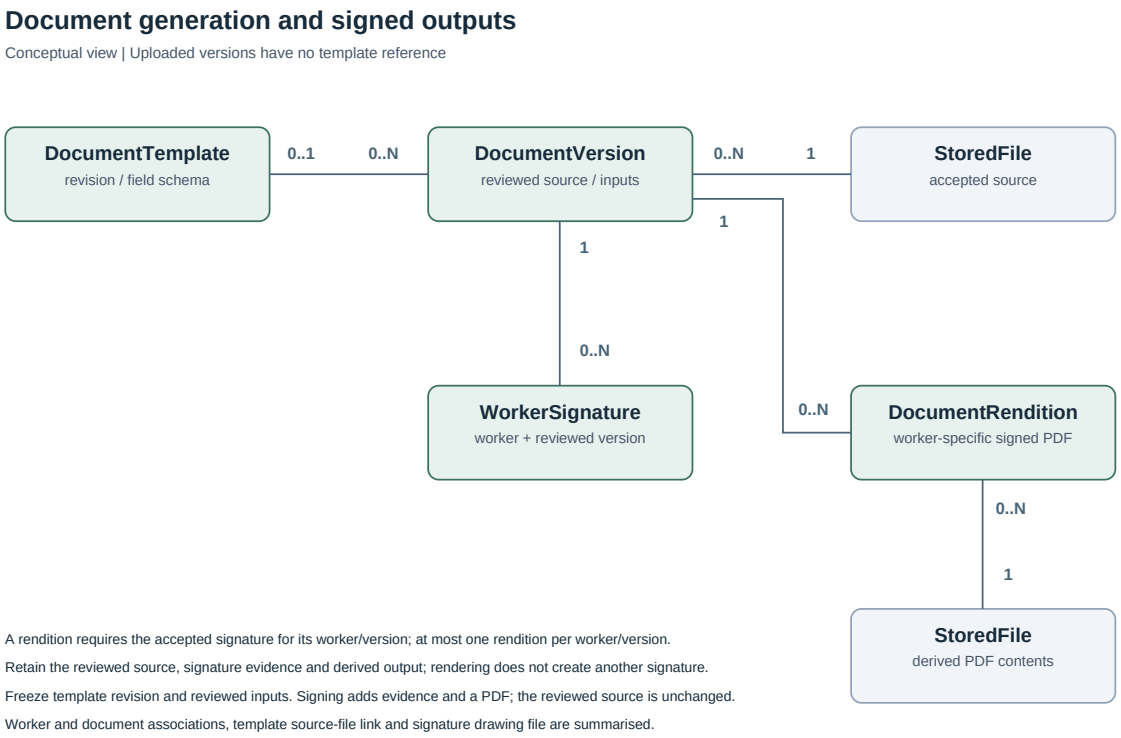


Figure 9: Template-based generation and worker-specific derived signed PDFs. Uploaded versions have no template reference.

remains pending, including review of proposed safety/training forms. Identity/permit originals are uploaded evidence; generation does not issue them. A dedicated attendance export is not implied. Source generation/acceptance requires connectivity, while an accepted prepared version may be signed offline.

9.1.7 Stored Files and Source Evidence

StoredFile holds storage metadata shared by document contents, captured signatures and source evidence. SourceAttachment links one stored file to exactly one ticket, analysis or valuation; the three target relationships are exclusive alternatives. This keeps source evidence independent of worker-document version retention. Figure 10 shows the exclusive attachment targets.

File access is authorised through the associated business record, never the storage key alone. There is no global deduplication or cross-workspace reuse. Remove content only after releasing retained references. SourceAttachment retention/upload recovery remains under Q-AT-02; prepared-file protection remains under Q-DOC-07.

Apply the complete-file consultation and upload grants in Section 2.4. A complete ticket may expose printed monetary values; consultation grants must therefore be configured consistently. Changing or removing a link cannot grant another account access to its file. Worker-document upload remains a separate operation.

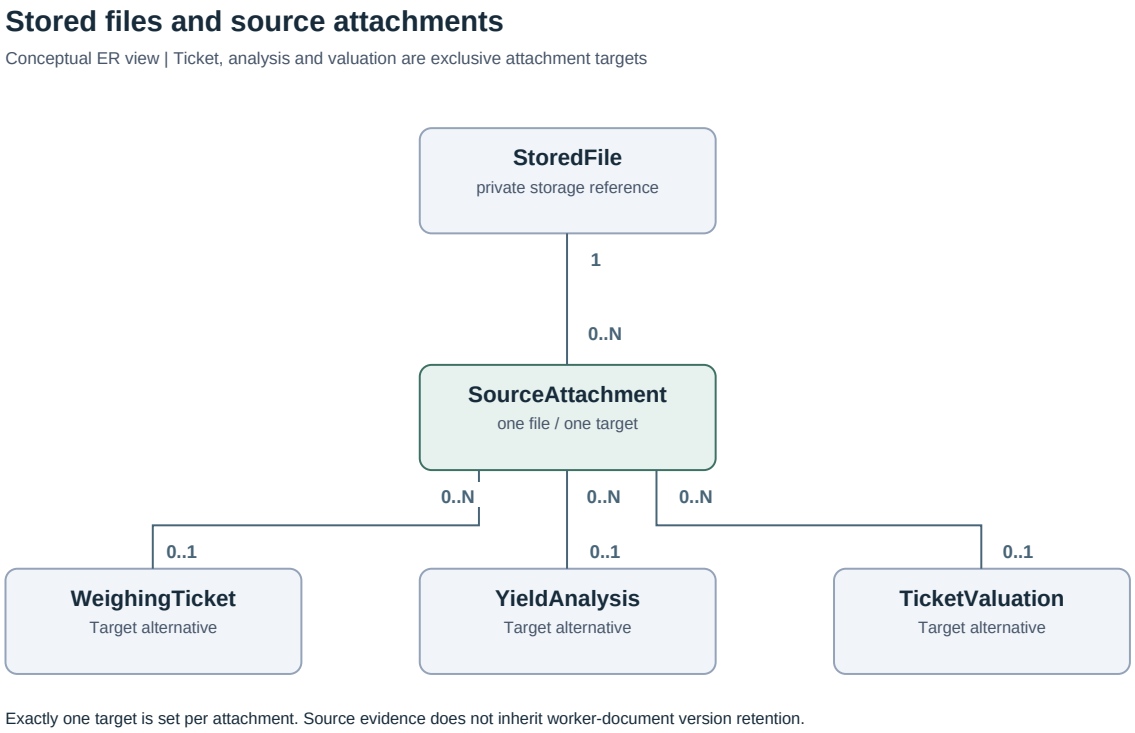


Figure 10: Source attachments and stored content. Each attachment targets exactly one source record; the three target links are exclusive alternatives.

9.2 Logical Data Model

Conventions. Each table has a stable UUID id. Business-scoped tables carry workspace identity; parent, worker, area and file references must stay in that workspace. Shared server metadata comprises creation/update times, author where relevant and a revision counter for concurrency, not a full field-change history. Accepted records satisfy requiredness; a locally pending proposal is not central acceptance. Optional parcel means whole-property scope; area uniqueness must handle NULL explicitly. The table lists principal fields, not the full dictionary. Work timestamps are complete instants; the business timezone determines the start-based working date.

Table 14: Principal entities, fields and relationships in the logical data model.

Entity / table	Principal fields	Relationships / key
User `app_user`	`name`, `normalised_email`, `account_status`	Global identity; unique normalised email. Verified changes preserve id.
Workspace `workspace`	`name`, `business_timezone`, `status`	Business boundary and timezone; Owner through membership.
WorkspaceMembership `workspace_membership`	`user_id`, `workspace_role`, `access_status`, `revoked_at`	User; unique workspace/User pair.

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Entity / table	Principal fields	Relationships / key
ManagerInvitation `manager_invitation`	`recipient_email`, `invited_by_user_id`, `invitation_status`, `expires_at`	Intended verified recipient; issuer and accepted membership.
AccessGrant `access_grant`	`membership_id`, `operation_code`, `property_id`	Required property scope; parcels/current/future campaigns inherit it. Unique membership/operation/property; insurance constraint.
WorkerAvailability `worker_availability`	`membership_id`, `worker_id`	Owner-explicit worker availability; unique membership/worker, same workspace; no other-property work authority.
Property `property`	`name`, `textual_location`, `declared_area_ha`, `olive_varieties`	Workspace; parent of Parcels.
Parcel `parcel`	`property_id`, `name`, `declared_area_ha`, `olive_varieties`	Property in the same workspace.
HarvestCampaign `harvest_campaign`	`name`, `planned_start_date`, `actual_start_date`, `actual_end_date`, `lifecycle_status`	Workspace; Planned/In progress/Closed; one or more coverage entries.
CampaignCoverage `campaign_coverage`	`campaign_id`, `property_id`, `parcel_id`	Campaign, Property, optional Parcel; unique selected area.
Worker `worker`	`full_name`, `telephone`, `official_id_type`, `official_id_country`, `official_id_number`, `can_operate_machinery`, `capability_description`	Workspace; no required User FK.
WorkPlan `work_plan`	`campaign_id`, `work_date`	Campaign; unique campaign/date.
PlannedAssignment `planned_assignment`	`plan_id`, `worker_id`, `property_id`, `parcel_id`	Plan, Worker, area; unique plan/worker/area.
PlannedWorkPeriod `planned_work_period`	`assignment_id`, `worker_id`, `planned_start`, `planned_end`	PlannedAssignment; checked worker reference.
PlannedBreak `planned_break`	`assignment_id`, `break_start`, `break_end`, `counts_as_work`	PlannedAssignment.

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Entity / table	Principal fields	Relationships / key
DailyAttendance daily_attendance	worker_id, work_date, attendance_status	Worker; unique workspace/worker/date.
ActualWorkAssignment actual_work_assignment	attendance_id, worker_id, campaign_id, property_id	Attendance, Campaign, area; optional source plan assignment. Unique attendance/campaign/area.
WorkPeriod work_period	assignment_id, worker_id, actual_start, actual_end, task_code	ActualWorkAssignment; the worker reference must match the associated actual assignment.
WorkBreak work_break	attendance_id, break_start, break_end, counts_as_work	DailyAttendance.
Delivery delivery	campaign_id, destination_name, delivery_status, delivered_at, cancellation_reason	Campaign; one transport and one or more source batches; cancellation preserves dependants.
TransportRecord transport_record	delivery_id, driver_worker_id, vehicle_reference, departure_at, transport_document_reference, declared_origin_property_id, declared_origin_parcel_id	Unique Delivery; Worker driver. Declared origin is separate from OliveBatch.
OliveBatch olive_batch	delivery_id, property_id, parcel_id, collection_origin	Delivery; source area within campaign coverage.
WeighingTicket weighing_ticket	delivery_id, clean_net_weight_kg, reported_weights, weighing_started_at, weighing_ended_at, excess_time_value, excess_time_unit	Unique delivery; canonical reported clean net kg plus optional source timing.
TicketAdjustment ticket_adjustment	ticket_id, adjustment_type, reported_value, unit	WeighingTicket.
YieldAnalysis yield_analysis	delivery_id, yield_percentage, analysis_date	Unique delivery. Weight through that delivery's ticket; no copied reference weight.
TicketValuation ticket_valuation	ticket_id, amount, currency, valuation_status	Unique WeighingTicket; current value only.
Document document	title, document_category	Workspace; worker associations and versions.
DocumentWorker document_worker	document_id, worker_id	Unique Document/Worker pair.

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Entity / table	Principal fields	Relationships / key
DocumentVersion `document_version`	`document_id`, `version_number`, `source_file_id`, `origin_kind`, `template_id`, `template_revision`, `is_withdrawn`	Document/sequence unique; source StoredFile; optional generation snapshot.
WorkerSignature `worker_signature`	`worker_id`, `version_id`, `initiated_by_user_id`, `client_signed_at`	Worker, Version, User, StoredFile; unique worker/version.
StoredFile `stored_file`	`storage_key`, `original_filename`, `media_type`, `byte_size`	Private content; unique storage key.
SourceAttachment `source_attachment`	`stored_file_id`, `ticket_id`, `analysis_id`, `valuation_id`	StoredFile; exactly one populated target FK.
DocumentTemplate `document_template`	`name`, `document_category`, `template_file_id`, `revision`, `field_schema`	Workspace; template StoredFile and reviewed field schema.
DocumentRendition `document_rendition`	`version_id`, `worker_id`, `stored_file_id`, `generated_at`	Unique version/worker; derived PDF StoredFile; associated accepted signature.

Other ticket weights and template/invitation/input schemas use validated JSON; core relationships remain foreign keys. Clean net kg is a canonical typed ticket column, not duplicated in JSON or analysis. A monetary penalty is a reported TicketAdjustment with EUR unit and known application status. Unique delivery references enforce one ticket and one analysis even under concurrent requests. The template revision and input snapshot freeze the generated source context; the rendition never replaces that source. Repeated rendering does not create another worker/version signature.

9.3 Data Dictionary

The complete current field inventory is maintained separately in `technical/data-architecture/data-dictionary.tex`. The following meanings are essential to interpreting the condensed model.

Table 15: Key field groups and their meanings.

Field group	Meaning
Identity and scope	User identifies an account; Worker identifies a person within a business, using a separate UUID. Required worker fields: full name, telephone and official identification type/country/number. Workspace references and composite keys prevent cross-business links.

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Field group	Meaning
Dates and timezone	Business timezone governs work date/display. Full instants preserve overnight and clock-change meaning. Enter minutes; report hours/minutes. Entry, event and acceptance times differ. Only selected/expected workers await attendance; complete confirmed periods require a task.
Delivery state	In transport, delivered or cancelled. Campaign lifecycle is separately Planned, In progress or Closed. Cancellation has reason/actor/time, preserves source records and excludes their contributions.
Reported weights	clean_net_weight_kg is the ticket's canonical cooperative-reported clean net kg. Other weights preserve meanings/units. Unknown is NULL, not zero.
Yield reference	yield_percentage is the reported result. Its weighted-average reference is read through the delivery ticket, not stored again on the analysis.
Weighing and penalties	Start/end and excess duration/unit reproduce the ticket. Monetary penalties are reported adjustments; no scale-rate calculation or second deduction follows.
Load provenance	OliveBatch stores actual origins and known quantities separately from declared transport origin. It has no source monetary amounts; mixed-load money remains unallocated by area. No proportional or dominant-parcel allocation is inferred.
Version protection	Retain all contracts and signed/external-signed versions. Unsigned insurance/other cleanup protects server-known pending signing. Exclusively local signatures may need Q-DOC-04 resolution.
Generation and renditions	Template/revision and reviewed input snapshot identify generated sources. A rendition is a worker-specific derived signed PDF linked to its source version.
Financial/local state	Reported provisional/final status is distinct from pending/accepted synchronisation. Offline financial content requires financial authorisation.
Files	PDF/JPG/PNG uploads have the initial 10 MB limit. Availability/upload state does not prove server acceptance of dependent records.

Source integrity. An accepted document source has a protected `content_hash` under SEC-08; the full dictionary records the conditional requiredness by file purpose. This is integrity metadata, not signer authentication or a replacement for version identity.

9.4 Integrity Constraints

Identity and row-level checks are insufficient for cross-record invariants. Enforce the following at central transactional acceptance, including concurrent requests.

Table 16: Cross-record integrity constraints.

Scope	Essential constraint	Rules
Ownership/access	One owner per active workspace; all business parent/file links in scope. Apply owner-reviewed grants and mandatory insurance access.	BR-001–BR-004
Coverage/work	Parcel belongs to property; assignments fit coverage. Preserve unique attendance and non-overlapping same-worker planned/actual intervals, checked separately. Reopening precedes confirmed changes where required.	BR-005–BR-021
Delivery/results	Accepted delivery has transport and source batch(es). Unique delivery FK on ticket and analysis. Analysis weighting reads the same delivery ticket. Cancelled parents accept no new source results.	BR-031–BR-033
Corrections and cancellation	Validate affected campaign scopes, chronology, coverage and dependent data; no cascade deletion. Cancellation preserves sources and removes contributions. Hard deletion requires no dependent tickets, analyses, valuations or evidence; remove composed transport/batch rows together.	BR-031, BR-038
Values/evidence	One current valuation per ticket. Exactly one target per source attachment; combine all applicable association and complete-file consultation permissions, without mandatory monetary-content classification or automatic redaction.	BR-024, BR-034–BR-036
Versions/signatures	Unique document/version sequence and accepted worker/version signature. Signer is associated with the document. Protect signed/declared-signed versions and pending signing operations known to the server against cleanup. Exclusively local signatures require preservation and possible resolution under Q-DOC-04.	BR-023–BR-030
Generation/rendition	Generated source has reviewed template/revision/input snapshot; accepted rendition uses associated worker/version signature and same-workspace files. Source identity remains immutable.	BR-041
Files/acceptance	Apply upload format/limit checks. No current source before file readiness. Revision/idempotency checks prevent duplicate effects; cleanup respects all retained references.	BR-021, BR-044

Coverage reservation, intervals, sequence allocation and minimum-child counts need transactional enforcement. File cleanup is recoverable across database/object-store boundaries. Signed retention durations, detailed break-entry rules and technical recovery mechanisms still require validation against the approved NFR targets.

Planned campaigns may overlap. Serialise competing Planned-to-In-progress starts, Closed-to-In-progress reopens and execution coverage changes over every relevant property/parcel key, including whole-property versus child conflicts. Actual-date checks remain independent of planned dates and archival. Two accepted competing executions must not result from concurrent offline requests.

Worker name, telephone and official type/country/number are required before accepting a new worker. The internal UUID is the primary key; same-business duplicate matches produce review, not an automatic merge or mandatory unique-document rejection. Apply parent identity and authorised document associations before dependent work, driving or signatures. Every confirmed complete WorkPeriod has its actual task. Do not create unresolved attendance alerts for unrelated unselected workers.

For accepted document sources, enforce the immutable file reference/digest required by SEC-08. Integrity metadata is protected with the associated source; a digest does not establish the worker's identity or make a drawn mark a certified signature.

9.5 Data Lifecycle and History

Archival/inactivation preserves identities and references; campaign archival is independent of closure. Delivery cancellation retains its reason and dependencies under BR-031. Ordinary profile, work, ticket, analysis and amount corrections retain current values rather than full field-change history.

Apply the category-specific retention rule in BR-026 to shared source versions, signature evidence and derived PDFs as a linked graph. Release files only after all protected references permit it. Generated snapshots remain unchanged when profiles/templates change. Exclusively local signatures whose source becomes unavailable remain recoverable under Q-DOC-04.

Deletion markers and access changes propagate through the protocol in Section 8.4; offline expiry is an independent access gate, not delivery of those changes. Restoring backups must not bypass current access or deletion decisions. Signed/source-evidence retention periods, account/business deletion remain pending; backup/recovery targets are approved under REL-04; ownership transfer is excluded.

Apply PRV-04/PRV-07 and REL-04 to rights handling, local deletion propagation and restore suppression without erasing protected pending evidence. Security-event records under SEC-07 use a separate controlled schedule, not business field-change history. The approved technical retention/recovery targets in Section 7 do not settle signed-document retention under Q-DOC-03.

9.6 Local and Central Data

The central database holds accepted records; private object storage holds accepted files. IndexedDB is selected to store account/workspace-isolated prepared subsets and durable proposals. Synchronisation follows Section 8.4.

Table 17: Local preparation and central storage responsibilities.

Data group	Prepared / pending locally	Central responsibility
Reference/plans	Assigned properties, coverage, workers and prepared plans; supported planning proposals.	Authoritative identity, grants, availability, campaign state and plan acceptance. Structure and existing-profile administration remain online; explicit starts and basic worker creation have pending dependencies.

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Data group	Prepared / pending locally	Central responsibility
Attendance/work	Prepared attendance, assignments, intervals/breaks and authorised correction proposals.	Unique daily state, intervals, dates, scopes, revision/conflict validation.
Documents/signing	Current/previous prepared insurance, explicitly selected retained files and accepted versions for signing; pending drawings and worker-specific renditions.	Version/file readiness, signer eligibility, evidence, protected signed retention and accepted rendition. Source generation remains online.
Deliveries/results	Prepared deliveries, tickets/clean net kg, analyses and supported entry/correction/cancellation proposals; separate attachment status.	Single ticket/analysis constraints, provenance, cancellation/dependency validation and source files.
Financial information	Permitted prepared values, penalties, financial/mixed files and consultation/entry/correction proposals.	Financial authorisation, current valuation and accepted source amounts. No unrestricted preparation or automatic redaction.
Indicators/PDF	Available baseline plus identifiable proposals; local arrival may enable provisional results. Financial results require grants. Record scope, freshness, exclusions and pending status.	Server-accepted arrivals and source values; only Delivered, non-cancelled parents contribute under BR-038.

Operation envelopes, revisions, change cursors, deletion markers and dependency ordering follow Section 8.4. The local store retains complete pending worker identification and stable UUID references across work, driving, document links and signatures. Accept parent identities, authorised associations and campaign starts before dependants, preserving proposals needing resolution.

Local loss may destroy proposals never received centrally. Offline access follows BR-002; local-only/cloud-only selectable modes are outside scope. Routine interaction follows Section 7.6.

9.7 Physical Data Model

PostgreSQL is the selected central relational store; exact DDL, indexes and transactional enforcement still require implementation and verification.

Table 18: Proposed physical data representations.

Family	Proposed representation
Identity/scope	UUID primary and operation ids; composite scope keys support same-workspace references. Integer revision counters support concurrency.

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Family	Proposed representation
Time	Full event/signing/acceptance instants use timestamptz; date for work/campaign/expiry dates. Workspace business_timezone stores a validated timezone identifier. This type preserves the instant but not the originally supplied timezone/offset. The business timezone is stored separately and governs local display and working-date assignment. Minute-level input does not reduce timestamps to time-of-day values.
Quantities/money	numeric for clean net kg, other weights, areas, yield and amounts. Monetary amounts use two decimal places; other precision/ranges remain for review.
Source/configuration	Schema-validated jsonb for other reported weights, invitation configuration, template fields and reviewed generation inputs. Core associations remain FKs.
Files	Private object storage for source, template, signature and rendition bytes; StoredFile metadata and server-side format/size/access validation. Initial upload limit: 10 MB; exact byte convention pending.

The instant/timezone distinction follows the PostgreSQL date/time documentation.

Essential keys and indexes. Use unique membership, coverage/assignment, attendance, plan, document-sequence, worker/version signature and ticket-valuation keys. Unique delivery references enforce one ticket/analysis; a unique version/worker key identifies a rendition. Index campaign/area/date queries, worker intervals, document associations and attachment targets. Owner and whole-property/parcel uniqueness need explicit partial/NULL handling; minimum-child counts and dependency checks are transactional.

Enforce planned and actual same-worker overlaps separately through exclusion or equivalent serialised validation. Serialise execution reservations across a property and its parcels, including start/reopening and coverage changes. Unlocked read-then-insert checks or ordinary indexes alone are insufficient.

Store required official identification as text, preserving leading zeroes; use a non-unique workspace/type/country/normalised-number lookup index. Worker and operation UUIDs support client creation. Campaign lifecycle values are `planned`, `in_progress` and `closed`.

The physical design also requires an operation-outcome ledger, authorised change/deletion feed and protected owner-review recovery store. Preserve template snapshots and retained signed/cancelled-record references. Local prepared subsets, outbox and file metadata need not mirror every central table. Local authorisation metadata stores online-validation and expiry instants; its approved policies and unresolved implementation mechanisms follow Sections 6–8.

SEC-08 requires an integrity digest for each accepted document source, stored with protected metadata and verified before association. Keep client confirmation and server acceptance timestamps distinct. Salted password hashes, session/token verifiers, security-event logs, deletion/restriction recovery metadata and deployment key controls are technical persistence concerns, not additional business actors/entities. Their implementation and approved technical retention limits follow Section 7 and Section 7.

AccessGrant has required property scope and a unique membership/operation/property key; there is no independent campaign/parcel scope discriminator. WorkerAvailability uses same-workspace membership/worker FKs and a unique pair. Validate invitation scope at creation and acceptance,

including revoked/archived property context. Security events use a 90-day server-event policy; minimal operation outcomes and change/deletion markers use their separate 90-day policies. Store originator and accepting owner separately for reviewed recovery. These mechanisms do not add ordinary field-change history or settle signed-document retention.

10 Risk Management

The working assessment contains twelve risks and proposed responses. It follows the STRA identification/analysis stages and continuous review cycle. It is a desk-based consolidation of the team's contribution; no facilitated workshop, complete nine-person vote or completed mitigation is claimed. Ratings and named response ownership still require team validation under Q-PM-01.

10.1 Approach and Assessment Scales

Identify a present condition and its possible adverse consequence, keeping the statement short and recording context/triggers separately. Review three project-specific source groups: technical design/integration, delivery/capacity, and stakeholder validation. This grouping organises the register; it is not a claim that a full SEI taxonomy questionnaire was completed. An actual defect or missed milestone enters the issue log, while its possible wider effects remain risks. Communication accompanies identification, analysis, planning, tracking and control throughout delivery.

The supplied probability bands are judgement categories, not measured frequencies: Low below 40%, Medium 40% through 70%, High above 70%. Impact is measured against the candidate threshold of success: Marginal permits success with limited rework; Critical permits it only with substantial effort or adjustment; Catastrophic prevents it within the agreed baseline. Timeframe means the time until action is needed: Short within one month, Medium one to three months, Long beyond three months. It is not the expected duration of an incident. Confirm timeframes against the actual remaining project horizon.

Exposure uses the categorical lookup in Table 19, not an unexplained multiplication of ordinal labels. The matrix contains the current provisional risk placements and is consistent with the register.

Table 19: Qualitative exposure matrix and provisional risk placement.

Probability	Marginal impact	Critical impact	Catastrophic impact
High	Medium: RS-04, RS-12	High: RS-01, RS-07, RS-08, RS-09	Critical: RS-06
Medium	Low: RS-03, RS-05, RS-11	Medium: RS-02, RS-10	High: none identified
Low	Low: none identified	Low: none identified	Medium: none identified

10.2 Candidate Threshold of Success

The STRA process uses a threshold of success (ToS): a small set of measurable, time-bound conditions that must all be met. Table 20 makes the proposed conditions and evidence explicit. They do not yet constitute an agreed ToS: the MVP, final delivery date, severity rules and resource commitments remain approval inputs. The intermediate submission on 6 November 2026 and

the prototype presentation on 10 November 2026 are separate milestones. The final delivery and defence dates remain to be agreed under Q-PM-01.

Table 20: Candidate success conditions and remaining baseline inputs.

ID / area	Condition and evidence at the agreed delivery	Input still to agree
ToS-1 Scope	All selected Must-Have criteria in the end-to-end owner/manager workflow have recorded acceptance evidence, including the selected field/document paths.	Selected criteria, approver and final milestone; Q-SCOPE-01.
ToS-2 Schedule	Required software/prototype, repository, report and presentation are delivered by their agreed milestone dates, with dated submission evidence.	Final calendar and responsibilities; Q-PM-01.
ToS-3 Quality	No unresolved defect blocks an agreed acceptance scenario, permits unauthorised access, silently loses accepted data, duplicates effects or invalidates accepted totals. Evidence identifies criteria checked and defects remaining.	Severity/acceptance rules and test scope; Section 11.
ToS-4 Field use	Selected prepared phone/tablet tasks pass the agreed offline/reconnection and usability criteria, with durable proposals, dependency validation and no silent conflict overwrite.	Approved scenario/configuration subset; Section 11.1.
ToS-5 Resources	Actual effort and service expenditure remain within the approved effort/infrastructure budget or a recorded authorised revision.	Applicable per-member effort and service limits; Section 10.6.

10.3 Risk Register

Statements separate the present source of concern from its possible loss. Context, trigger and proposed role owner accompany the short statement; probability, impact and timeframe are inherited from the contribution pending team confirmation. *ToS* numbers identify potentially affected success conditions, not proof of approved priorities. Re-rate impacts when the selected scope and evidence are known.

Table 21: Working risk register and qualitative assessment.

ID	Condition, consequence and trigger	Assessment / proposed owner
RS-01	**Architecture validation.** The selected stack and candidate services still require an integration slice; incompatible choices may delay implementation or cause rework. *Source:* Technical / integration. *ToS:* 1, 2, 5. *Trigger:* First integration slice blocked by an unresolved contract, dependency or service choice.	High / Critical / Short. **Exposure: High.** Backend/architecture lead.
RS-02	**Offline synchronisation.** Field workflows require durable queues, dependency ordering and concurrent updates; integration complexity may cause reconciliation failures or delay the agreed delivery. *Source:* Technical / integration. *ToS:* 1, 3, 4. *Trigger:* Retries duplicate effects, pending work disappears, or stale revisions overwrite accepted data in tests.	Medium / Critical / Short. **Exposure: Medium.** Backend/synchronisation lead.

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ID	Condition, consequence and trigger	Assessment / proposed owner
RS-03	**Outdoor interaction.** Phone/tablet field use is in scope, but representative outdoor validation is not evidenced; unsuitable controls may prevent efficient logging. *Source:* Technical / usability. *ToS:* 1, 4. *Trigger:* Representative phone/tablet users cannot complete the selected tasks under agreed conditions.	Medium / Marginal / Medium. **Exposure: Low.** UI/usability lead.
RS-04	**Regression readiness.** The report contains planned checks but no verified CI/regression evidence; missing automated checks may allow defects to reach a milestone demonstration. *Source:* Technical / integration. *ToS:* 2, 3. *Trigger:* A change reaches integration without passing the agreed build and core-rule checks.	High / Marginal / Short. **Exposure: Medium.** Testing/integration lead.
RS-05	**Contribution consistency.** Multiple contributors share client, server and rule contracts; inconsistent implementation or unreviewed generated code may create incompatible dependencies and rework. *Source:* Technical / integration. *ToS:* 2, 3, 5. *Trigger:* An unreviewed dependency, incompatible interface or unexplained generated block enters a proposed merge.	Medium / Marginal / Short. **Exposure: Low.** Technical reviewer.
RS-06	**Academic capacity.** Nine students share project work with other academic commitments; reduced availability may leave insufficient effort for implementation and acceptance. *Source:* Delivery / capacity. *ToS:* 1, 2, 5. *Trigger:* Forecast work exceeds available person-hours or an owner cannot meet a dependent task date.	High / Catastrophic / Short. **Exposure: Critical.** Project coordinator.
RS-07	**Scope expansion.** The catalogue is not yet prioritised for implementation; uncontrolled additions may displace essential development and validation work. *Source:* Delivery / scope. *ToS:* 1, 2, 5. *Trigger:* An unestimated feature or excluded payroll/payment/mapping capability enters the delivery backlog.	High / Critical / Short. **Exposure: High.** Project coordinator.
RS-08	**Client clarification.** Ticket/document interpretation depends on client clarification through an intermediary; delayed answers may block correct implementation or acceptance. *Source:* Stakeholder validation. *ToS:* 1, 2, 3. *Trigger:* A decision needed by a scheduled task has no validated answer or agreed fallback by its review date.	High / Critical / Short. **Exposure: High.** Client intermediary.
RS-09	**Task ownership.** Nine members contribute across connected deliverables; unclear ownership or stale tracking may create gaps, duplicated effort and uneven workload. *Source:* Delivery / coordination. *ToS:* 1, 2, 5. *Trigger:* A deliverable lacks one accountable owner/date or completed work is absent from the shared tracking.	High / Critical / Short. **Exposure: High.** Project coordinator.

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ID	Condition, consequence and trigger	Assessment / proposed owner
RS-10	**Integration friction.** Parallel client, API and schema changes share contracts; incompatible changes may break integration and consume the planned validation period. *Source:* Technical / integration. *ToS:* 2, 3, 5. *Trigger:* A dependent branch cannot merge/build or a contract/migration change has no coordinated consumer update.	Medium / Critical / Short. **Exposure: Medium.** Integration lead.
RS-11	**Handover continuity.** Delivery tasks need continuing ownership despite changing availability; incomplete handovers may leave decisions or unfinished work without an accountable owner. *Source:* Delivery / coordination. *ToS:* 1, 2. *Trigger:* A departing/unavailable owner has no documented handover, backup or accessible work.	Medium / Marginal / Medium. **Exposure: Low.** Project coordinator.
RS-12	**Representative validation.** Acceptance depends on representative users, devices and source examples that are not yet fully evidenced; synthetic scenarios alone may conceal unsuitable workflows. *Source:* Stakeholder validation. *ToS:* 1, 3, 4. *Trigger:* Planned validation lacks a representative user, device or sanitised source example.	High / Marginal / Medium. **Exposure: Medium.** Client intermediary / usability lead.

The assessment column lists probability / impact / timeframe, followed by exposure and proposed role owner.

10.4 Priorities and Response Plans

The source's six focus risks are retained as a provisional intervention list. Exposure puts RS-06 first and the four High-exposure risks next; RS-02 also receives early attention because offline behaviour is a prerequisite for the field workflow. This explains its inclusion alongside, rather than automatically above, other Medium-exposure risks. Reassess this tie-breaker once the ToS and implementation plan are approved. A low vote never removes a mandatory security/integrity check.

Multi-voting is optional. The supplied eight-column ballot does not establish a complete nine-member vote. If repeated with four votes per participant, nine participants would cast 36 votes across the twelve risk statements. Record actual participation, discuss ties and discrepancies, and agree the ranking before treating it as a team result.

Table 22: Initial intervention priorities and affected project work.

Risk	Response strategy	Why act / linked work
RS-06	Reduction	Critical exposure affects deliverable capacity; reconcile actual/remaining person-hours and owner availability (Section 10.6).
RS-01	Avoidance / reduction	High exposure affects implementation dependencies; validate Q-ARCH-01/Q-ARCH-02 with the authenticated vertical slice.

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Risk	Response strategy	Why act / linked work
RS-07	Avoidance	High exposure threatens scope/effort; apply Q-SCOPE-01 and the change process in Section 3.6.
RS-08	Reduction	High exposure blocks source/template acceptance; obtain examples for Q-DA-02, Q-DOC-01 and Q-DOC-02.
RS-09	Reduction	High exposure threatens ownership and dependencies; reconcile task board, contribution review and hours evidence.
RS-02	Reduction	Medium exposure with core field dependency; implement/test REL-01–REL-03 and AOC-02/AOC-03 before claiming verified offline behaviour.

10.4.1 RS-06 — Academic Workload and Capacity

Prevent. Collect availability and assessment dates, split delivery into small owned tasks, reserve integration/test time and review remaining effort against actual capacity. Identify backups for dependent work. Do not assume nine people can all work simultaneously or replace elapsed lead time with person-hours.

Respond. When capacity is insufficient, rebalance tasks and propose a reviewed scope reduction under Q-SCOPE-01. Do not silently remove required permissions, integrity checks or evidence. Record the effect on estimates and milestones. **Owner/review:** Project coordinator; weekly and immediately when a task dependency is at risk.

10.4.2 RS-01 — Architecture Validation

Prevent. Validate the selected stack and API/data contracts before expanding implementation. Build one minimal path: authenticate, prepare permitted data, save an identified work proposal durably, reconnect and accept it idempotently in PostgreSQL. Include a stale-revision case. Record deployment/service choices under Q-ARCH-01 and Q-ARCH-02.

Respond. If a service or library blocks that path, evaluate a simpler compatible implementation with a recorded decision and migration impact. Avoid an unreviewed stack switch or bypassing central authority. **Owner/review:** Backend/architecture lead; next implementation planning session and at each blocked integration checkpoint until the slice passes.

10.4.3 RS-07 — Scope Expansion

Prevent. Maintain the included/excluded scope in Section 3, select the first functional delivery explicitly and estimate additions before accepting them. Payroll, wage calculations, payments, invoicing and automatic ticket extraction remain excluded. Keep the agreed cooperative-reported values and campaign indicators; they are not payroll or accounting.

Respond. Defer optional additions or exchange scope through a reviewed decision with effort, dependencies and acceptance impact. Do not promise new CSV/Excel exports as a mitigation; use only outputs already selected for delivery. **Owner/review:** Project coordinator; each new request and milestone review.

10.4.4 RS-08 — Client Clarification

Prevent. The client intermediary collects focused questions, requests sanitised ticket/document examples and schedules concise validation sessions. Connect each blocking question to Section 11.3, an owner and the task that needs the answer. Use synthetic examples while protecting real worker data.

Respond. If the client is unavailable, record an explicitly provisional team assumption and develop only reversible parts that do not depend on an unvalidated business/legal conclusion. Escalate blocking decisions; the intermediary does not acquire binding product/legal approval authority. **Owner/review:** Client intermediary; weekly and before dependent-task dates.

10.4.5 RS-09 — Task Coordination

Prevent. Maintain the shared task board with one accountable owner, reviewer where needed, due date, dependencies and completion evidence. Record actual person-hours in the existing shared sheet. Use brief asynchronous updates for completed work, next work and blockers; create a handover for absences.

Respond. Rebalance overloaded or unowned work in the team review and update the plan and estimates. Reconcile document contributions before final report integration. **Owner/review:** Project coordinator; weekly and whenever ownership or availability changes.

10.4.6 RS-02 — Offline Synchronisation

Prevent. Implement and test the protocol in Section 8.4 early: stable record/operation IDs, atomic local saves, ordered dependencies, idempotent server effects, optimistic revisions and explicit conflict review. Test normal restart, uncertain outcomes, concurrent starts/edits, worker creation, authorisation expiry and protected pending data. A transport operation log is not an append-only restriction on editable business records.

Respond. If automatic replay is unreliable, expose a clear authorised retry/review action while fixing it. A manual trigger uses the same durable queue, permissions and conflict validation; it does not introduce Last Write Wins or discard pending proposals. If necessary, revise first-delivery scope explicitly. **Owner/review:** Backend/synchronisation lead; each vertical-slice increment and before demonstrating verified offline behaviour.

10.4.7 Responses for the Remaining Risks

The remaining risks retain monitoring and proportionate actions even when not in the first six priorities.

Table 23: Responses for other monitored risks.

ID	Prevention and response
RS-03	Review phone/tablet workflows under the approved USA criteria and representative outdoor conditions. If tasks fail, simplify the affected screen and repeat the affected task checks.
RS-04	Establish build and core-rule regression checks before broad integration. Block/review a failing merge, fix the cause and repeat affected regression checks.

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ID	Prevention and response
RS-05	Require shared contracts, dependency review and peer review for all code regardless of authoring tool. Rework incompatible contributions before merge and record required follow-up.
RS-10	Integrate small changes against reviewed contracts/migrations. Resolve conflicting branches with the relevant owners and rerun affected build/integration checks.
RS-11	Record a backup and handover for changed ownership. Assign an interim owner and recover outstanding decisions/work when the trigger occurs.
RS-12	Obtain sanitised examples and remote walkthroughs on representative devices. If field validation is unavailable, state the limitation and keep dependent assumptions open.

10.5 Monitoring, Ownership and Escalation

Pepe and Mateo maintain the risk assessment contribution. The role owners in the register are proposed response responsibilities; assign a named person and backup before treating an action as scheduled. Review priority risks weekly and the full register at agreed milestones, scope changes and significant incidents. Escalate active triggers immediately when they threaten delivery, access or data integrity.

Each review records date, named owner, status (Open, Monitoring, Materialised or Closed), rating, action/due date, trigger evidence and residual exposure. These are fields to maintain, not fabricated current results. Materialised risks become issues with actions; closure needs a reason and evidence. Link mitigation work to the backlog, remaining estimates and the hours sheet. Reassess exposure after action; planning mitigation alone does not demonstrate reduced residual risk.

Client questions are coordinated through the intermediary without granting binding business/legal approval authority. Communicate assumptions and changes at planned checkpoints, seeking explicit feedback on scope, schedule, resources and quality. Retain detailed votes/action history in the project artefacts; update the report when the working baseline changes.

10.6 Planning and Effort Summary

Detailed estimates remain in the separately submitted *Estimations.xlsx*; actual work remains in the shared hours sheet. This summary refers to the workbook snapshot reviewed on 5 November 2026, not a newly approved resource baseline.

Its 56 tasks use three-point expected effort $E = (B + 4M + W)/6$, totalling approximately 872.1 person-hours. Against the workbook's assumed 810 hours (nine members at 90 hours), the gap is 62.1 hours, or 7.7%. Applicability of that course guidance per programme and the alternative 950-hour ceiling still need confirmation.

Reconcile outdated scope assumptions and actual/remaining effort before committing the plan. Forecast total effort is actual plus remaining effort, without double counting. Link tasks to requirements, one accountable owner, availability, dependencies and completion evidence; effort divided by nine does not establish dates or a critical path.

Track task evidence on the board and person-hours in the shared sheet (meetings: duration times actual attendees). Review remaining effort, risks and changes together. Detailed estimates and assignments remain separate project artefacts; MVP, final dates, capacity and risk commitments remain under Q-SCOPE-01/Q-PM-01/Q-PM-02.

11 Validation and Key Decisions

11.1 Validation Approach

Validate each functional and non-functional requirement against its lettered criteria, recording configuration, preconditions, expected/observed result, reviewer, date and evidence. Use Section 5.2 for the existing use-case mapping rather than repeating a traceability matrix here.

Use functional tests and client review for business behaviour; authorisation and cross-workspace tests for access; interruption/retry/concurrency tests for offline integrity; measured workloads for PERF; representative tasks for USA; and a database-plus-files restore rehearsal for recovery. Test property-scoped campaign views, owner recovery as new authority, signing-source conflicts and expired synchronisation context explicitly. The required architecture slice is described in Section 8.

Documented decisions, demonstrated interactions, implemented functionality and verified results are distinct. Approval of a target does not establish measured feasibility. Preserve detailed test cases and execution evidence in the repository; the source file `sections/validation/nfr-scenarios.tex` remains a supporting planned checklist, outside this report's compiled body.

11.2 Key Decisions and Rationale

The table summarises the current specification and the decisions approved in the author's review. It does not assert separate whole-team/client, procurement or legal approval. Detailed behaviour belongs to the referenced requirements/rules. Historical decision/question anchors remain as link targets for compatibility; the full discussion history is a separate project working artefact.

Table 24: Current key decisions and rationale.

Topic	Decision / rationale
Business identity and access D-01, D-09, D-16, D-21	One owner workspace; distinct account/worker identities; owner-invited managers, independent operation grants and mandatory scoped insurance. Financial consultation is initially enabled and revocable. See Section 2.4.
Campaign lifecycle D-02, D-03, D-04, D-05, D-06, D-23, D-24	Explicit execution and exclusive area coverage; complete valid missing work after closure, reopen for confirmed-work corrections. Existing late ticket/analysis/value updates do not require reopening. See BR-008–BR-012.
Source and indicator meaning D-10, D-11, D-12, D-13, D-30	At most one ticket/analysis per delivery; Delivered-only eligible totals, recorded clean net kg and weighted known yields; no invented mixed-source monetary allocation. Registered drivers and authorised delivery correction/cancellation follow BR-031. See Section 6.4.
Documents and signatures D-07, D-14	Category-specific retention, immutable signed source/evidence and worker-specific output. Withdrawal is distinct from replacement; generation is included but concrete templates remain pending. See BR-026–BR-030.

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Topic	Decision / rationale
Field records and local boundaries D-15, D-17, D-19, D-20, D-22, D-25, D-26, D-27, D-28	Optional planning, daily presence and task-classified actual work are distinct. Support prepared starts/identified new workers offline; 15-day authority, separate 30-day cache cleanup and 10 MB upload policy remain. See Sections 6 and 7.6.
Specification and evidence D-18, D-29, D-31, D-32, D-33, D-34	The navigable prototype, selected design, implementation and verified results are distinct. Quality targets are now approved as below; procurement, measured feasibility, risk/resource commitments and first functional scope are not implied.
Recovery authority D-08	Owner acceptance is new explicit authority after restricted origin-account submission; preserve origin/reviewer and validate domain constraints. No client-clock proof of pre-revocation creation is claimed. See FR-008, BR-004, REL-08.
Property delegation Q-AA-02	Property-based grants cover parcels/current/future campaigns; shared responsibility is permitted. Filter partial campaign records/indicators and other-property worker work. Explicit worker availability is represented separately. See Section 2.4.
Selected stack Q-ARCH-01	React/TypeScript, Vite, Tailwind, installable PWA, IndexedDB, one modular NestJS backend and PostgreSQL. The stack is selected; the offline-to-central technical slice remains unverified. See Section 8.
Authentication policy Q-NFR-01	Application-managed Argon2id; 15-character minimum, with support for passphrases of at least 64 characters; five failures/15 minutes, 15-minute access and seven-day absolute session; 24-hour verification, 30-minute reset, seven-day invite. See SEC-01/02/09/10.
Recovery and continuity targets Q-NFR-02	Daily encrypted backups retained 14 days, RPO at most 24 hours, RTO at most four hours; one full restore rehearsal. Monthly availability at least 99.5%, planned downtime included. Preceding-client support: 30 days; operation/feed metadata: 90 days. See REL/AOC requirements.
Security audit Q-NFR-04	90-day server-event retention, restricted access and minimal data; offline file openings are excluded. No permanent ordinary field-change history. See SEC-07.
Performance and interaction Q-NFR-07, Q-NFR-08	The reference fixtures, samples, timing limits and selected accessibility targets in PERF/USA are adopted for this working specification. English is approved for academic artefacts; operational language remains to validate. No benchmark/usability result is claimed.
Offline signing conflicts Q-DOC-04	Exact reviewed source only; eligible retained older source may be accepted with a newer-version notice. Withdrawn/deleted sources require protected inspection/rejection; owner approval cannot validate them. Replacement requires fresh worker review/signing.

11.3 Pending Validations

Only residual choices and verification tasks are listed below. Assign a responsible person and target date when agreed; no dates or resources are invented here.

Table 25: Residual validations and their impact.

Topic	Remaining validation	Responsibility
Hosting and operations Q-ARCH-02	Select affordable HTTPS application/API hosting, private file storage, database region/backup configuration, quotas, credentials and named monitoring/incident ownership. EU region is a reviewed preference, not proof of compliance or confirmed resources.	Team
Transactional email Q-NFR-05	Backend transactional sending and protected credentials, authenticated domain, minimal messages, controlled resend and technical monitoring are approved. Select provider, sender domain and configuration; validate terms/transfers and controlled test sending.	Team
Local security proof Q-DOC-07	Encryption, account/business isolation, logout locking, bounded offline authority and fail-closed untrusted time are required. Validate concrete encryption/key lifecycle, trusted time, lost-device and recovery behaviour on supported devices.	Team
Configuration and technical evidence Q-AA-04, Q-AA-05, Q-TECH-01	Benchmark Argon2id parameters; define progressive delays, non-login rate limits, TLS/HSTS/session mechanism and warning/retry timing. Verify owner-recovery isolation, expired-context anti-duplication, signature quarantine and 90-day metadata safeguards. Targets are approved; execution evidence is pending.	Team
Privacy and closure Q-NFR-03, Q-AA-07	Validate purposes/lawful bases, notices, provider roles/transfers, incident/rights responsibility and account/business closure/deletion with retained data and pending proposals.	Client/controller; team
Protected retention Q-DOC-03	Agree signed/source/evidence retention durations and permitted erasure, including rejected signing proposals. No automatic pending/signed cleanup is introduced.	Client; team
Ticket interpretation Q-DA-02, Q-DA-08	Obtain representative tickets; confirm weight labels, deduction order, units, ranges and completeness. Clean net kg remains entered, never inferred.	Client; team
Delivery/value exceptions Q-DA-09, Q-DA-10	Validate final-to-provisional reclassification and the one-associated-transport model with real examples; interrupted/refused/reinstated trips are not silently added workflows.	Client; team
Templates and signing use Q-DOC-01, Q-DOC-02	Confirm template inventory/fields, operational signing suitability and whether a dedicated attendance export is included; generation remains in scope.	Client; team

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Topic	Remaining validation	Responsibility
File policy details Q-DOC-06, Q-AT-02	Check representative files, exact 10 MB byte convention, source-attachment retention and protected upload recovery.	Client; team
Work/recovery detail Q-WF-03, Q-WF-04	Validate break-entry details and destination context/revision recovery without changing agreed counted-time/correction rules.	Team; client
Filters and precision Q-IND-01	Confirm ticket/analysis/financial date-filter bases and percentage display precision; working-date and arrival-date meanings remain fixed.	Client; team
Identification vocabulary Q-ID-01	Confirm official document types, issuing-country vocabulary and safe normalisation/validation; required entry does not imply legal verification or a scan.	Client; team
First functional increment Q-SCOPE-01	Select implementation MVP/priorities against estimates, capacity and acceptance expectations. The navigable demonstration is not a functional release.	Team; client
Management baseline Q-PM-01, Q-PM-02	Confirm risk ratings/owners, applicable effort budget, scope cutoff, actual/remaining estimates, task ownership/dependencies, final dates, process and AI-use inventory. Keep workbook, hours and detailed planning separately.	Team; course constraints

Operational language/locale and exact usability devices still need client/team confirmation; the academic interface is English. Reference hardware/network/server profiles, actual performance/usability measurements, restore rehearsal and monthly availability evidence remain to be recorded against approved targets. Assign the maintenance/incident channel and alert recipient before operational use.